Rinse both hands for 5 seconds under tap water at the completion of the scrub.

Discard the brush.

Note: This same procedure will be followed exactly in steps 4, 6, and 8 of figure 44.1.

Step 3 With a *fresh* sterile brush, scrub the hands into Basin B in a manner that is identical to step 1. Don't use soap. Notify Group B when this basin is ready.

Note: Exactly the same procedure is used in steps 5, 7, and 9 of figure 44.1, using Basins C, D, and E.

Remember: It is important to use a fresh sterile brush for the preparation of each of these basins.

After Scrubbing After all scrubbing has been completed, the scrubber should dry his or her hands and apply hand lotion.

Making the Pour Plates

While the scrub is being performed, the rest of the class will be divided into five groups (A, B, C, D, and E) by the instructor. Each group will make six plate inoculations from one of the five basins (A, B, C, D, or E). It is the function of these groups to determine the bacterial count per milliliter in each basin. In this way we hope to determine, in a relative way, the effectiveness of scrubbing in bringing down the total bacterial count of the skin.

Materials:

30 veal infusion agar pours—6 per group
1 ml pipettes
30 sterile Petri plates—6 per group
70% alcohol
L-shaped glass stirring rod (optional)

1. Liquefy six pours of veal infusion agar and cool to 50° C. While the medium is being liquefied, label two plates each: 0.1 ml, 0.2 ml, and 0.4 ml. Also, indicate your group designation on the plate.
2. As soon as the scrubber has prepared your basin, take it to your table and make your inoculations as follows:
   a. Stir the water in the basin with a pipette or an L-shaped stirring rod for 15 seconds. If the stirring rod is used (figure 44.2), sterilize it before using by immersing it in 70% alcohol and flaming. *For consistency of results all groups should use the same method of stirring.*
   b. Deliver the proper amounts of water from the basin to the six Petri plates with a sterile serological pipette. Refer to figure 44.3. If a pipette was used for stirring, it may be used for the deliveries.
   c. Pour a tube of veal infusion agar, cooled to 50° C, into each plate, rotate to get good distribution of organisms, and allow to cool.
   d. Incubate the plates at 37° C for 24 hours.
3. After the plates have been incubated, select the pair that has the best colony distribution with no fewer than 30 or more than 300 colonies. Count the colonies on the two plates and record your counts on the chart on the chalkboard.
4. After all data are on the chalkboard, complete the table and graph on the Laboratory Report.

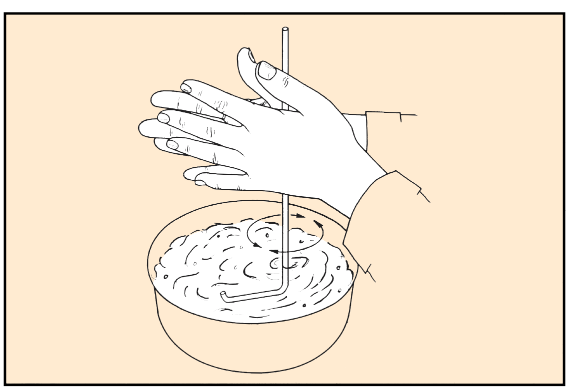


Figure 44.2 An alternative method of stirring utilizes an L-shaped glass stirring rod.

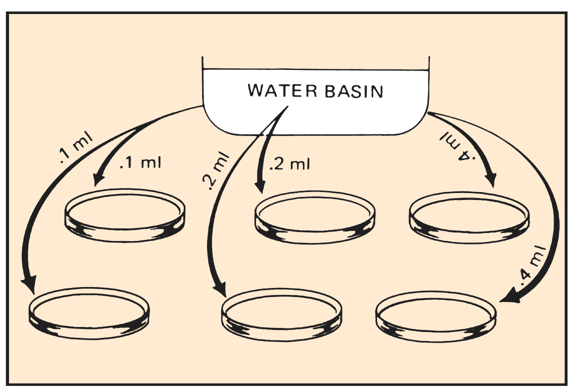


Figure 44.3 Scrub water for count is distributed to six Petri plates in amounts as shown.

PART 8 Identification of Unknown Bacteria

One of the most interesting experiences in introductory microbiology is to attempt to identify an unknown microorganism that has been assigned to you as a laboratory problem. The next seven exercises pertain to this phase of microbiological work. You will be given one or more cultures of bacteria to identify. The only information that might be given to you about your unknowns will pertain to their sources and habitats. All the information needed for identification will have to be acquired by you through independent study.

Although you will be engrossed in trying to identify an unknown organism, there is a more fundamental underlying objective of this series of exercises that goes far beyond simply identifying an unknown. That objective is to gain an understanding of the cultural and physiological characteristics of bacteria. Physiological characteristics will be determined with a series of biochemical tests that you will perform on the organisms. Although correctly identifying the unknowns that are given to you is very important, it is just as important that you thoroughly understand the chemistry of the tests that you perform on the organisms.

The first step in the identification procedure is to accumulate information that pertains to the organisms' morphological, cultural, and physiological (biochemical) characteristics. This involves making different kinds of slides for cellular studies and the inoculation of various types of media to note the growth characteristics and types of enzymes produced. As this information is accumulated, it is recorded in an orderly manner on **Descriptive Charts,** which are located toward the back of the manual with the Laboratory Reports.

After sufficient information has been recorded, the next step is to consult a taxonomic key, which enables one to identify the organism. For this final step, *Bergey's Manual of Systematic Bacteriology* will be used. Copies of volumes 1 and 2 of this book will be available in the laboratory, library, or both. In addition, a CD-ROM computer simulation program called *Identibacter interactus* may be available, which can be used for identifying and reporting your unknown. Exercise 51 pertains to the use of *Bergey's Manual* and *Identibacter interactus.*

Success in this endeavor will require meticulous techniques, intelligent interpretation, and careful recordkeeping. Your mastery of aseptic methods in the handling of cultures and the performance of inoculations will show up clearly in your results. Contamination of your cultures with unwanted organisms will yield false results, making identification hazardous speculation. If you have reason to doubt the validity of the results of a specific test, repeat it; *don't rely on chance!* As soon as you have made an observation or completed a test, record the information on the Descriptive Chart. Do not trust your memory—record data immediately.

45 Preparation and Care of Stock Cultures

Your unknown cultures will be used for making many different kinds of slides and inoculations. Despite meticulous aseptic practice on your part, the chance of contamination of these cultures increases with frequency of use. If you were to attempt to make all your inoculations from the single tube given to you, it is very likely that somewhere along the way contamination would result.

Another problem that will arise is aging of the culture. Two or three weeks may be necessary for the performance of all tests. In this period of time, the organisms in the broth culture may die, particularly if the culture is kept very long at room temperature. To ensure against the hazards of contamination or death of your organisms, it is essential that you prepare stock cultures before any slides or routine inoculations are made.

Different types of organisms require different kinds of stock media, but for those used in this unit, nutrient agar slants will suffice. For each unknown, you will inoculate two slants. One of these will be your reserve stock and the other one will be your working stock.

The **reserve stock culture** will *not* be used for making slides or routine inoculations; instead, it will be stored in the refrigerator after incubation until some time later when a transfer may be made from it to another reserve stock or working stock culture.

The **working stock culture** will be used for making slides and routine inoculations. When it becomes too old to use or has been damaged in some way, replace it with a fresh culture that is made from the reserve stock.

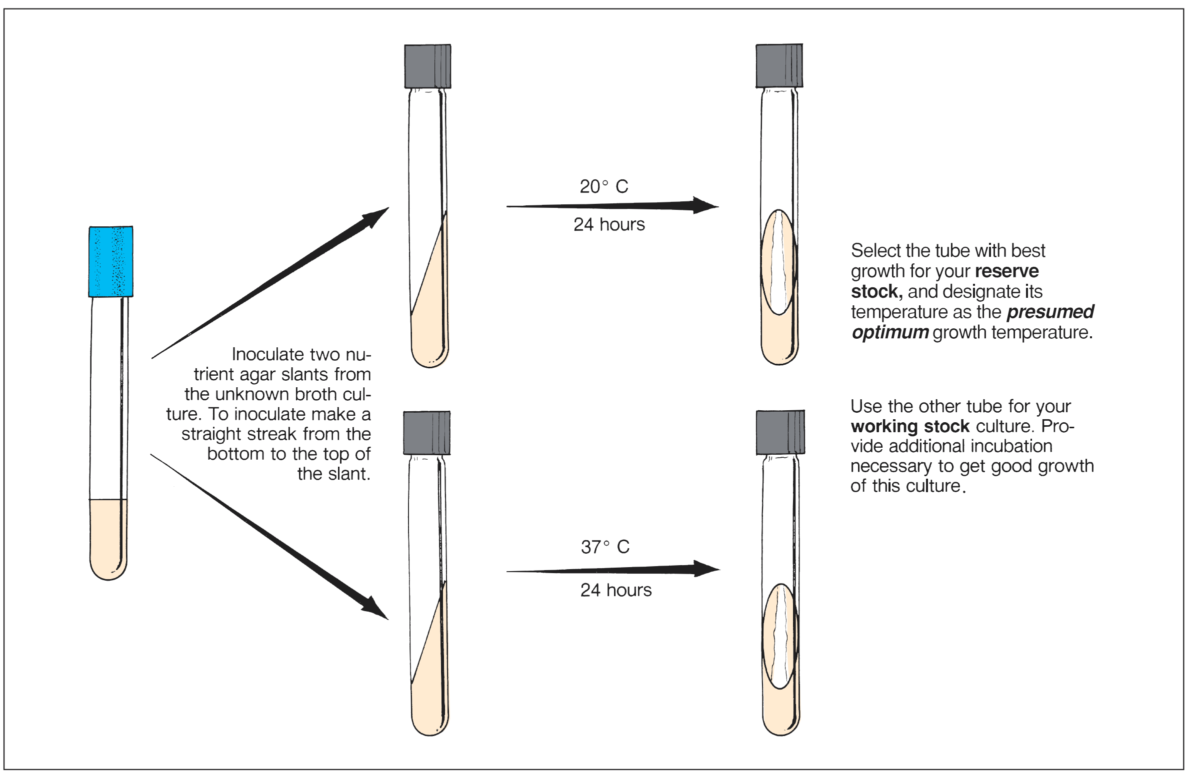


Figure 45.1 Stock culture procedure

Note in figure 45.1 that one slant will be incubated at 20° C and the other at 37° C. This will enable you to learn something about the optimum growth temperature of your unknown, which will be pertinent in Exercise 47. Proceed as follows:

First Period

Inoculate two nutrient agar slants from each of your unknowns as follows:

Materials:
for each unknown:
2 nutrient agar slants (screw-cap type)
gummed labels

1. Label two slants with the code number of the unknown and your initials. Use gummed labels. Also, mark one tube 20° C and the other 37° C.
2. With a loop, inoculate each slant with a straight streak *from the bottom to the top.* Since these slants will be used for your cultural study in Exercise 47, a straight streak is more useful than one that is spread over the entire surface.
3. Place the two slants in separate baskets on the demonstration table that are designated with labels for the two temperatures (20° C and 37° C).

   Although the 20° C temperature is thought of as "room temperature," it should be incubated in a biological incubator instead of leaving it out at laboratory room temperature. Laboratory temperatures are often quite variable in a 24-hour period.

Second Period

After 24 hours incubation, evaluate the slants made from each of your unknowns, as follows:

1. Examine the slants to note the extent of growth. Some organisms require close examination to see the growth, especially if the growth is thin and translucent.
2. Determine which temperature seems to promote the best growth.
3. Record on the Descriptive Chart the *presumed* optimum temperature. (Obviously, this may not be the actual optimum growth temperature, but for all practical purposes, it will suffice for this exercise.)
4. If there is no growth visible on either slant, there are several possible explanations:
   - It may be that the culture you were issued was not viable.
   - Another possibility might be that the organism grows too slowly to be visible at this time.
   - Or, possibly, neither temperature was suitable!

   Think through these possibilities and decide what you should do to circumvent the problem.
5. Label the tube with the best growth **reserve stock.** Label the other tube **working stock.**
6. If both tubes have good growth, place them in the refrigerator until needed.

   If one tube has very scanty growth, refrigerate the good one (reserve stock) and incubate the other one at the more desirable temperature for another 24 hours, then refrigerate.
7. Remember these points concerning your stock cultures:
   - Most stock cultures will keep for 4 weeks in the refrigerator. Some fastidious pathogens will survive for only a few days. Although none of the organisms issued in this unit are of the extremely delicate type, don't wait 4 weeks to make a new reserve stock culture; instead, make fresh transfers every 10 days.
   - Don't use your reserve stock culture for making slides or routine inoculations.
   - Don't store either of your stock cultures in your desk drawer or a cupboard. After the initial incubation period cultures must be refrigerated. After 2 or 3 days at room temperature, cultures begin to deteriorate. Some die out completely.
8. Answer the questions on the Laboratory Report.

46 Morphological Study of Unknown

The first step in the identification of an unknown bacterial organism is to learn as much as possible about its morphological characteristics. One needs to know whether the organism is rod-, coccus-, or spiral-shaped; whether or not it is pleomorphic; its reaction to gram staining; and the presence or absence of endospores, capsules, or granules. All this morphological information provides a starting point in the categorization of an unknown.

Figure 46.1 illustrates the steps that will be followed in determining morphological characteristics of your unknown. Note that fresh broth and slant cultures will be needed to make the various slides and perform motility tests. Since most of the slide techniques were covered in Part 3, you will find it necessary to refer to that section from time to time. Note that gram staining, motility testing, and measurements will be made from the broth culture; gram staining and other stained slides will also be made from the agar slant. The rationale as to the choice of broth or agar slants will be explained as each technique is performed.

As soon as morphological information is acquired be sure to record your observations on the Descriptive Chart at the back of the manual. Proceed as follows:

Materials:

- gram-staining kit
- spore-staining kit
- acid-fast staining kit
- Loeffler's methylene blue stain
- nigrosine or india ink
- tubes of nutrient broth and nutrient agar
- gummed labels for test tubes

New Inoculations

For all of these staining techniques you will need 24–48 hour cultures of your unknown. If your working stock slant is a fresh culture, use it. If you don't have a fresh broth culture of your unknown inoculate a tube of nutrient broth and incubate it at its estimated optimum temperature for 24 hours.

Gram's Stain

Since a good gram-stained slide will provide you with more valuable information than any other slide, this is the place to start. Make gram-stained slides from both the broth and agar slants, and compare them under oil immersion.

Two questions must be answered at this time: (1) Is the organism gram-positive, or is it gram-negative? and (2) Is the organism rod- or coccus-shaped? If your staining technique is correct, you should have no problem with the Gram reaction. If the organism is a long rod, the morphology question is easily settled; however, if your organism is a very short rod, you may incorrectly decide it is coccus-shaped.

Keep in mind that short rods with round ends (coccobacilli) look like cocci. If you have what seems to be a coccobacillus, examine many cells before you make a final decision. Also, keep in mind that *while rod-shaped organisms frequently appear as cocci under certain growth conditions, cocci rarely appear as rods.* (*Streptococcus mutans* is unique in forming rods under certain conditions.) Thus, it is generally safe to assume that if you have a slide on which you see both cocculike cells and short rods, the organism is probably rod-shaped. This assumption is valid, however, only if you are not working with a contaminated culture!

Record the shape of the organism and its reaction to the stain on the Descriptive Chart.

Cell Size

Once you have a good gram-stained slide, determine the size of the organism with an ocular micrometer. Refer to Exercise 5. If the size is variable, determine the size range. Record this information on the Descriptive Chart.

Motility and Cellular Arrangement

If your organism is a nonpathogen make a wet mount or hanging drop slide from the broth culture. Refer to Exercise 19. This will enable you to determine whether the organism is motile, and it will allow you to confirm the cellular arrangement. By making this

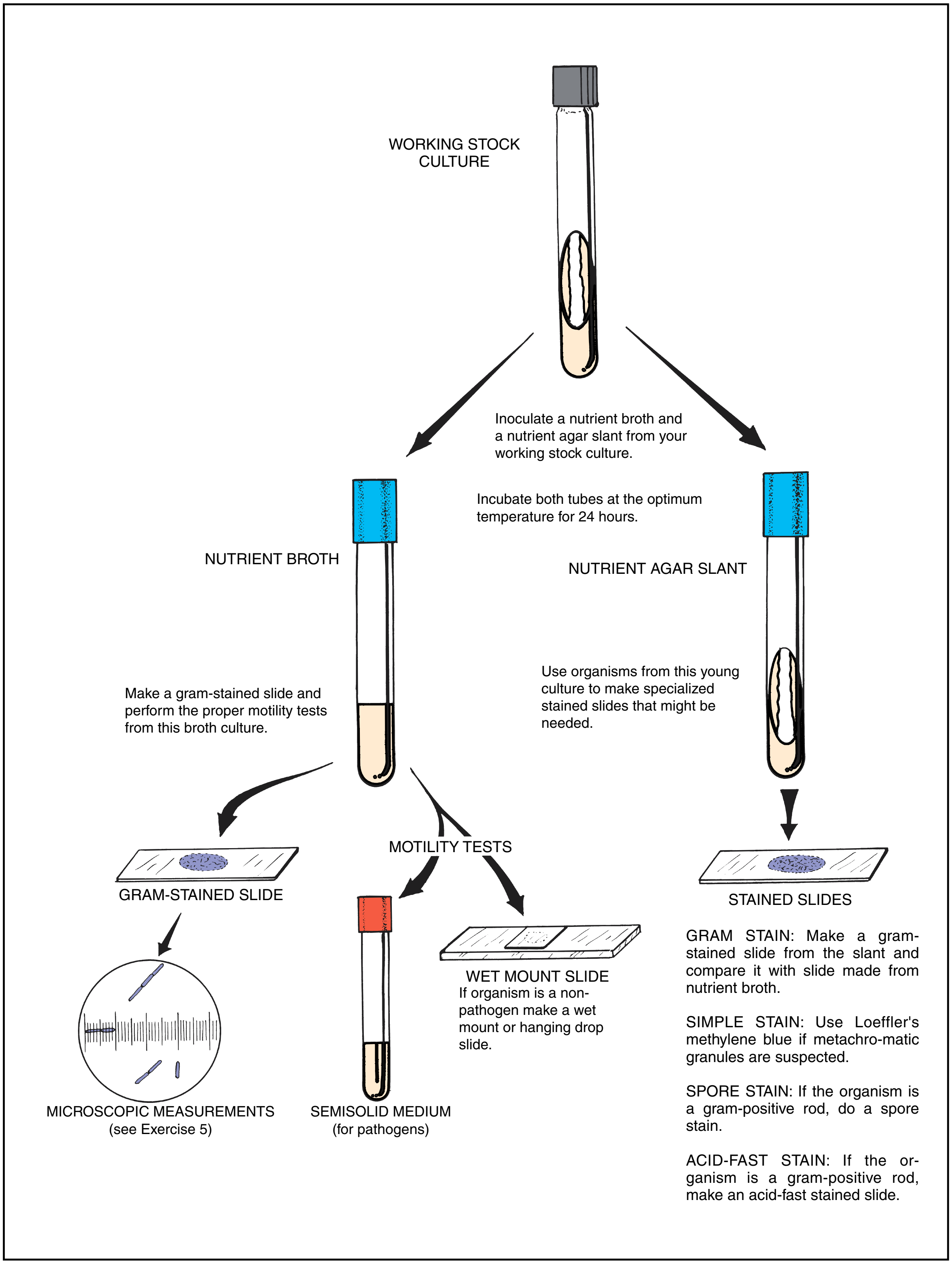


Figure 46.1 Procedure for morphological study

slide from broth instead of the agar slant, the cells will be well dispersed in natural clumps. Note whether the cells occur singly, in pairs, masses, or chains. *Remember to place the slide preparation in a beaker of disinfectant when finished with it.*

If your organism happens to be a pathogen do not make a slide preparation of the organisms; instead, stab the organism into a tube of semisolid or SIM medium to determine motility (Exercise 19). Incubate for 48 hours.

Be sure to record your observations on the Descriptive Chart.

Endospores

If your unknown is a gram-positive rod, check for endospores. *Only rarely is a coccus or gram-negative rod a spore-former.* Examination of your gram-stained slide made from the agar slant should provide a clue, since endospores show up as transparent holes in gram-stained spore-formers. Endospores can also be seen on unstained organisms if studied with phase-contrast optics.

If there seems to be evidence that the organism is a spore-former, make a slide using one of the spore-staining techniques you used in Exercise 16. *Since some spore-formers require at least a week's time of incubation before forming spores, it is prudent to double-check for spores in older cultures.*

Record on the Descriptive Chart whether the spore is terminal, subterminal, or in the middle of the rod.

Acid-Fast Staining

If your organism is a gram-positive, non-spore-forming rod, you should determine whether or not it is acid-fast. Although some bacteria require 4 or 5 days growth to exhibit acid-fastness, most species become acid-fast within 2 days. For best results, therefore, do not use cultures that are too old.

Another point to keep in mind is that most acid-fast bacteria do not produce cells that are 100% acid-fast. An organism is considered acid-fast if only portions of the cells exhibit this characteristic. Refer to Exercise 17 for this staining technique.

A final bit of advice: If you feel insecure about your adeptness at Gram staining and think that you might *possibly* have a gram-positive organism, even though your organism seems to be gram-negative, make an acid-fast stained slide. Many students find (much to their chagrin later) that they didn't do acid-fast staining because their organism seemed to be gram-negative. An improperly gram-stained slide can be very misleading when it comes to unknown identification.

Other Structures

If the protoplast in gram-stained slides stains unevenly, you might wish to do a simple stain with Loeffler's methylene blue (Exercise 13) for evidence of metachromatic granules.

Although a capsule stain (Exercise 14) may be performed at this time, it might be better to wait until a later date when you have the organism growing on blood agar. Capsules usually are more apparent when the organisms are grown on this medium.

Laboratory Report

There is no Laboratory Report to fill out for this exercise. All information is recorded on the Descriptive Chart.

Cultural Characteristics

47

The cultural characteristics of an organism pertain to its macroscopic appearance on different kinds of media. Descriptive terms, which are familiar to all bacteriologists, and are used in *Bergey's Manual,* must be used in recording cultural characteristics. The most frequently used media for a cultural study are nutrient agar, nutrient broth, and nutrient gelatin. For certain types of unknowns it is also desirable to inoculate a blood agar plate; if necessary, this plate can be inoculated later. In addition to these media, you will be inoculating a fluid thioglycollate medium to determine the oxygen requirements of your unknown.

First Period

(Inoculations)

During this period one nutrient agar plate, one nutrient gelatin deep, two nutrient broths, and one tube of fluid thioglycollate medium will be inoculated. Inoculations will be made with the original broth culture of your unknown. The reason for inoculating two tubes of nutrient broth here is to recheck the optimum growth temperature of your unknown. In Exercise 45 you incubated your nutrient agar slants at 20° C and 37° C. It may well be that the optimum growth temperature is closer to 30° C. It is to check out this intermediate temperature that an extra nutrient broth is being inoculated. Proceed as follows:

Materials:

for each unknown:

- 1 nutrient agar pour
- 1 nutrient gelatin deep
- 2 nutrient broths
- 1 fluid thioglycollate medium (FTM)
- 1 Petri plate

1. Pour a Petri plate of nutrient agar for each unknown and streak it with a method that will give good isolation of colonies. Use the original broth culture for streaking.
2. Inoculate the tubes of nutrient broth with a loop.
3. Make a stab inoculation into the gelatin deep by stabbing the inoculating needle (straight wire) directly down into the medium to the bottom of the tube and pulling it straight out. The medium must not be disturbed laterally.
4. Inoculate the tube of FTM with a loopful of your unknown. Mix the organisms throughout the tube by rolling the tube between your palms.

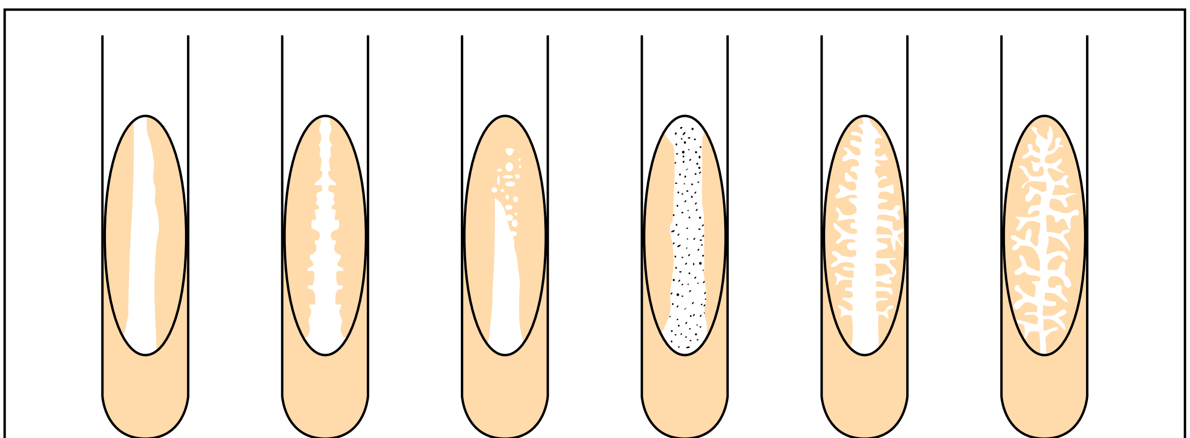


Figure 47.1 Types of bacterial growth on nutrient agar slants

5. Place all tubes except one nutrient broth into a basket and incubate for 24 hours at the temperature that seemed best in Exercise 45. Incubate the remaining tube of nutrient broth separately at 30° C. Incubate the agar plate, inverted, at the presumed best temperature.

Second Period
(Evaluation)

After the cultures have been properly incubated, *carry them to your desk in a careful manner* to avoid disturbing the growth pattern in the nutrient broths and FTM. Before studying any of the tubes or plates, place the tube of nutrient gelatin in an ice water bath. It will be studied later. Proceed as follows to study each type of medium and record the proper descriptive terminology on the Descriptive Chart.

Materials:

reserve stock agar slant of unknown
spectrophotometer and cuvettes
hand lens
ice water bath near sink

Nutrient Agar Slant (Reserve Stock)

Examine your reserve stock agar slant of your unknown that has been stored in the refrigerator since the last laboratory period. Evaluate it in terms of the following criteria:

Amount of Growth The abundance of growth may be described as *none, slight, moderate, and abundant.*

Color Pigmentation should be looked for on the organisms and within the medium. Most organisms will lack chromogenesis, exhibiting a white growth; others are various shades of different colors. Some bacteria produce soluble pigments that diffuse into the medium. Hold the slant up to a strong light to examine it for diffused pigmentation.

Opacity Organisms that grow prolifically on the surface of a medium will appear more opaque than those that exhibit a small amount of growth. Degrees of opacity may be expressed in terms of *opaque, transparent, and translucent* (partially transparent).

Form The gross appearance of different types of growth are illustrated in figure 47.1. The following descriptions of each type will help in differentiation:

Filiform: characterized by uniform growth along the line of inoculation
Echinulate: margins of growth exhibit toothed appearance
Beaded: separate or semiconfluent colonies along the line of inoculation
Effuse: growth is thin, veil-like, unusually spreading
Arborescent: branched, treelike growth
Rhizoid: rootlike appearance

Nutrient Broth

The nature of growth on the surface, subsurface, and bottom of the tube is significant in nutrient broth cultures. Describe your cultures as thoroughly as possible on the Descriptive Chart with respect to these characteristics:

Surface Figure 47.2 illustrates different types of surface growth. A *pellicle* type of surface differs from the *membranous* type in that the latter is much thinner. A *flocculent* surface is made up of floating adherent masses of bacteria.

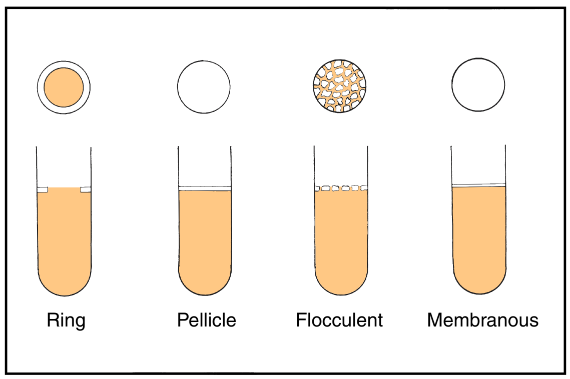


Figure 47.2 Types of surface growth in nutrient broth

Subsurface Below the surface, the broth may be described as *turbid* if it is cloudy; *granular* if specific small particles can be seen; *flocculent* if small masses are floating around; and *flaky* if large particles are in suspension.

Sediment The amount of sediment in the bottom of the tube may vary from none to a great deal. To describe the type of sediment, agitate the tube, putting the material in suspension. The type of sediment can be described as *granular, flocculent, flaky,* and *viscid.* Test for viscosity by probing the bottom of the tube with a sterile inoculating loop.

Amount of Growth To determine the amount of growth, it is necessary to shake the tube to disperse the organisms. Terms such as *slight* (scanty), *moderate,* and *abundant adequately* describe the amount.

Optimum Temperature To determine which temperature produced the best growth, pour the contents from each tube of nutrient broth into separate cuvettes and measure their percent transmittances on the spectrophotometer. If the percent transmittance is less at 30° C than at the other presumed optimum temperature, revise the optimum temperature on your Descriptive Chart.

Fluid Thioglycollate Medium

Since the primary purpose of inoculating a tube of fluid thioglycollate medium is to determine oxygen requirements of your unknown, examine the tube to note the position of growth in the tube. Compare your tube with figure 22.5 on page 92 to make your analysis. Designate your organism as being *aerobic, microaerophilic, facultative,* or *anaerobic* on the Descriptive Chart.

Gelatin Stab Culture

Remove your tube of nutrient gelatin from the ice water bath and examine it. Check first to see if liquefaction has occurred. Organisms that are able to liquefy gelatin produce the enzyme *gelatinase.*

Liquefaction Tilt the tube from side to side to see if a portion of the medium is still liquid. If liquefaction has occurred, check the configuration with figure 47.3 to see if any of the illustrations match your tube. A description of each type follows:

Crateriform: saucer-shaped liquefaction
Napiform: turniplike
Infundibular: funnel-like or inverted cone
Saccate: elongate sac, tubular, cylindrical
Stratiform: liquefied to the walls of the tube in the upper region

Note: The configuration of liquefaction is not as significant as the mere fact that liquefaction takes place. If your organism liquefies gelatin, but you are unable to determine the exact configuration, don't worry about it. However, be sure to record on the Descriptive Chart the *presence* or *absence* of gelatinase production.

Another important point: Some organisms produce gelatinase at a very slow rate. Tubes that are negative should be incubated for another 4 or 5 days to see if gelatinase is produced slowly.

Type of Growth (No Liquefaction) If no liquefaction has occurred, check the tube to see if the organism grows in nutrient gelatin (some do, some don't). If growth has occurred compare the growth with the left-hand illustration in figure 47.3. It should be pointed out, however, that, from a categorization standpoint, the nature of growth in gelatin is not very important.

Nutrient Agar Plate

Colonies grown on plates of nutrient agar should be studied with respect to size, color, opacity, form, elevation, and margin. With a dissecting microscope or hand lens study individual colonies carefully. Refer to figure 47.4 for descriptive terminology. Record your observations on the Descriptive Chart.

Laboratory Report

There is no Laboratory Report for this exercise. Record all information on the Descriptive Chart.

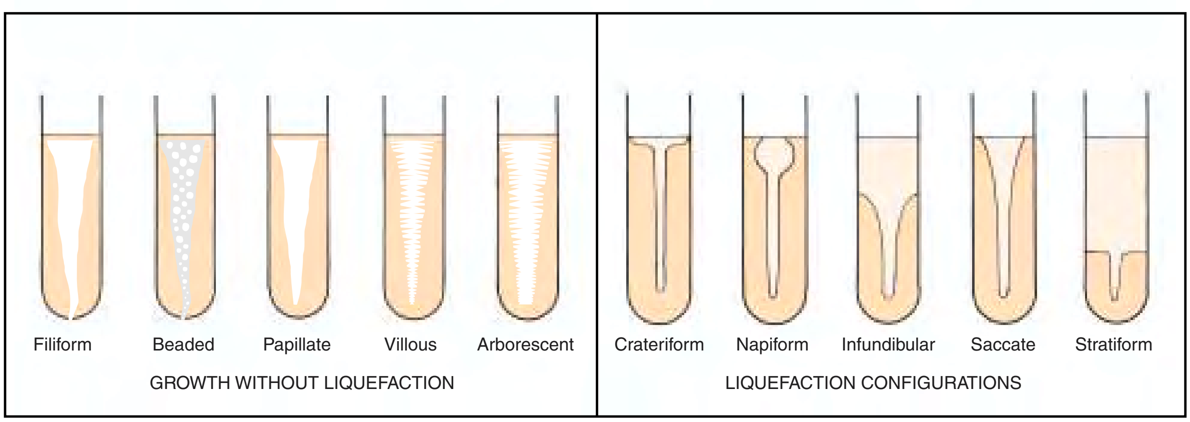


Figure 47.3 Growth in gelatin stabs

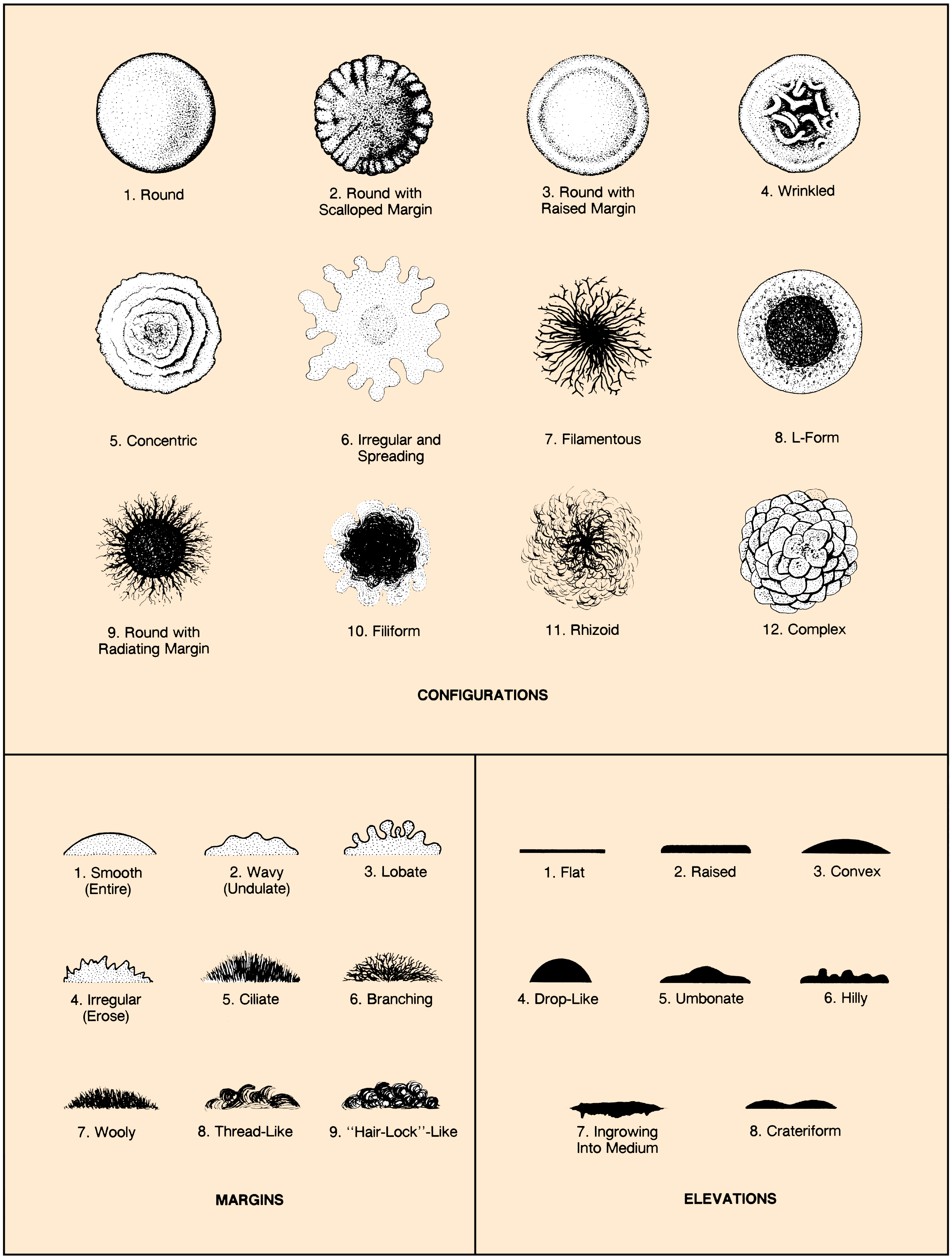


Figure 47.4 Colony characteristics

Physiological Characteristics: Oxidation and Fermentation Tests 48

The assemblage of morphological and cultural characteristics on your Descriptive Chart during the past few laboratory periods may be leading you to believe that you already know the name of your unknown. Students at this stage often begin to draw premature conclusions. To provide you with a clearer perspective of where you are in the categorization process, refer to the separation outlines in figures 51.1 and 51.2, pages 178 and 179. Note that morphological and cultural characteristics can lead you to 11 separate groups of genera. It is very likely that one of these groups contains the genus that includes your unknown.

Although morphological and cultural characteristics are essential in getting to the genus, species determination requires a good deal more information. The physiological information that will be accumulated here and in the next two exercises will make species identification possible.

Before we get into the details of the various inoculations and tests, let's review some of the essentials of microbial metabolism.

Metabolic Reactions

The chemical reactions that occur within the cells of all living organisms are referred to as **metabolism.** These reactions are catalyzed by protein molecules called **enzymes.** The majority of enzymes function within the cell and are called **endoenzymes.** Many bacteria also produce **exoenzymes,** which are released by the cell to catalyze reactions outside of the cell. Figure 48.1 illustrates how these enzymes function.

In deriving energy from food, bacteria may be either oxidative or fermentative. **Oxidative** bacteria utilize oxygen to yield carbon dioxide and water. These bacteria have a *cytochrome enzyme system.* By utilizing organic compounds as electron donors, with oxygen as the ultimate electron (and hydrogen) acceptor, they produce CO_2 and water as end-products. **Fermentative** bacteria, on the other hand, also utilize organic compounds for energy, but they lack a cytochrome system. Instead of producing only CO_2 and water, they produce complex end-products, such as acids, aldehydes, and alcohols. Various gases, such as carbon dioxide, hydrogen, and methane, are also produced. In fermentative bacteria, the organic compounds act both as electron donors and electron acceptors.

Sugars, particularly glucose, are the compounds most widely used by fermenting organisms. Other substances such as organic acids, amino acids, purines, and pyrimidines also can be fermented by some bacteria. The end-products of a particular fermentation are determined by the nature of the organism, the characteristics of the substrate, and environmental conditions such as temperature and pH.

Although fermentation and oxidation represent two different types of energy-yielding reactions, they can both be present in the same organism, as is true of facultative anaerobes. It was pointed out in Exercise

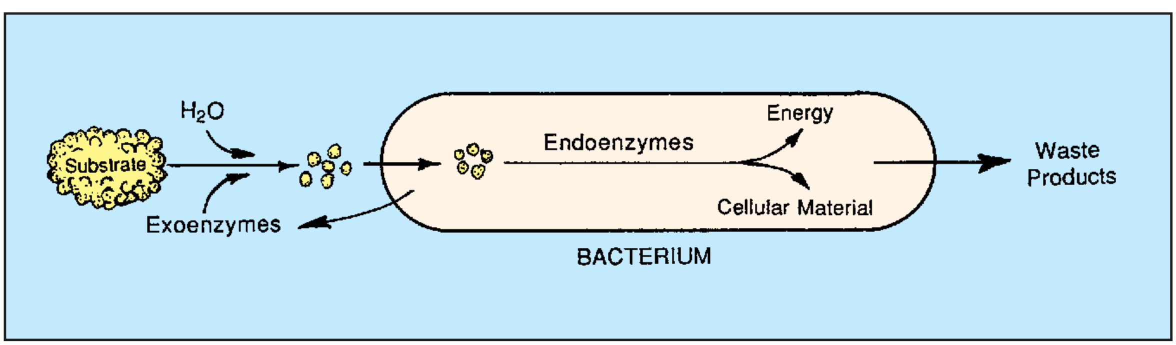


Figure 48.1 Note that the hydrolytic exoenzymes split larger molecules into smaller ones, utilizing water in the process. The smaller molecules are then assimilated by the cell to be acted upon by endoenzymes to produce energy and cellular material

22 that in the presence of molecular oxygen these organisms shift from fermentation to oxidation. An exception, however, is seen in the lactic acid bacteria where fermentation occurs in the presence of air (O_2).

Tests to Be Performed

Six types of reactions will be studied in this exercise: (1) Durham-tube sugar fermentations, (2) mixed acid fermentation, (3) butanediol fermentation, (4) catalase production, (5) oxidase production, and (6) nitrate reduction. The performance of all these tests on your unknown will involve a considerable number of inoculations because a set of positive test controls will also be needed. Although photographs of positive test results are provided in this exercise, seeing the actual test results in test tubes will make it more meaningful.

As you perform these various tests, attempt to keep in mind what groups of bacteria relate to each test. Although some tests are not very specific in pointing the way to unknown identification, others are very narrow in application.

One last comment of importance: *It is not routine practice to perform all these tests in identifying every unknown.* Remember that although it might appear that our prime concern here is to identify an organism, our most important goal is to learn about the various types of tests for enzymes that are available. The use of unknown bacteria to learn about them simply makes it more of a challenge. In actual practice, physiological tests are used very selectively. The "shotgun approach" employed here is used to expose you to the multitude of tests that are available.

First Period

(Inoculations)

The following two sets of inoculations (unknown and test controls) may be done separately or combined into one operation. The media for each set of inoculations are listed separately under each heading.

Unknown Inoculations

Figure 48.2 illustrates the procedure for inoculating seven test tubes and one Petri plate with your unknown. Since your instructor may want you to inoculate some different sugar broths, blanks have been provided in the materials list for write-ins. *If different media are distinguished from each other with different-colored tube caps, write down the colors after each medium below.*

Materials: *(for each unknown)*

Durham tubes with phenol red indicator
- 1 glucose broth
- 1 lactose broth
- 1 mannitol broth
- __________ __________
- __________ __________

2 MR-VP medium
1 nitrate broth
1 nutrient agar slant
1 Petri plate of trypticase soy agar (TSA)

1. Label each tube with the number of your unknown and an identifying letter as designated in figure 48.2.
2. Label one half of the Petri plate UNKNOWN and the other half *P. AERUGINOSA*.
3. Inoculate all broths and the slant with a loop. Inoculate one half of the TSA plate with your unknown, using an isolation technique.

Test Control Inoculations

Figure 48.4 on page 165 illustrates the procedure that will be used for inoculating five test tubes to be used for positive test controls. The Petri plate shown on the right side is the same one that is shown in figure 48.2; thus, it will not be listed in the materials below.

Materials:

1 glucose broth (Durham tube)
2 MR-VP medium
1 nitrate broth
1 nutrient agar slant
nutrient broth cultures of *Escherichia coli, Enterobacter aerogenes, Staphylococcus aureus,* and *Pseudomonas aeruginosa*

1. Label each tube with the code letter assigned to it as listed:

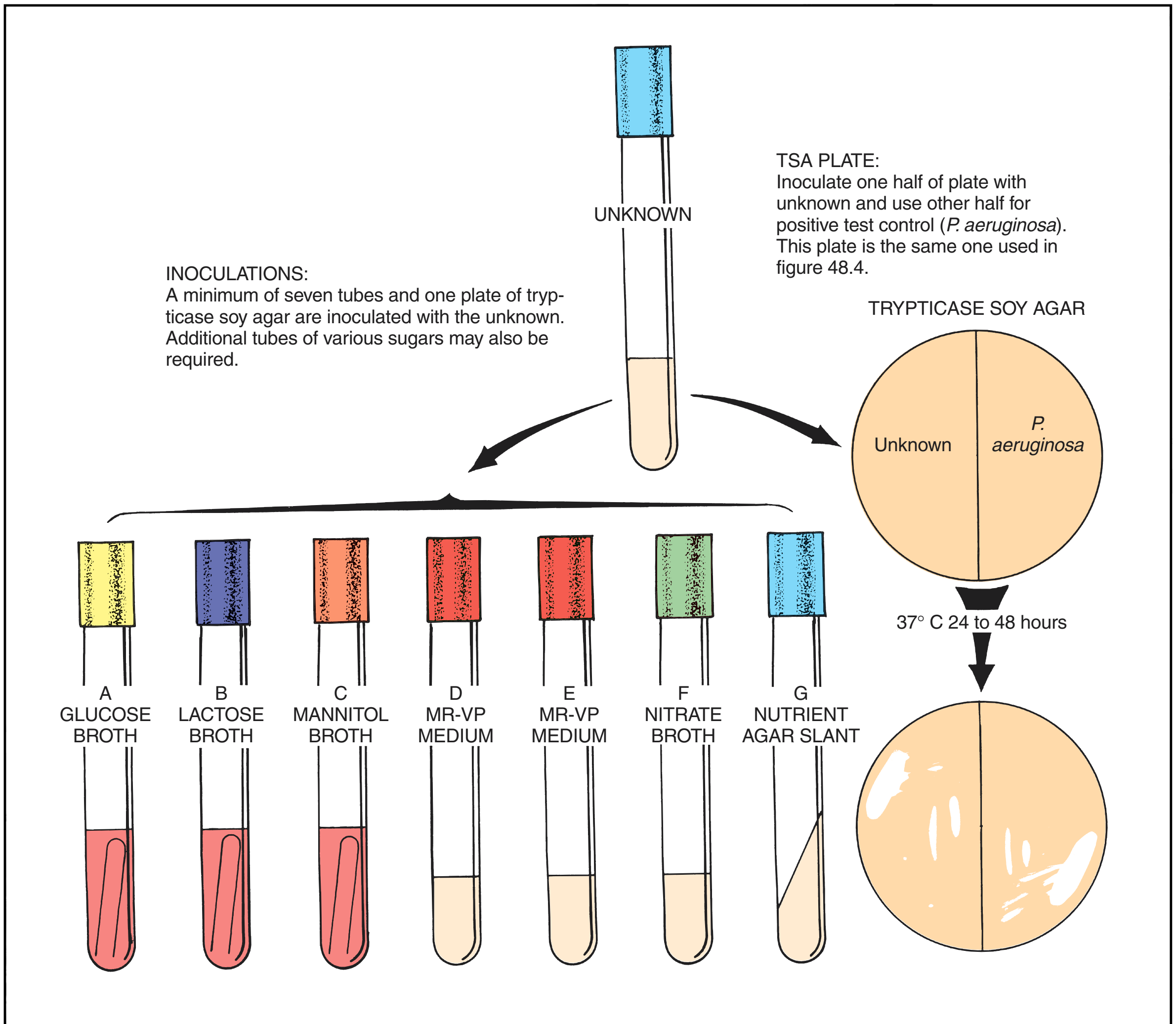
glucose broth	A^1
MR-VP medium	D^1
MR-VP medium	E^1
nitrate broth	F^1
nutrient agar slant	G^1

2. Inoculate each of these tubes with a loopful of the appropriate test organism according to figure 48.4.
3. Inoculate the other half of the TSA plate with *P. aeruginosa.*

Incubation

Except for tube E (MR-VP), all the unknown inoculations should be incubated for 24–48 hours at the unknown's optimum temperature. Tube E should be incubated for 3–5 days at the optimum temperature.

Except for Tube E^1 of the test controls, incubate all the test-control tubes and the TSA plate at 37° C for 24–48 hours. Tube E^1 should be incubated at 37° C for 3–5 days.



INCUBATION	EVALUATION AND TESTS	POSITIVE TEST RESULTS
Except for tube E, incubate all tubes for 24–48 hours at the optimum temperature for the unknown. Tube E should be incubated for 3–5 days at the optimum temperature.	Tubes A, B, and C: If broth turns yellow, acid has been produced. If a bubble is present in the inverted vial, gas has been produced. Tube D: Do methyl red test. Tube E: Do Voges-Proskauer test. Tube F: Do nitrite test. Tube G: Do catalase test. TSA plate: Do oxidase test.	Tube D: **Red color** is positive for mixed acids production. Tube E: **Pink** or **red color** is positive for butanediol production. Tube F: **Red color** is positive for nitrate reduction to nitrite. Tube G: **Bubbles** effervescing from streak indicate that catalase is produced. TSA plate: **Black colonies** indicate that oxidase is produced.

Figure 48.2 Procedure for performing oxidation and fermentation tests

Second Period

(Test Evaluations)

After 24 to 48 hours incubation, arrange all your tubes (except tubes E and E[1]) in a test-tube rack in alphabetical order, with the unknown tubes in one row and the test controls in another row. As you interpret the results, record the information on the Descriptive Chart immediately. Don't trust your memory. Any result that is not properly recorded will have to be repeated.

Durham Tube Sugar Fermentations

When we use a bank of Durham tubes containing various sugars, we are able to determine what sugars an organism is able to ferment. If an organism is able to ferment a particular sugar, acid will be produced and gas *may* be produced. The presence of acid is detectable with the color change of a pH indicator in the medium. Gas production is revealed by the formation of a void in the inverted vial of the Durham tube. If it were important to know the composition of the gas, we would have to use a Smith tube as shown in figure 48.3. For our purposes here, the Durham tube is preferable.

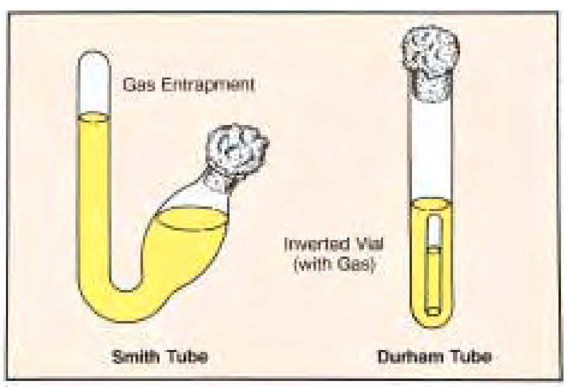


Figure 48.3 Two types of fermentation tubes

Media The sugar broths used here contain 0.5% of the specific carbohydrate plus sufficient amounts of beef extract and peptone to satisfy the nitrogen and mineral needs of most bacteria. The pH indicator phenol red is included for acid detection. This indicator is red when the pH is above 7 and yellow below this point.

Although there are many sugars that one might use, glucose, lactose, and mannitol are logical ones to begin with. Your instructor may have had you include one or more additional kinds, and it is very likely that you may wish to use some others later.

Interpretation Examine the glucose test control tube (tube A[1]), which you inoculated with *E. coli.* Note that the phenol red has turned yellow, indicating acid production. Also, note that the inverted vial has a gas bubble in it. These observations tell us that *E. coli* ferments glucose to produce acid and gas. The left-hand illustration in figure 48.5, page 167, illustrates how this positive tube compares with a negative tube and an uninoculated one.

Now examine the three sugar broths (tubes A, B, and C) that were inoculated with your unknown and record your observations on the Descriptive Chart. If there is no color change, record NONE after the specific sugar. If the tube is yellow with no gas, record ACID. If the inverted vial contains gas and the tube is yellow, record ACID AND GAS.

An important point to keep in mind at this time is that *a negative result on an unknown is as important as a positive result.* Don't feel that you have failed in your technique if many of your tubes are negative!

Mixed Acid Fermentation

(Methyl-Red Test)

A considerable number of gram-negative intestinal bacteria can be differentiated on the basis of the end-products produced when they ferment the glucose in MR-VP medium. Genera of bacteria such as *Escherichia, Salmonella, Proteus,* and *Aeromonas* ferment glucose to produce large amounts of lactic, acetic, succinic, and formic acids, plus CO_2, H_2, and ethanol. The accumulation of these acids lowers the pH of the medium to 5.0 and less.

If methyl red is added to such a culture, the indicator turns red, an indication that the organism is a *mixed acid fermenter.* These organisms are generally great gas producers, too, because they produce the enzyme *formic hydrogenylase,* which splits formic acid into equal parts of CO_2 and H_2.

$$HCOOH \xrightarrow{\text{formic hydrogenylase}} CO_2 + H_2$$

Medium MR-VP medium is essentially a glucose broth with some buffered peptone and dipotassium phosphate.

Test Procedure Perform the methyl-red test first on your test-control tube (D[1]) and then on your unknown (tube D). Proceed as follows:

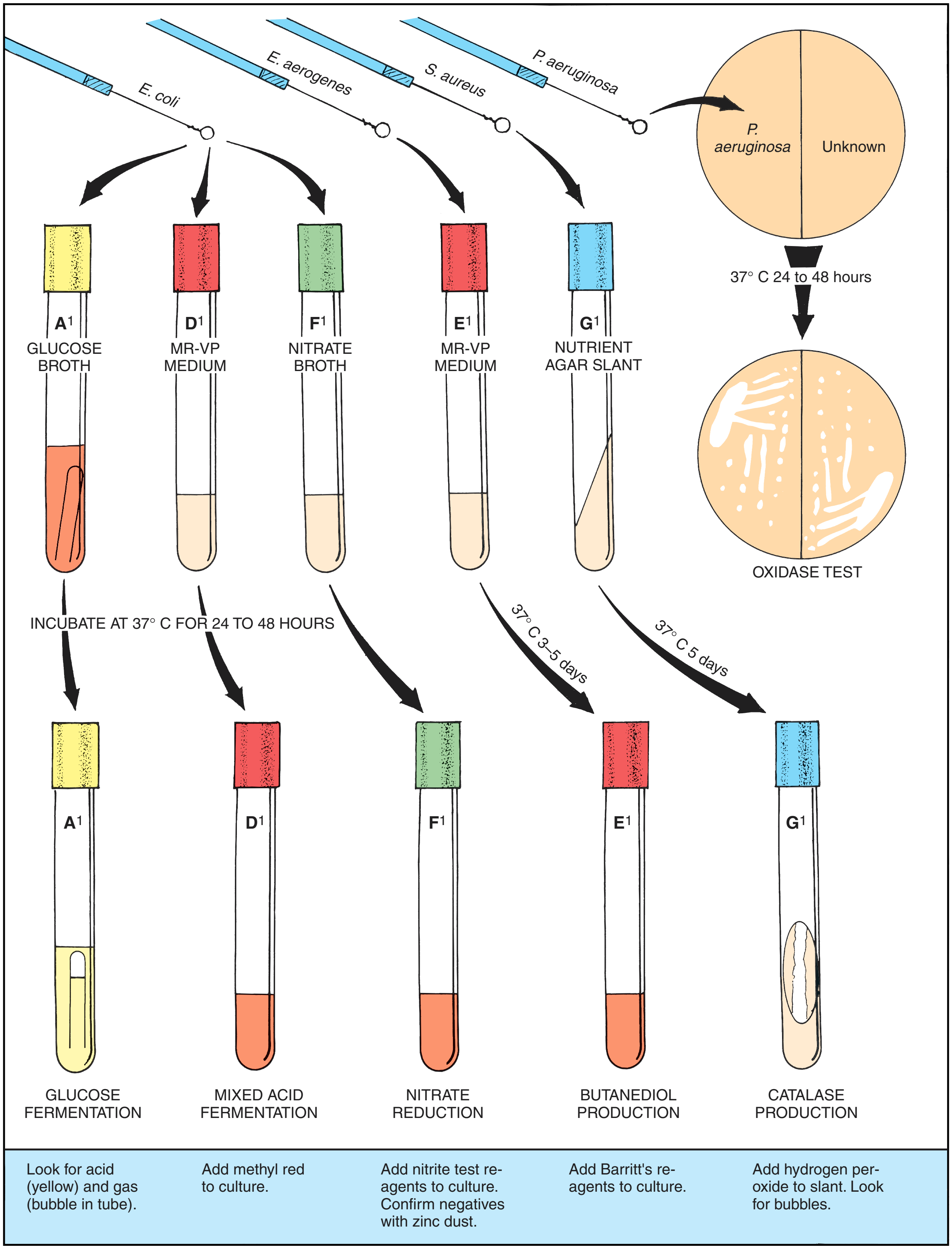


Figure 48.4 Procedure for doing positive test controls

Materials:

dropping bottle of methyl-red indicator

1. Add 3 or 4 drops of methyl red to test-control tube D^1, which was inoculated with *E. coli.* The tube should become red immediately. A **reddish color,** as shown in the left-hand tube of the middle illustration of figure 48.5 is a positive methyl-red test.
2. Repeat the same procedure with your unknown culture (tube D) of MR-VP medium. If your unknown culture becomes yellow like the right-hand tube in figure 47.5, your unknown is negative for this test.
3. Record your results on the Descriptive Chart.

Butanediol Fermentation

(Voges-Proskauer Test)

A negative methyl-red test may indicate that the organism being tested produced a lot of 2,3 butanediol and ethanol instead of acids. All species of *Enterobacter* and *Serratia,* as well as some species of *Erwinia, Bacillus,* and *Aeromonas,* do just that. The production of these nonacid end-products results in less lowering of the pH in MR-VP medium, causing the methyl-red test to be negative.

Unfortunately, there is no satisfactory test for 2,3 butanediol; however, acetoin (acetylmethylcarbinol), a precursor of 2,3 butanediol, is easily detected with Barritt's reagent.

Barritt's reagent consists of alpha naphthol and KOH. When added to a 3 to 5 day culture of MR-VP medium, and allowed to stand for some time, the medium changes to pink or red in the presence of acetoin. Since acetoin and 2,3 butanediol are always simultaneously present, the test is valid. This indirect method of testing for 2,3 butanediol is called the *Voges-Proskauer* test.

Test Procedure Perform the Voges-Proskauer test on your unknown and test-control tubes of MR-VP medium (tubes E and E^1). Note that the test-control tube was inoculated with *E. aerogenes.* Follow this procedure:

Materials:

Barritt's reagents
2 pipettes (1 ml size)
2 empty test tubes

1. Label one empty test tube E (for unknown) and the other E^1 (for control).
2. Pipette 1 ml from culture tube E to the empty tube E and 1 ml from culture tube E^1 to the empty tube E^1. Use separate pipettes for each tube.
3. Add 18 drops (about 0.5 ml) of Barritt's solution A (alpha naphthol) to each of the tubes that contain 1 ml of culture.
4. Add an equal amount of Barritt's solution B (KOH) to the same tubes.
5. Shake the tubes vigorously every 20 seconds until the control tube (E^1) turns pink or red. Let the tubes stand for 1 or 2 hours to see if the unknown turns red. *Vigorous shaking is very important* to achieve complete aeration.

   A positive Voges-Proskauer reaction is **pink** or **red.** The left-hand tube in the right-hand illustration of figure 48.5 shows what a positive result looks like.
6. Record your results on the Descriptive Chart.

DURHAM TUBES	METHYL RED TEST	VOGES-PROSKAUER TEST
From left to right: uninoculated positive, and negative.	Tube on left is positive (*E. coli*); tube on right is negative.	Tube on left is positive (*E. aerogenes*); tube on right is negative.

Figure 48.5 Durham tubes, mixed acid, and butanediol fermentation tests

Catalase Production

Most aerobes and facultatives that utilize oxygen produce hydrogen peroxide, which is toxic to their own enzyme systems. Their survival in the presence of this antimetabolite is possible because they produce an enzyme called *catalase,* which converts the hydrogen peroxide to water and oxygen:

$$2H_2O_2 \xrightarrow{\text{catalase}} 2H_2O + O_2$$

It has been postulated that the death of strict anaerobes in the presence of oxygen may be due to the suicidal act of H_2O_2 production in the absence of catalase production. The presence or absence of catalase production is an important means of differentiation between certain groups of bacteria.

Test Procedure To determine whether or not catalase is produced, all that is necessary is to place a few drops of 3% hydrogen peroxide on the organisms of a slant culture. If the hydrogen peroxide effervesces, the organism is catalase-positive.

Materials:

3% hydrogen peroxide
test-control tube G^1 with *S. aureus* growth and unknown tube G

1. While holding test-control tube G^1 at an angle, allow a few drops of H_2O_2 to flow slowly down over the *S. aureus* growth on the slant. Note how bubbles emerge from the organisms.
2. Repeat the test on your unknown (tube G) and record your results on the Descriptive Chart.

Oxidase Production

The production of oxidase is one of the most significant tests we have for differentiating certain groups of bacteria. For example, all the Enterobacteriaceae are oxidase-negative and most species of *Pseudomonas* are oxidase-positive. Another important group, the *Neisseria,* are oxidase producers.

Two methods are described here for performing this test. The first method utilizes the entire TSA plate; the second method is less demanding in that only a loopful of organisms from the plate is used. Both methods are equally reliable.

Materials:

TSA plate streaked with unknown and *P. aeruginosa*
oxidase test reagents (1% solution of dimethyl-*p*-phenylenediamine hydrochloride)
Whatman no. 2 filter paper
Petri dish

Entire Plate Method Onto the TSA plate where you streaked your unknown and *P. aeruginosa,* pour some of the oxidase test reagent, covering the colonies of both organisms.

Observe that the *Pseudomonas* colonies first become **pink,** then change to **maroon, dark red,** and finally **black.** Refer to figure 48.6. If your unknown follows the same color sequence, it, too, is oxidase-positive. Record your results on the Descriptive Chart.

Filter Paper Method On a piece of Whatman no. 2 filter paper in a Petri dish, place several drops of oxidase test reagent. Remove a loopful of the organisms from one of the colonies and smear the organisms over a small area of the paper. The positive color reaction described above will show up within 10–15 seconds. Record your results on the Descriptive Chart.

Nitrate Reduction

Many facultative bacteria are able to use the oxygen in nitrate as a hydrogen acceptor in anaerobic respiration, thus converting nitrate to nitrite. This enzymatic reaction is controlled by an inducible enzyme called nitratase.

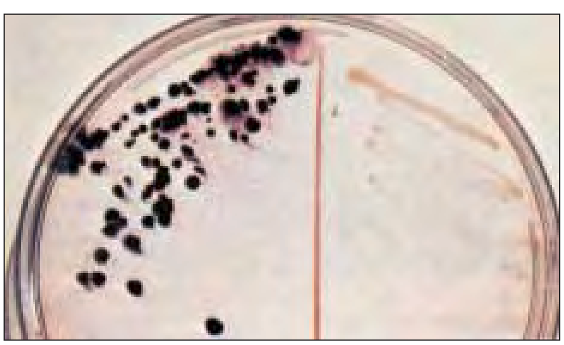


Figure 48.6 Oxidase Test: The colonies on the left are positive; the ones on the right are negative.

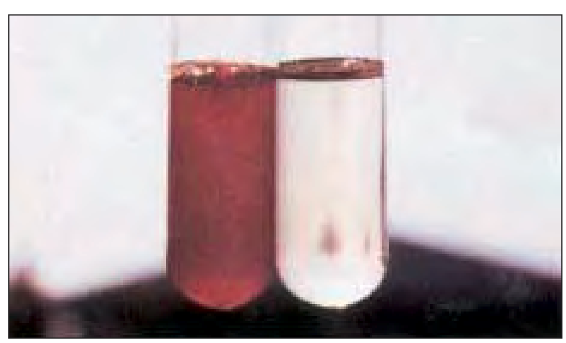


Figure 48.7 Nitrate Reduction Test Tube on left is positive (*E. coli*); tube on right is negative.

The chemical reaction for this enzymatic reaction is as follows:

$$NO_3^- + 2e^- + 2H^+ \xrightarrow{\text{nitratase}} NO_2^- + H_2O$$

Since the presence of free oxygen prevents nitrate reduction, actively multiplying organisms will use up the oxygen first and then utilize the nitrate. In culturing some organisms, it is desirable to use anaerobic methods to ensure nitrate reduction.

Test Procedure The nitrate broth used in this test consists of beef extract, peptone, and potassium nitrate. To test for nitrite after incubation, we use two reagents designated as A and B.

Reagent A contains sulfanilic acid and reagent B contains dimethyl-alpha-naphthylamine. In the presence of nitrite, these reagents cause the culture to turn red. Negative results must be confirmed as negative with zinc dust.

Materials:

nitrate broth cultures of unknown (tube F) and test control *E. coli* (tube F[1])
nitrite test reagents (solutions A and B)
zinc dust

1. Add 2 or 3 drops of nitrite test solution A (sulfanilic acid) and an equal amount of solution B (dimethyl-alpha-naphthylamine) to the nitrate broth culture of *E. coli* (tube F[1]). A **red color** should appear almost immediately (see figure 48.7), indicating that nitrate reduction has occurred.

CAUTION

Avoid skin contact with solution B. Dimethyl-alpha-naphthylamine is carcinogenic.

2. Repeat this procedure with your unknown (tube F). If the red color does not develop, your unknown is negative for nitrate reduction. All negative results should be confirmed as being negative as follows:

   Negative Confirmation: Add a pinch of zinc dust to the tube and shake it vigorously. If the tube becomes red, the test is confirmed as being negative. Zinc causes this reaction by reducing nitrate to nitrite; the newly formed nitrite reacts with the reagents to produce the red color.
3. Record your results on the Descriptive Chart.

Laboratory Report

Answer the questions on Laboratory Report 48–50 that pertain to this exercise.

49 Physiological Characteristics: Hydrolytic Reactions

As indicated in the last exercise, many bacteria produce *hydrolases,* which split complex organic compounds into smaller units. These exoenzymes accomplish molecular splitting in the presence of water. We have already observed one example of protein hydrolysis in Exercise 47: gelatin hydrolysis by gelatinase. In this exercise we will observe the hydrolysis of starch, casein, fat, tryptophan, and urea. Each test plays an important role in the identification of certain types of bacteria. This exercise will be performed in the same manner as the previous one with test controls being made for comparisons.

Figure 49.1 illustrates the general procedure to be used. Three agar plates and four test tubes will be inoculated. After incubation, some of the plates and tubes will have test reagents added to them; others will reveal the presence of hydrolysis by changes that have occurred during incubation. Proceed as follows:

First Period

(Inoculations)

If each student is working with only one unknown, students can work in pairs to share Petri plates. Note in figure 49.1 how each plate can serve for two unknowns with the test-control organism streaked down the middle. If each student is working with two unknowns, the plates will not be shared. Whether or not the two tubes for test controls will be shared depends on the availability of materials.

Materials:

per *pair* of students with one unknown each, or
for *one* student with two unknowns:
- 1 starch agar plate
- 1 skim milk agar plate
- 1 spirit blue agar plate
- 3 urea broths
- 3 tryptone broths

nutrient broth cultures of *B. subtilis, E. coli, S. aureus,* and *P. vulgaris.*

1. Label and streak the three different agar plates in the manner shown in figure 49.1. Note that straight line streaks are made on each plate. Indicate, also, the type of medium in each plate.
2. Label a tube of urea broth *P. VULGARIS* and a tube of tryptone broth *E. COLI.* These will be your test controls for urea and tryptophan hydrolysis. Inoculate each tube accordingly.
3. For each unknown, label one tube of urea broth and one tube of tryptone broth with the code number of your unknown. Inoculate each tube with the appropriate unknown.
4. Incubate the plates and two test-control tubes at 37° C. Incubate the unknown tubes of urea broth and tryptone broth at the optimum temperatures for the unknowns.

Second Period

(Evaluation of Tests)

After 24 to 48 hours incubation of unknowns and test controls, compare your unknowns with the test controls, recording all data on the Descriptive Chart.

Starch Hydrolysis

Since many bacteria are capable of hydrolyzing starch, this test has fairly wide application. The starch molecule is a large one consisting of two constituents: amylose, a straight chain polymer of 200 to 300 glucose units, and amylopectin, a larger branched polymer with phosphate groups. Bacteria that hydrolyze starch produce *amylases* that yield molecules of maltose, glucose, and dextrins.

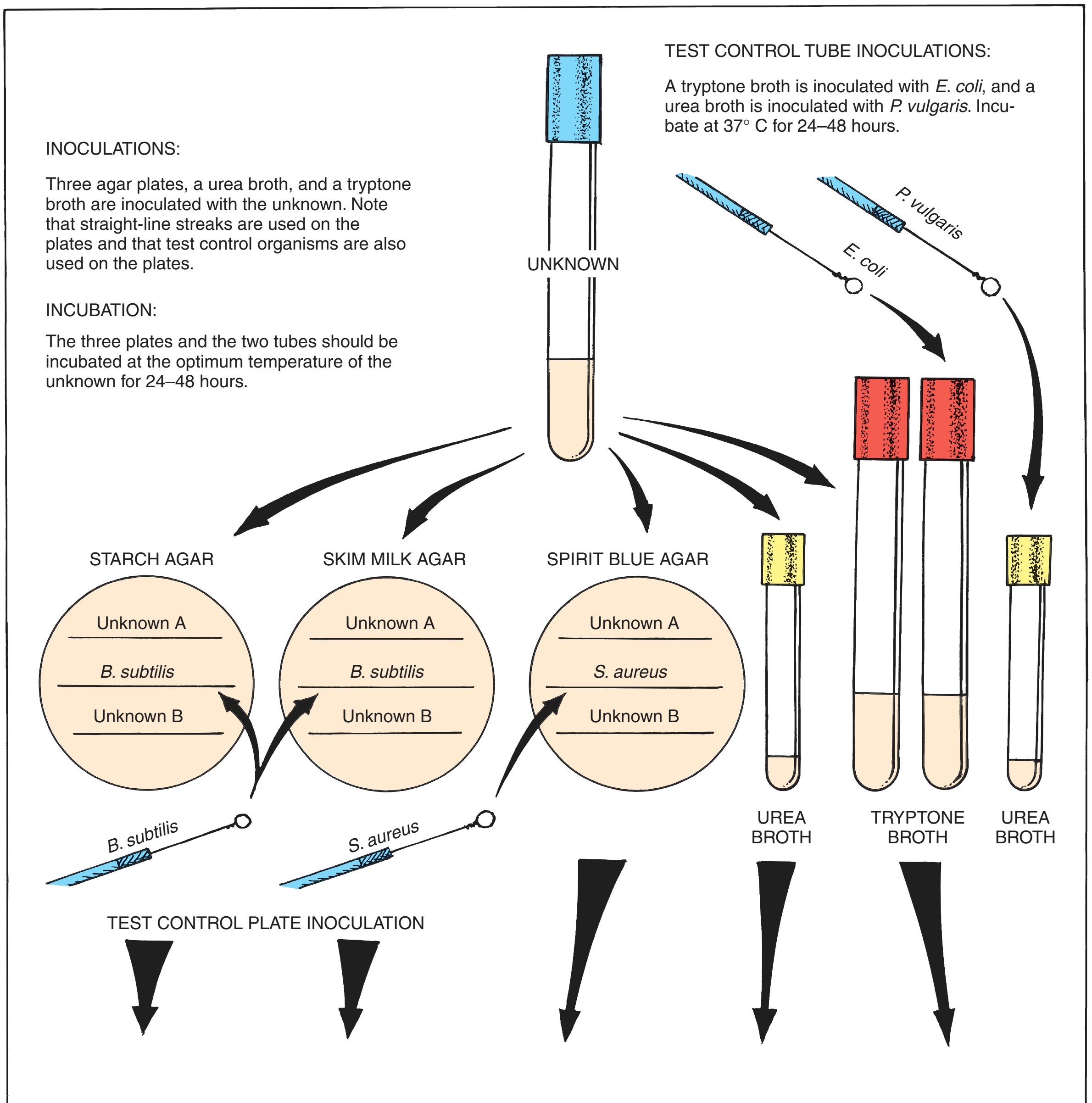
Materials:

Gram's iodine
starch agar culture plate

Iodine solution (Gram's) is an indicator of starch. When iodine comes in contact with a medium containing starch, it turns blue. If starch is hydrolyzed and starch is no longer present, the medium will have a **clear zone** next to the growth.

By pouring Gram's iodine over the growth on the medium, one can see clearly where starch has been hydrolyzed. If the area immediately adjacent to the growth is clear, amylase is produced.

Pour enough iodine over each streak to completely wet the entire surface of the plate. Rotate and tilt the plate gently to spread the iodine. Compare



STARCH HYDROLYSIS	CASEIN HYDROLYSIS	FAT HYDROLYSIS	UREA HYDROLYSIS	TRYPTOPHAN HYDROLYSIS
Add several drops of Gram's iodine to growth on plate. Clear areas next to growth indicate that starch is hydrolyzed.	If clear zone is seen adjacent to growth on medium, casein is hydrolyzed.	If growth streak exhibits a dark blue precipitate, the organism is positive for fat hydrolysis.	If broth exhibits a red or cerise color, the organism can hydrolyze urea.	If after adding Kovacs' reagent, a red ring appears on the surface of the broth, the organism is able to hydrolyze tryptophan (indole-positive).

Figure 49.1 Procedure for doing hydrolysis tests on unknowns

your unknowns with the positive result seen along the growth of *B. subtilis.* The left-hand illustration of figure 49.2 illustrates what it looks like.

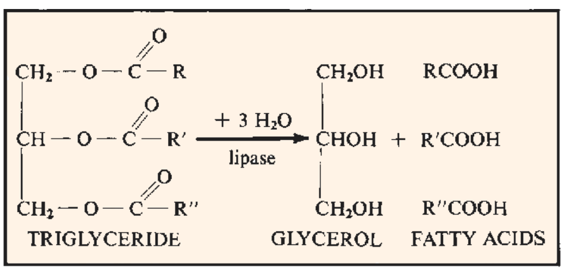
Casein Hydrolysis

Casein is the predominant protein in milk. Its presence causes milk to have its characteristic white appearance. Many bacteria produce the exoenzyme *caseinase,* which hydrolyzes casein to produce more soluble, transparent derivatives. Protein hydrolysis also is referred to as *proteolysis,* or *peptonization.*

Examine the streaks on the skim milk agar plates. Note that a **clear zone** exists adjacent to the growth of *B. subtilis.* This is evidence of casein hydrolysis. The middle illustration in figure 49.2 shows what it looks like. Compare your unknown with this positive result and record the results on the Descriptive Chart.

Fat Hydrolysis

The ability of organisms to hydrolyze fat is accomplished with the enzyme *lipase.* In this reaction the fat molecule is split to form one molecule of glycerol and three fatty acid molecules. The glycerol and fatty acids can be used by the organism to synthesize bacterial fats and other cell components. In many instances they are even oxidized to yield energy under aerobic conditions. This ability of bacteria to decompose fats plays a role in the rancidity of certain foods, such as margarine.



Spirit blue agar contains a vegetable oil that, when hydrolyzed by most organisms, results in the lowering of the pH sufficiently to produce a **dark blue precipitate.** Unfortunately, the hydrolytic action of some organisms on this medium does not produce a blue precipitate because the pH is not lowered sufficiently.

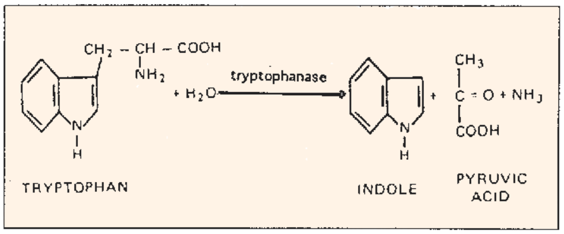
Examine the *S. aureus* growth carefully. You should be able to see this dark blue reaction. The right-hand illustration in figure 49.2 exhibits what it should look like.

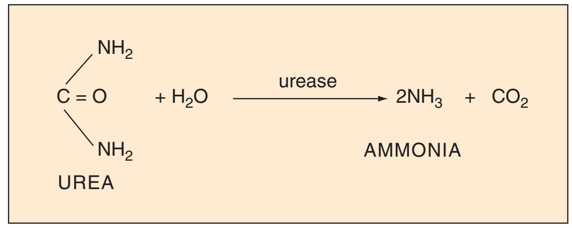
Compare the positive reaction of *S. aureus* with the reaction of your unknown. *If your unknown appears to be negative, hold the plate up toward the light and look for a region near growth where oil droplets are depleted.* If you see depletion of oil drops, consider your organism to be positive for this test. Record the results on the Descriptive Chart.

Tryptophan Hydrolysis

Certain bacteria, such as *E. coli,* have the ability to split the amino acid tryptophan into indole and pyruvic acid. The enzyme that causes this hydrolysis is *tryptophanase.* Indole can be easily detected with Kovacs' reagent. This test is particularly useful in differentiating *E. coli* from some closely related enteric bacteria.

Figure 49.2 **Hydrolysis test plates:** Starch, casein, fat





Tryptone broth (1%) is used for this test because it contains a great deal of tryptophan. Tryptone is a peptone derived from casein by pancreatic digestion.

Materials:

Kovacs' reagent
tryptone broth cultures of unknown and *E. coli*

To test for indole, add 10 or 12 drops of Kovacs' reagent to the *E. coli* culture in tryptone broth. A **red layer** should form at the top of the culture, as shown in figure 49.3. Repeat the test on your unknown and record the results on the Descriptive Chart.

Urea Hydrolysis

The differentiation of gram-negative enteric bacteria is greatly helped if one can demonstrate that the unknown can produce *urease.* This enzyme splits off ammonia from the urea molecule, as shown nearby. Note in the separation outline in figure 51.3 that three genera (*Proteus, Providencia, and Morganella*) are positive for the production of this hydrolytic enzyme.

Urea broth is a buffered solution of yeast extract and urea. It also contains phenol red as a pH indicator. Since urea is unstable and breaks down in the autoclave at 15 psi steam pressure, it is usually sterilized by filtration. It is tubed in small amounts to hasten the visibility of the reaction.

When urease is produced by an organism in this medium, the ammonia that is released raises the pH. As the pH becomes higher, the phenol red changes from a yellow color (pH 6.8) to a **red** or **cerise color** (pH 8.1 or more).

Examine your tube of urea broth that was inoculated with *Proteus vulgaris.* Compare your unknown with this standard. Figure 49.4 reveals how positive and negative results of the test should appear. *If your unknown is negative, incubate the tube for a total of 7 days to check for a slow urease producer.* Record your result on the Descriptive Chart.

Figure 49.3 **Indole Test:** Tube on the left is positive (*E. coli*); tube on the right is negative.

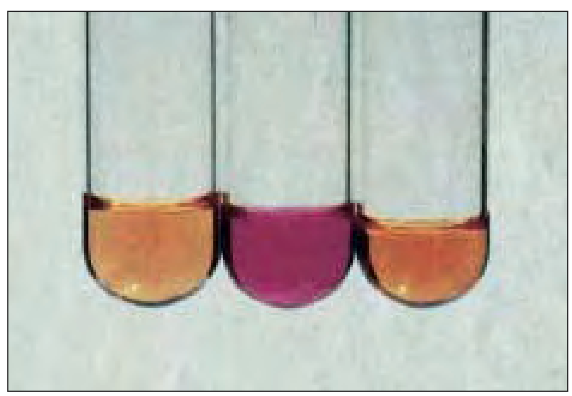


Figure 49.4 **Urease Test:** From left to right—uninoculated, positive (*Proteus*) and negative

50 Physiological Characteristics: Miscellaneous Tests

There are several additional physiological tests used in unknown identification that are best grouped separately as "miscellaneous tests." They include tests for hydrogen sulfide production, citrate utilization, phenylalanine deaminization, and litmus milk reactions. During the first period, inoculations of four kinds of media will be made for these tests. An explanation of the value of the IMViC tests will also be included.

First Period

(Inoculations)

Since test controls are included in this exercise, two sets of inoculations will be made. For economy of materials, one set of test controls will be made by students working in pairs.

Materials:

for test controls, per pair of students:

- 1 Kligler's iron agar deep or SIM medium
- 1 Simmons citrate agar slant
- 1 phenylalanine agar slant
- nutrient broth cultures of *Proteus vulgaris, Staphylococcus aureus,* and *Enterobacter aerogenes*

per unknown, per student:

- 1 Kligler's iron agar deep or SIM medium
- 1 Simmons citrate agar slant
- 1 phenylalanine agar slant
- 1 litmus milk

1. Label one tube of Kligler's iron agar (or SIM medium) *P. VULGARIS* and additional tubes with your unknown numbers. Inoculate each tube by stabbing with a straight wire.
2. Label one tube of Simmons citrate agar *E. AEROGENES* and additional tubes with your unknown numbers. Use a straight wire to streak-stab each slant; i.e., streak the slant first, and then stab into the middle of the slant.
3. Label one tube of phenylalanine agar slant *P. VULGARIS* and the other with your unknown code number. Streak each slant with the appropriate organisms.
4. With a loop, inoculate one tube of litmus milk with your unknown. (**Note:** A test control for this medium will not be made. Figure 50.2 will take its place.)
5. Incubate the unknowns at their optimum temperatures. Incubate the test controls at 37° C for 24–48 hours.

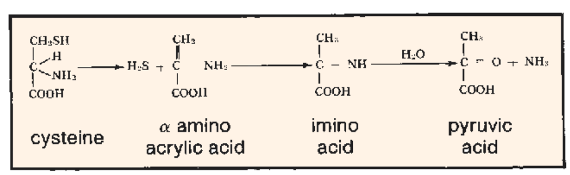
Second Period

(Evaluation of Tests)

After 24 to 48 hours incubation, examine the tubes to evaluate according to the following discussion. Record all results on the Descriptive Chart.

Hydrogen Sulfide Production

Certain bacteria, such as *Proteus vulgaris,* produce hydrogen sulfide from the amino acid cysteine. These organisms produce the enzyme *cysteine desulfurase,* which works in conjunction with the coenzyme pyridoxyl phosphate. The production of H_2S is the initial step in the deamination of cysteine as indicated below:



Kligler's iron agar or SIM medium is used here to detect hydrogen sulfide production. Both of these media contain iron salts that react with H_2S to form a **dark precipitate** of iron sulfide.

Kligler's iron agar also contains glucose, lactose, and phenol red. When this medium is used in slants it is an excellent medium for detecting glucose and lactose fermentation. SIM medium, on the other hand, can also be used for determining motility and testing for indole production.

Examine the tube of one of these media that was inoculated with *P. vulgaris.* If it is Kligler's iron agar it will look like the left-hand tube in figure 50.1. A positive reaction in SIM medium will look like the small tube on the right.

Compare your unknown with this control tube and record your results on the Descriptive Chart.

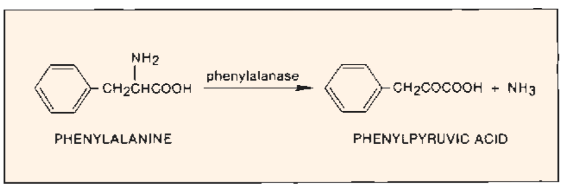
Citrate Utilization

The ability of some organisms, such as *E. aerogenes* and *Salmonella typhimurium,* to utilize citrate as a sole source of carbon can be a very useful differentiation characteristic in working with intestinal bacteria. Koser's citrate medium and Simmons citrate agar are two media that are used to detect this ability in bacteria. In both of these synthetic media sodium citrate is the sole carbon source; nitrogen is supplied by ammonium salts instead of amino acids.

Examine the test control slant of this medium that was inoculated with *E. aerogenes.* Note the distinct **Prussian blue color change** that has occurred. Refer to the right-hand illustration in figure 50.1. Record your results on the Descriptive Chart.

Phenylalanine Deamination

A few bacteria, such as *Proteus, Morganella,* and *Providencia,* produce the deaminase *phenylalanase,* that deaminizes the amino acid phenylalanine to produce phenylpyruvic acid (PPA). This characteristic is used to help differentiate these three genera from other genera of the Enterobacteriaceae. The reaction is as follows:



Proceed as follows to test for the production of phenylpyruvic acid, which is evidence that the enzyme phenylalanase has been produced:

Materials:

dropping bottle of 10% ferric chloride

Allow 5–10 drops of 10% ferric chloride to flow down over the slants of the test control (*P. vulgaris*) and your unknowns. To hasten the reaction, use a loop to emulsify the organisms into solution. A deep **green color** should appear on the test control slant in 1–5 minutes. Refer to the middle illustration in figure 50.1. Compare your unknown with the control and record your results on the Laboratory Report.

The IMViC Tests

In the differentiation of *E. aerogenes* and *E. coli,* as well as some other related species, four physiological tests have been grouped together into what are called the IMViC tests. The *I* stands for indole; the *M* and *V* stand for methyl red and Voges-Proskauer tests; *i* simply facilitates pronunciation; and the *C* signifies citrate utilization. In the differentiation of the two coliforms *E. coli* and *E. aerogenes,* the test results appear as charted below, revealing completely opposite reactions for the two organisms on all tests.

	I	M	V	C
E. coli	+	+	–	–
E. aerogenes	–	–	+	+

The significance of these tests is that when testing drinking water for the presence of the sewage indicator *E. coli,* one must be able to rule out *E. aerogenes,* which has many of the morphological and physiological characteristics of *E. coli.* Since *E. aerogenes* is not always associated with sewage, its

HYDROGEN SULFIDE TEST	PPA TEST	CITRATE UTILIZATION
Positive tubes have black precipitate. Large tubes: Kligler; Small tube is SIM.	Left-hand tube exhibits a positive reaction (green). Other tube is negative.	Left to right: uninoculated, positive (*E. aerogenes*), and negative.

Figure 50.1 Hydrogen sulfide, PPA, and citrate utilization tests

presence in water would not necessarily indicate sewage contamination.

If you are attempting to identify a gram-negative, facultative, rod-shaped bacterial organism, group these series of tests together in this manner to see how your unknown fits this combination of tests.

Litmus Milk Reactions

Litmus milk contains 10% powdered skim milk and a small amount of litmus as a pH indicator. When the medium is made up, its pH is adjusted to 6.8. It is an excellent growth medium for many organisms and can be very helpful in unknown characterization. In addition to revealing the presence or absence of fermentation, it can detect certain proteolytic characteristics in bacteria. A number of facultative bacteria with strong reducing powers are able to utilize litmus as an alternative electron acceptor to render it colorless. Figure 50.2 reveals the color changes that cover the spectrum of litmus milk changes. Since some of the reactions take 4 to 5 days to occur, the cultures should be incubated for at least this period of time; they should be examined every 24 hours, however. Look for the following reactions:

Acid Reaction Litmus becomes pink. Typical of fermentative bacteria.

Alkaline Reaction Litmus turns blue or purple. Many proteolytic bacteria cause this reaction in the first 24 hours.

Litmus Reduction Culture becomes white; actively reproducing bacteria reduce the O/R potential of medium.

Coagulation Curd formation. Solidification is due to protein coagulation. Tilting tube at 45° will indicate whether or not this has occurred.

Peptonization Medium becomes translucent. It often turns brown at this stage. Caused by proteolytic bacteria.

Ropiness Thick, slimy residue in bottom of tube. Ropiness can be demonstrated with sterile loop.

Record the litmus milk reactions of your unknown on the Descriptive Chart.

Laboratory Report

Complete Laboratory Report 48–50, which reviews all physiological tests performed in the last three exercises.

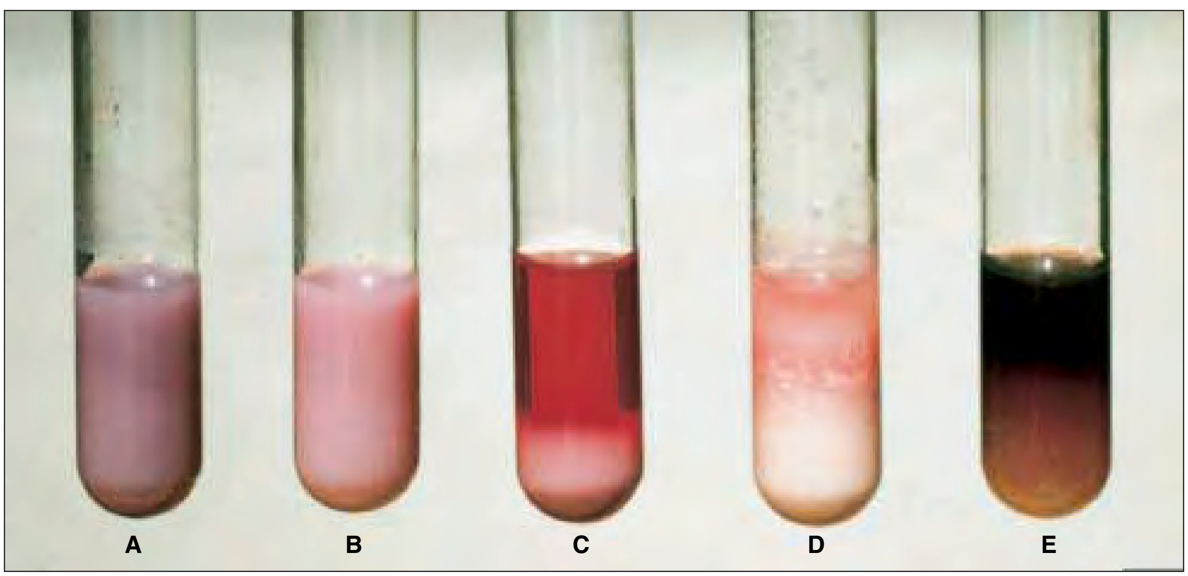


Figure 50.2 Litmus milk reactions: (A) Alkaline. (B) Acid. (C) Upper transparent portion is peptonization; solid white portion in bottom is coagulation and litmus reduction; overall redness is interpreted as acid. (D) Coagulation and litmus reduction in lower half; some peptonization (transparency) and acid in top portion. (E) Litmus indicator is masked by production of soluble pigment (*Pseudomonas*); some peptonization is present but difficult to see in photo.

Use of *Bergey's Manual* and *Identibacter Interactus* 51

Once you have recorded all the data on your Descriptive Chart pertaining to morphological, cultural, and physiological characteristics of your unknown, you are ready to determine its genus and species. Determination of the genus should be relatively easy; species differentiation, however, is considerably more difficult.

The most important single source of information we have for the identification of bacteria is *Bergey's Manual of Systematic Bacteriology.* This monumental achievement, which consists of four volumes, replaced a single-volume eighth edition of *Bergey's Manual of Determinative Bacteriology.* Although the more recent publication consists of four volumes, only volumes 1 and 2 will be used for the identification of the unknowns in this course.

In addition to using *Bergey's Manual,* you may have an opportunity to use a computer simulation program called *Identibacter interactus,* which is available on a CD-ROM disc. Details of the application of this computer program are discussed on page 181 and in Appendix F.

Bergey's Manual is a worldwide collaborative effort that has an editorial board of 13 trustees. Over 200 specialists from 19 countries are listed as contributors to the first two volumes. One of the purposes of this exercise is to help you glean the information from these two volumes that is needed to identify your unknown. Before we get into the mechanics of using *Bergey's Manual,* a few comments are in order pertaining to the problems of bacterial classification.

Classification Problems

Compared with the classification of bacteria, the classification of plants and animals has been relatively easy. In these higher forms, a hierarchy of orders, families, and genera is based, primarily, on evolutionary evidence revealed by fossils laid down in sedimentary layers of Earth's crust. Some of the earlier editions of *Bergey's Manual* attempted to use the same hierarchial system, but the attempt had to be abandoned when the eighth edition was published; without paleontological information to support the system it literally fell apart.

The present system of classification in *Bergey's Manual* uses a list of "Sections" that separate the various groups. Each section is described in common terms so that it is easily understood (even for beginners). For example, Section 1 is entitled **The Spirochaetes.** Section 4 pertains to **Gram-Negative Aerobic Rods and Cocci.** If one scans the Table of Contents in each volume after having completed all tests, it is possible, usually, to find a section that contains the unknown being studied.

A perusal of these sections will reveal that some sections have a semblance of hierarchy in the form of orders, families, and genera. Other sections list only genera.

Thus, we see that the classification system of bacteria, as developed in *Bergey's Manual,* is not the tidy system we see in higher forms of life. The important thing is that it works.

Our dependency over the years on *Bergey's Manual* has led many to think of its classification system as the "Official Classification." Staley and Krieg in their Overview in Volume 1 emphasize that no official classification of bacteria exists; in other words, the system offered in *Bergey's Manual* is simply a workable system, but in no sense of the word should it be designated as the official classification system.

Presumptive Identification

The place to start in identifying your unknown is to determine what genus it fits into. If *Bergey's Manual* is available, scan the Tables of Contents in Volumes 1 and 2 to find the section that seems to describe your unknown. If these books are not immediately available you can determine the genus by referring to the separation outlines in figures 51.1 and 51.2. Note that seven groups of gram-positive bacteria are winnowed out in figure 51.1 and four groups of gram-negative bacteria in figure 51.2.

To determine which genus in the group best fits the description of your unknown, compare the genera descriptions provided below. Note that each group has a section designation to identify its position in *Bergey's Manual.*

Group I (Section 13, Vol. 2) Although there are only three genera listed in this group, Section 13 in *Bergey's Manual* lists three additional genera, one of which is *Sporosarcina,* a coccus-shaped organism (see Group V). Most members of Group I are motile and differentiation is based primarily on oxygen needs.

Bacillus Although most of these organisms are aerobic, some are facultative anaerobes. Catalase is usually produced. For comparative characteristics of the 34 species in this genus refer to Table 13.4 on pages 1122 and 1123.

Clostridium While most of members of this genus are strict anaerobes, some may grow in the presence of oxygen. Catalase is not usually produced. An excellent key for presumptive species identification is provided on pages 1143–1148. Species characterization tables are also provided on pages 1149–1154.

Sporolactobacillus Microaerophilic and catalase-negative. Nitrates are not reduced and indole is not formed. Spore formation occurs very infrequently (1% of cells).

Since there is only one species in this genus, one needs only to be certain that the unknown is definitely of this genus. Table 13.11 on page 1140 can be used to compare other genera that are similar to this one.

Group II (Section 16, Vol. 2) This group consists of Family Mycobacteriaceae, with only one genus: *Mycobacterium.* Fifty-four species are listed in Section 16. Differentiation of species within this group depends to some extent on whether the organism is classified as a slow or a fast grower. Tables on pages 1439–1442 can be used for comparing the characteristics of the various species.

Group III (Section 14, Vol. 2) Of the seven diverse genera listed in Section 14, only three have been included here in this group.

Lactobacillus Non-spore-forming rods, varying from long and slender to coryneform (club-shaped) coccobacilli. Chain formation is common. Only rarely motile. Facultative anaerobic or microaerophilic. Catalase-negative. Nitrate usually not reduced. Gelatin not liquefied. Indole and H_2S not produced.

Listeria Regular, short rods with rounded ends; occur singly and in short chains. Aerobic and facultative anaerobic. Motile when grown at 20–25° C. Catalase-positive and oxidase-negative. Methyl red positive. Voges-Proskauer positive. Negative for citrate utilization, indole production, urea hydrolysis, gelatinase production, and casein hydrolysis. Table 14.12 on page 1241 provides information pertaining to species differentiation in this genus.

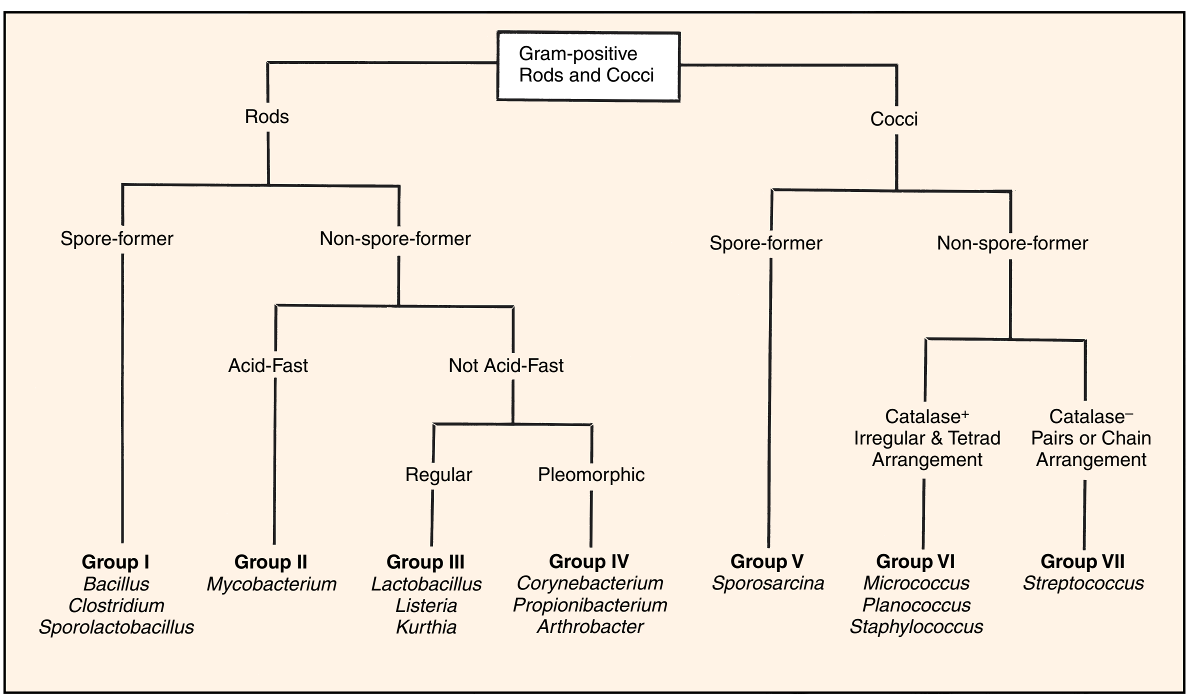


Figure 51.1 Separation outline for gram-positive rods and cocci

Kurthia Regular rods, 2–4 micrometers long with rounded ends; in chains in young cultures; coccoidal in older cultures. Strictly aerobic. Catalase-positive, oxidase-negative. Also negative for gelatinase production and nitrate reduction. Only two species in this genus.

Group IV (Section 15, Vol. 2) Although there are 21 genera listed in this section of *Bergey's Manual,* only three genera are described here.

Corynebacterium Straight to slightly curved rods with tapered ends. Sometimes club-shaped. Palisade arrangements common due to snapping division of cells. Metachromatic granules formed. Facultative anaerobic. Catalase-positive. Most species produce acid from glucose and some other sugars. Often produce pellicle in broth. Table 15.3 on page 1269 provides information for species characterization.

Proprionibacterium Pleomorphic rods, often diphtheroid or club-shaped with one end rounded and the other tapered or pointed. Cells may be coccoid, bifid (forked, divided), or even branched. Nonmotile. Some produce clumps of cells with "Chinese character" arrangements. Anaerobic to aerotolerant. Generally catalase-positive. Produce large amounts of proprionic and acetic acids. All produce acid from glucose.

Arthrobacter Gram-positive rod and coccoid forms. Pleomorphic. Growth often starts out as rods, followed by shortening as growth continues, and finally becoming coccoidal. Some V and angular forms; branching by some. Rods usually nonmotile; some motile. Oxidative, never fermentative. Catalase-positive. Little or no gas produced from glucose or other sugars. Type species is *Arthrobacter globiformis.* For species differentiation see tables on pages 1294 and 1295.

Group V (Section 13, Vol. 2) This group, which has only one genus in it, is closely related to genus *Bacillus.*

Sporosarcina Cells are spherical or oval when single. Cells may adhere to each other when dividing to produce tetrads or packets of eight or more. Endospores formed (see photomicrographs on page 1203). Strictly aerobic. Generally motile. Only two species: *S. ureae* and *S. halophila.*

Group VI (Section 12, Vol. 2) This section contains two families and 15 genera. Our concern here is with only three genera in this group. Oxygen requirements and cellular arrangement are the principal factors in differentiating the genera. Most of these genera are not closely related.

Micrococcus Spheres, occurring as singles, pairs, irregular clusters, tetrads, or cubical packets. Usually nonmotile. Strict aerobes (one species is facultative anaerobic). Catalase- and oxidase-positive. Most species produce carotenoid pigments. All species will grow in media containing 5% NaCl. For species differentiation see Table 12.4 on page 1007.

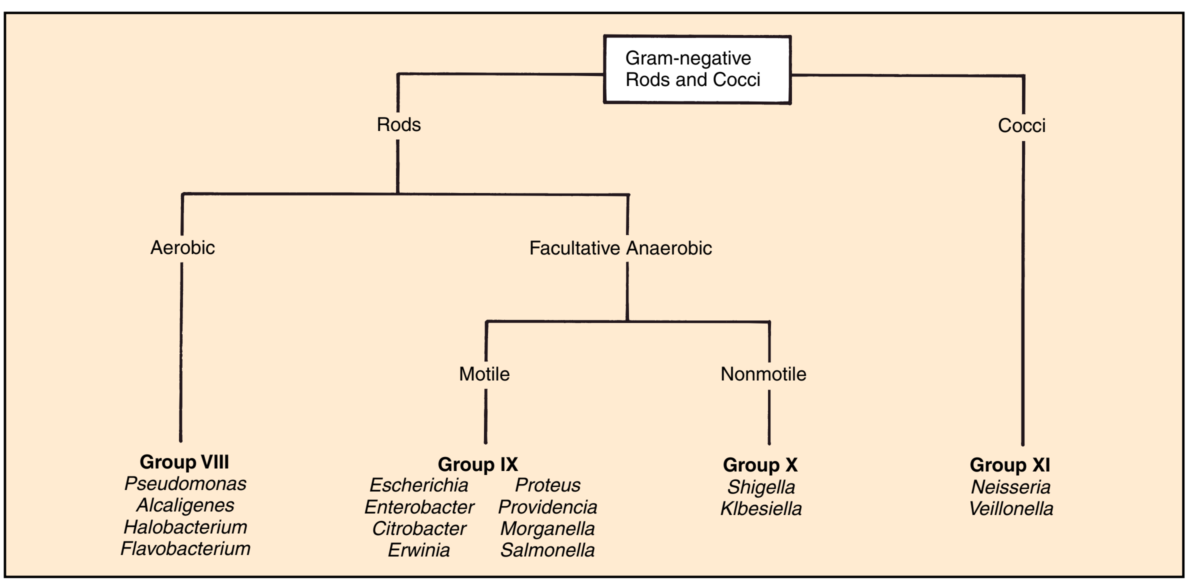


Figure 51.2 Separation outline for gram-negative rods and cocci

Planococcus Spheres, occurring singly, in pairs, in groups of three cells, occasionally in tetrads. Although cells are generally gram-positive, they may be gram-variable. Motility is present. Catalase- and gelatinase-positive. Carbohydrates not attacked. Do not hydrolyze starch or reduce nitrate. Refer to Table 12.9 on page 1013 for species differentiation.

Staphylococcus Spheres, occurring as singles, pairs, and irregular clusters. Nonmotile. Facultative anaerobes. Usually catalase-positive. Most strains grow in media with 10% NaCl. Susceptible to lysis by lysostaphin. Glucose fermentation: acid, no gas. Coagulase production by some. Refer to Exercise 78 for species differentiation, or to Table 12.10 on pages 1016 and 1017.

Group VII (Section 12, Vol. 2) Note that the single genus of this group is included in the same section of *Bergey's Manual* as the three genera in group VI. Members of the genus *Streptococcus* have spherical to ovoid cells that occur in pairs or chains when grown in liquid media. Some species, notably, *S. mutans,* will develop short rods when grown under certain circumstances. Facultative anaerobes. Catalase-negative. Carbohydrates are fermented to produce lactic acid without gas production. Many species are commensals or parasites of humans or animals. Refer to Exercise 79 for species differentiation of pathogens. Several tables in *Bergey's Manual* provide differentiation characteristics of all the streptococci.

Group VIII (Section 4, Vol. 1) Although there are many genera of gram-negative aerobic rod-shaped bacteria, only four genera are likely to be encountered here.

Pseudomonas Generally motile. Strict aerobes. Catalase-positive. Some species produce soluble fluorescent pigments that diffuse into the agar of a slant. Many tables are available in *Bergey's Manual* for species differentiation.

Alcaligenes Rods, coccal rods, or cocci. Motile. Obligate aerobes with some strains capable of anaerobic respiration in presence of nitrate or nitrite.

Halobacterium Cells may be rod- or disk-shaped. Cells divide by constriction. Most are strict aerobes; a few are facultative anaerobes. Catalase- and oxidase-positive. Colonies are pink, red, or red to orange. Gelatinase not produced. Most species require high NaCl concentrations in media. Cell lysis occurs in hypotonic solutions.

Flavobacterium Gram-negative rods with parallel sides and rounded ends. Nonmotile. Oxidative. Catalase-, oxidase-, and phosphatase-positive. Growth on solid media is typically pigmented yellow or orange. Nonpigmented strains do exist. For differentiation see tables on pages 356 and 357.

Groups IX and X (Section 5, Vol. 1) Section 5 in *Bergey's Manual* lists three families and 34 genera; of these 34 only 10 genera of Family *Enterobacteriaceae*

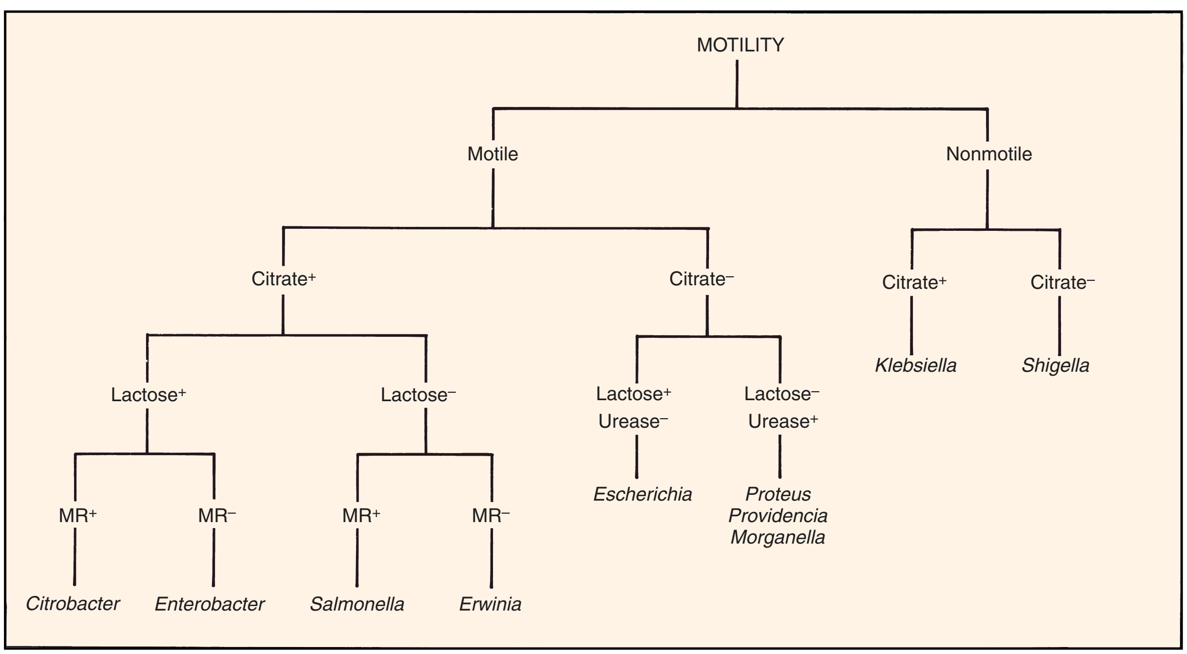


Figure 51.3 Separation outline for groups IX and X

have been included in these two groups. If your unknown appears to fall into one of these groups, use the separation outline in figure 51.3 to determine the genus. Another useful separation outline is provided in figure 80.1 on page 270. *Keep in mind, when using these separation outlines, that there are some minor exceptions in the applications of these tests.* The diversity of species within a particular genus often presents some problematical exceptions to the rule. Your final decision can be made only after checking the species characteristics tables for each genus in *Bergey's Manual.*

Group XI These genera are morphologically quite similar, yet physiologically quite different.

Neisseria (Section 4, Vol. 1) Cocci, occurring singly, but more often in pairs (diplococci); adjacent sides are flattened. One species (*N. elongata*) consists of short rods. Nonmotile. Except for *N. elongata,* all species are oxidase- and catalase-positive. Aerobic.

Veillonella (Section 8, Vol. 1) Cocci, appearing as diplococci, masses, and short chains. Diplococci have flattening at adjacent surfaces. Nonmotile. All are oxidase- and catalase-negative. Nitrate is reduced to nitrite. Anaerobic.

Problem Analysis

If you have identified your unknown by following the above procedures, congratulations! Not everyone succeeds at first attempt. If you are having difficulty, consider the following possibilities:

- You may have been given the wrong unknown! Although this is a remote possibility, it does happen at times. Occasionally, clerical errors are made when unknowns are put together.
- Your organism may be giving you a "false-negative" result on a test. This may be due to an incorrectly prepared medium, faulty test reagents, or improper testing technique.
- Your unknown organisms may not match the description *exactly* as stated in *Bergey's Manual.* By now you are aware that the words *generally, usually,* and *sometimes* are frequently used in the book. It is entirely possible for one of these words to be inadvertently left out in Bergey's assignment of certain test results to a species. *In other words, test results, as stated in the manual, may not always apply!*
- Your culture may be contaminated. If you are not working with a pure culture, all tests are unreliable.
- You may not have performed enough tests. Check the various tables in *Bergey's Manual* to see if there is some other test that will be helpful. In addition, double check the tables to make sure that you have read them correctly.

Confirmation of Results

There are several ways to confirm your presumptive identification. One method is to apply serological techniques, if your organism is one for which typing serum is available. Another alternative is to use one of the miniature multitest systems that are described in the next section of this manual. Your instructor will indicate which of these alternatives, if any, will be available.

Identibacter Interactus

Identibacter interactus is a computer simulation program on a CD-ROM disc that was developed by Allan Konopka, Paul Furbacher, and Clark Gedney at Purdue University. The program is copyrighted by the Purdue Research Foundation, and McGraw-Hill Co. has exclusive distribution rights.

If your laboratory is set up with a computer that has this program installed in it, use the program to confirm the identification of your unknown. A group of 50 tests can be pulled down from menus. The tests are shown in color exactly as you would see them when performed in the laboratory. It is your responsibility to interpret each test as applied to your unknown. Before you attempt to use this program, read the comments pertaining to it in Appendix F.

Laboratory Report

There is no Laboratory Report for this exercise.

PART 9 Miniaturized Multitest Systems

Having run a multitude of tests in Exercises 45 through 50 in an attempt to identify an unknown, you undoubtedly have become aware of the tremendous amount of media, glassware, and preparation time that is involved just to set up the tests. And then, after performing all of the tests and meticulously following all the instructions, you discover that finding the specific organism in "Encyclopedia Bergey" is not exactly the simplest task you have accomplished in this course. The question must arise occasionally: "There's got to be an easier way!" Fortunately, there is: *miniaturized multitest systems.*

Miniaturized systems have the following advantages over the macromethods you have used to study the physiological characteristics of your unknown: (1) minimum media preparation, (2) simplicity of performance, (3) reliability, (4) rapid results, and (5) uniform results. These advantages have resulted in widespread acceptance of these systems by microbiologists.

Since it is not possible to describe all of the systems that are available, only four have been selected here: two by Analytab Products and two by Becton-Dickinson. All four of these products are designed specifically to provide rapid identification of medically important organisms, often within 5 hours. Each method consists of a plastic tube or strip that contains many different media to be inoculated and incubated. To facilitate rapid identification, these systems utilize numerical coding systems that can be applied to charts or computer programs.

The four multitest systems described in this unit have been selected to provide several options. Exercises 52 and 53 pertain to the identification of gram-negative *oxidase-negative* bacteria (Enterobacteriaceae). Exercise 54 (Oxi/Ferm Tube) is used for identifying gram-negative *oxidase-positive* bacteria. Exercise 55 (Staph-Ident) is a rapid system for the differentiation of the staphylococci.

As convenient as these systems are, one must not assume that the conventional macromethods of Part 8 are becoming obsolete. Macromethods must still be used for culture studies and confirmatory tests; confirmatory tests by macromethods are often necessary when a particular test on a miniaturized system is in question. Another point to keep in mind is that all of the miniaturized multitest systems have been developed for the identification of *medically important* microorganisms. If one is trying to identify a saprophytic organism of the soil, water, or some other habitat, there is no substitute for the conventional methods.

If these systems are available to you in this laboratory, they may be used to confirm your conclusions that were drawn in Part 8 or they may be used in conjunction with some of the exercises in Part 14. Your instructor will indicate what applications will be made.

Enterobacteriaceae Identification: The API 20E System

52

The **API 20E System** is a miniaturized version of conventional tests that is used for the identification of members of the family Enterobacteriaceae and other gram-negative bacteria. It was developed by Analytab Products, of Plainview, New York. This system utilizes a plastic strip (figure 52.1) with 20 separate compartments. Each compartment consists of a depression, or *cupule,* and a small *tube* that contains a specific dehydrated medium (see illustration 4, figure 52.2). The system has a capacity of 23 biochemical tests.

To inoculate each compartment it is necessary to first make up a saline suspension of the unknown organism; then, with the aid of a Pasteur pipette, each compartment is filled with the bacterial suspension. The cupule receives the suspension and allows it to flow into the tube of medium. The dehydrated medium is reconstituted by the saline. To provide anaerobic conditions for some of the compartments it is necessary to add sterile mineral oil to them.

After incubation for 18–24 hours, the reactions are recorded, test reagents are added to some compartments, and test results are tabulated. Once the test results are tabulated, a *profile number* (7 or 9 digits) is computed. By finding the profile number in a code book, the *Analytical Profile Index,* one is able to determine the name of the organism. If no *Analytical Profile Index* is available, characterization can be done by using Chart III in Appendix D.

Although this system is intended for the identification of nonenterics, as well as the Enterobacteriaceae, only the identification of the latter will be pursued in this experiment. Proceed as follows to use the API 20E System to identify your unknown enteric.

First Period

Two things will be accomplished during this period: (1) the oxidase test will be performed if it has not been previously performed, and (2) the API 20E test strip will be inoculated. All steps are illustrated in figure 52.2. Proceed as follows to use this system:

Materials:

agar slant or plate culture of unknown
test tube of 5 ml 0.85% sterile saline

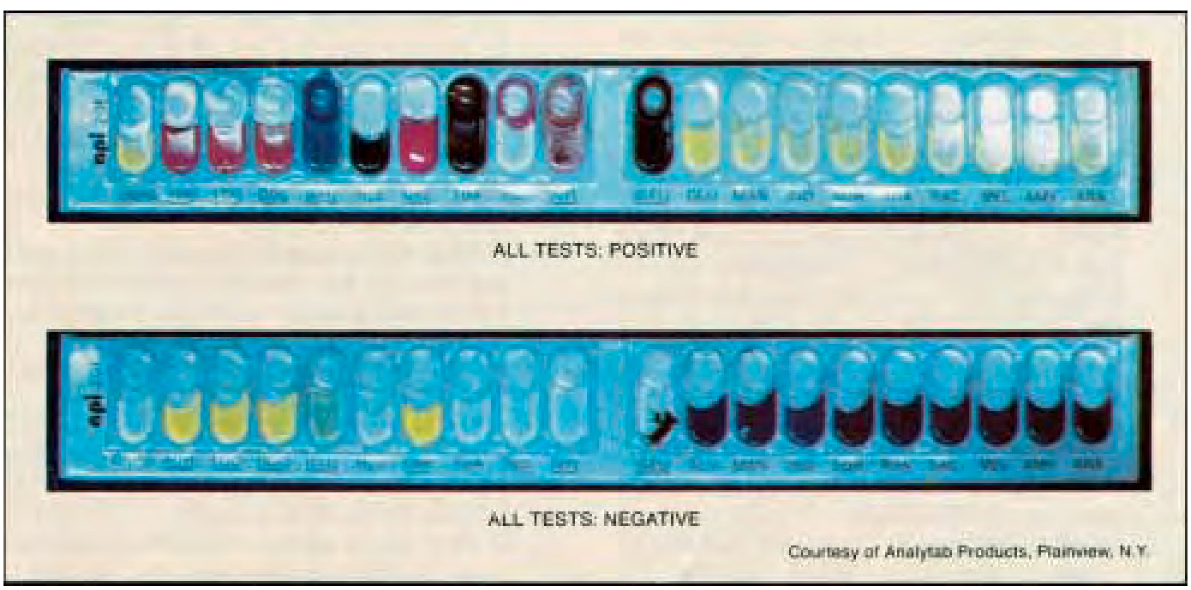


Figure 52.1 Positive and negative test results on API 20E test strips

API 20E test strip
API incubation tray and cover
squeeze bottle of tap water
test tube of 5 ml sterile mineral oil
Pasteur pipettes (5 ml size)
oxidase test reagent
Whatman no. 2 filter paper
empty Petri dish
Vortex mixer

1. If you haven't already done the **oxidase test** on your unknown, do so at this time. It must be established that your unknown is definitely oxidase-negative before using this system. Use the filter paper method that is described on page 168.
2. Prepare a **saline suspension** of your unknown by transferring organisms from the center of a well-established colony on an agar plate (or from a slant culture) to a tube of 0.85% saline solution. Disperse the organisms well throughout the saline.
3. Label the end strip of the API 20E tray with your name and unknown number. See illustration 2, figure 52.2.
4. Dispense about 5 ml of tap water into the tray with a squeeze bottle. Note that the bottom of the tray has numerous depressions to accept the water.
5. Remove an API 20E test strip from the sealed pouch and place it into the tray (see illustration 3). Be sure to reseal the pouch to protect the remaining strips.
6. Vortex mix the saline suspension to get uniform dispersal, and fill a sterile Pasteur pipette with the suspension. *Take care not to spill any of the organisms on the table or yourself. You may have a pathogen!*
7. Inoculate all the tubes on the test strip with the pipette by depositing the suspension into the cupules as you tilt the API tray (see illustration 4, figure 52.2).
   Important: Slightly *underfill* ADH, LDC, ODC, H_2S, and URE. (Note that the labels for these compartments are underlined on the strip.) Underfilling these compartments leaves room for oil to be added and facilitates interpretation of the results.
8. Since the media in |CIT|, |VP|, and |GEL| compartments require oxygen, *completely fill both the cupule and tube* of these compartments. Note that the labels on these three compartments are bracketed as shown here.
9. To provide anaerobic conditions for the ADH, LDC, ODC, H_2S, and URE compartments, dispense sterile **mineral oil** to the cupules of these compartments. Use another sterile Pasteur pipette for this step.
10. Place the lid on the incubation tray and incubate at 37° C for 18 to 24 hours. Refrigeration after incubation is not recommended.

Second Period

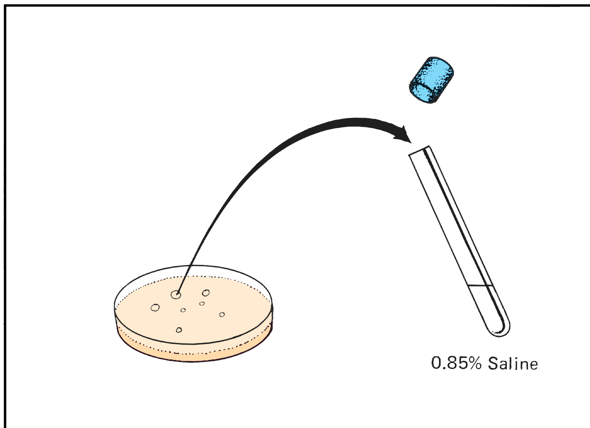
(Evaluation of Tests)

During this period all reactions will be recorded on the Laboratory Report, test reagents will be added to four compartments, and the seven-digit profile number will be determined so that the unknown can be looked up in the *API 20E Analytical Profile Index.* Proceed as follows:

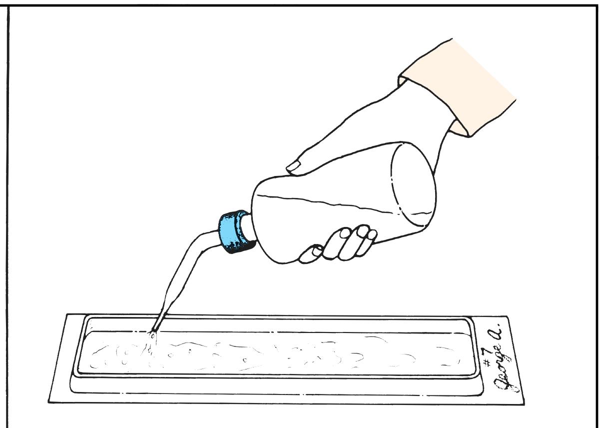
Materials:

incubation tray with API 20E test strip
10% ferric chloride
Barritt's reagents A and B
Kovacs' reagent
nitrite test reagents A and B
zinc dust or 20-mesh granular zinc
hydrogen peroxide (1.5%)
API 20E Analytical Profile Index
Pasteur pipettes

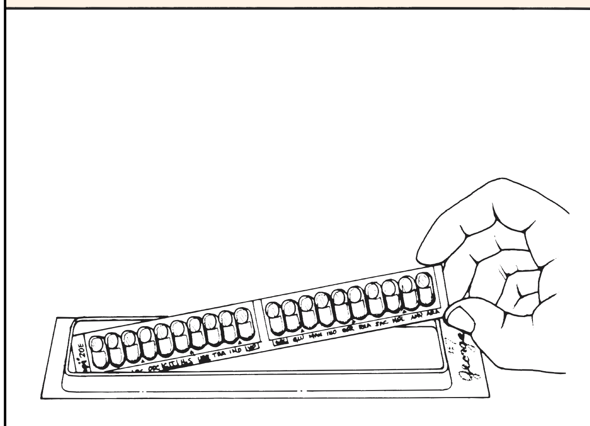
1. Before any test reagents are added to any of the compartments, consult Chart I, Appendix D, to determine the nature of positive reactions of each test, except TDA, VP, and IND.
2. Refer to Chart II, Appendix D, for an explanation of the 20 symbols that are used on the plastic test strip.
3. Record the results of these tests on the Laboratory Report.
4. **If GLU test is negative** (blue or blue-green), **and there are fewer than three positive reactions** before adding reagents, do not progress any further with this test as outlined here in this experiment. Organisms that are GLU-negative are nonenterics.
   For nonenterics, additional incubation time is required. If you wish to follow through on an organism of this type, consult your instructor for more information.
5. **If GLU test is positive** (yellow), **or there are more than three positive reactions,** proceed to add reagents as indicated in the following steps.
6. Add one drop of **10% ferric chloride** to the TDA tube. A positive reaction (brown-red), if it occurs, will occur immediately. A negative reaction color is yellow.
7. Add one drop each of **Barritt's A** and **B solutions** to the VP tube. Read the VP tube within 10 min-



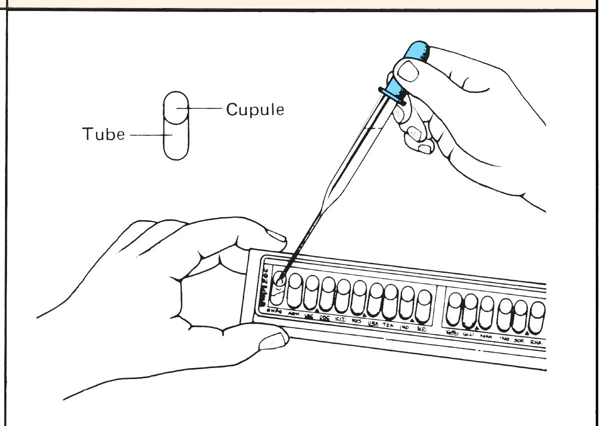
1 Select one well-isolated colony to make a saline suspension of the unknown organism. Suspension should be well dispersed with a Vortex mixer.



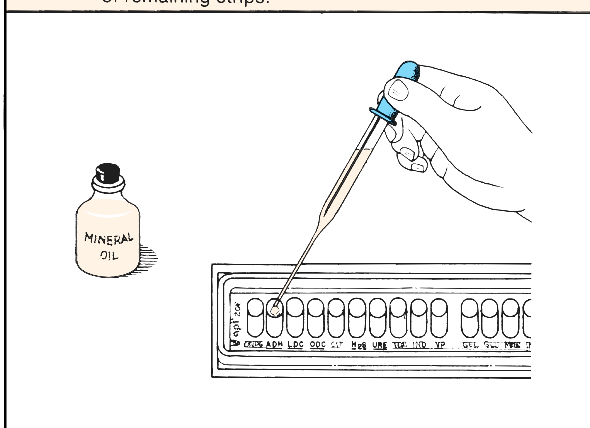
2 After labeling the end tab of a tray with your name and unknown number, dispense approximately 5 ml. of tap water into bottom of tray.



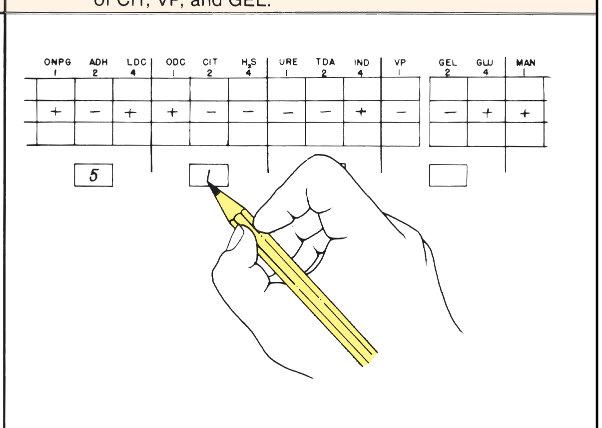
3 Place an API 20E test strip into the bottom of the moistened tray. Be sure to seal the pouch from which the test strip was removed to prevent contamination of remaining strips.



4 Dispense saline suspension of organisms into cupules of all twenty compartments. Slightly *underfill* ADH, LDC, ODC, H_2S, and URE. *Completely fill* cupules of CIT, VP, and GEL.



5 To provide anaerobic conditions for chambers ADH, LDC, ODC, H_2S, and URE, completely fill cupules of these chambers with sterile mineral oil. Use a fresh sterile Pasteur pipette.



6 After incubation and after adding test reagents to four compartments, record all results and total numbers to arrive at 7-digit code. Consult the *Analytical Profile Index* to find the unknown.

Figure 52.2 The API 20E procedure

utes. The pale pink color that occurs immediately has no significance. A positive reaction is dark pink or red and may take 10 minutes before it appears.

8. Add one drop of **Kovacs' reagent** to the IND tube. Look for a positive (red ring) reaction within 2 minutes.

   After several minutes the acid in the reagent reacts with the plastic cupule to produce a color change from yellow to brownish-red, which is considered negative.
9. Examine the GLU tube closely for evidence of bubbles. Bubbles indicate the reduction of nitrate and the formation of N_2 gas. Note on the Laboratory Report that there is a place to record the presence of this gas.
10. Add two drops of each **nitrite test reagent** to the GLU tube. A positive (red) reaction should show up within 2 to 3 minutes if nitrates are reduced.

    If this test is negative, confirm negativity with **zinc dust** or 20-mesh granular zinc. A pink-orange color after 10 minutes confirms that nitrate reduction did not occur. A yellow color results if N_2 was produced.
11. Add one drop of **hydrogen peroxide** to each of the MAN, INO, and SOR cupules. If catalase is produced, gas bubbles will appear within 2 minutes. Best results will be obtained in tubes that have no gas from fermentation.

Final Confirmation

After all test results have been recorded and the seven-digit profile number has been determined, according to the procedures outlined on the Laboratory Report, identify your unknown by looking up the profile number in the *API 20E Analytical Profile Index.*

Cleanup

When finished with the test strip be sure to place it in a container of disinfectant that has been designated for test strip disposal.

Enterobacteriaceae Identification: The Enterotube II System

53

The **Enterotube II** miniaturized multitest system was developed by Becton-Dickinson of Cockeysville, Maryland, for rapid identification of Enterobacteriaceae. It incorporates 12 different conventional media and 15 biochemical tests into a single ready-to-use tube that can be simultaneously inoculated in a moment's time with a minimum of equipment.

If you have an unknown gram-negative rod or coccobacillus that appears to be one of the Enterobacteriaceae, you may wish to try this system on it. Before applying this test, however, *make certain that your unknown is oxidase-negative,* since with only a few exceptions, all Enterobacteriaceae are oxidase-negative. If you have a gram-negative rod that is oxidase-positive you might try the *Oxi/Ferm Tube II* instead, which is featured in the next exercise.

Figure 53.1 illustrates an uninoculated tube (upper) and a tube with all positive reactions (lower). Figure 53.2 outlines the entire procedure for utilizing this system.

Each of the 12 compartments of an Enterotube II contains a different agar-based medium. Compartments that require aerobic conditions have openings for access to air. Those compartments that require anaerobic conditions have layers of paraffin wax over the media. Extending through all compartments of the entire tube is an inoculating wire. To inoculate the media, one simply picks up some organisms on the end of the wire and pulls the wire through each of the chambers in a single, rotating action.

After incubation, the reactions in all the compartments are noted and the indole test is performed. The Voges-Proskauer test may also be performed as a confirmation test. Positive reactions are given numerical values, which are totaled to arrive at a five-digit code. Identification of the unknown is achieved by consulting a coding manual, the *Enterotube II Interpretation Guide,* which lists these numerical codes for the Enterobacteriaceae. Proceed as follows to use an Enterotube II in the identification of your unknown.

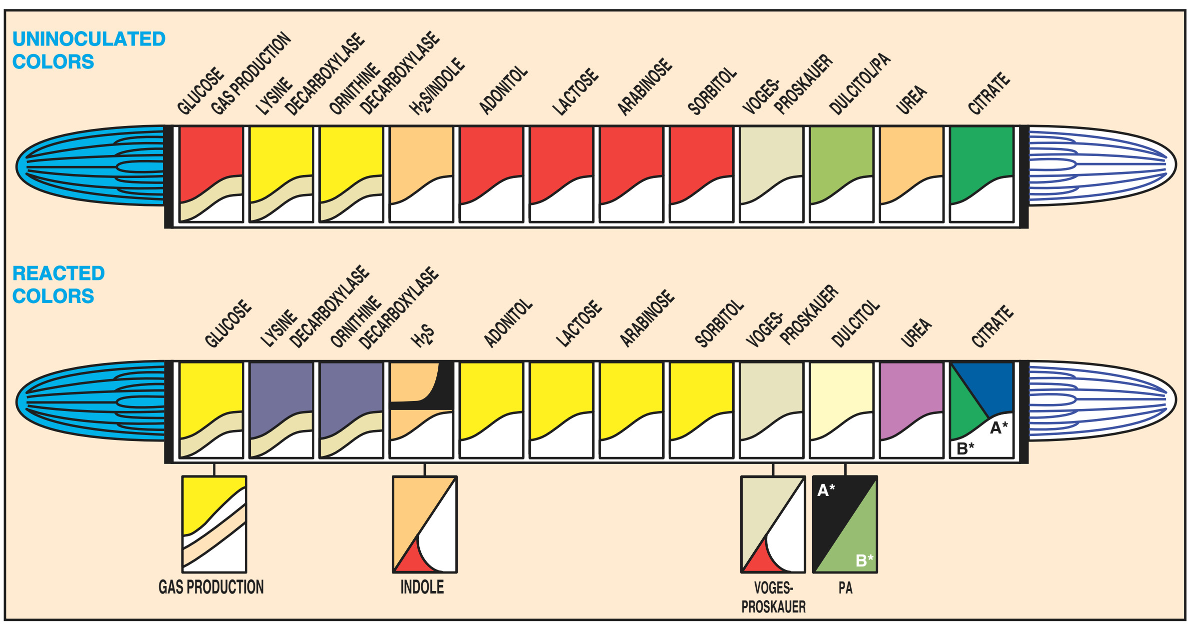


Figure 53.1 Enterotube II color differences between uninoculated and positive tests
Courtesy of Becton-Dickinson, Cockeysville, Maryland.

First Period

Inoculation and Incubation

The Enterotube II can be used to identify Enterobacteriaceae from colonies on agar that have been inoculated from urine, blood, sputum, etc. The culture may be taken from media such as MacConkey, EMB, SS, Hektoen enteric, or trypticase soy agar.

Materials:

culture plate of unknown
1 Enterotube II

1. Write your initials or unknown number on the white paper label on the side of the tube.
2. Unscrew both caps from the Enterotube II. The tip of the inoculating end is under the white cap.
3. *Without heat-sterilizing* the exposed inoculating wire, insert it into a well-isolated colony.
4. Inoculate each chamber by first twisting the wire and then withdrawing it through all 12 compartments. Rotate the wire as you pull it through. See illustration 2, figure 53.2.
5. Again, *without sterilizing,* reinsert the wire, and with a turning motion, force it through all 12 compartments until the notch on the wire is aligned with the opening of the tube. (The notch is about 1⅝″ from handle end of wire.) The tip of the wire should be visible in the citrate compartment. See illustration 3, figure 53.2.
6. Break the wire at the notch by bending, as shown in step 4, figure 53.2. The portion of the wire remaining in the tube maintains anaerobic conditions essential for fermentation of glucose, production of gas, and decarboxylation of lysine and ornithine.
7. With the retained portion of the needle, punch holes through the thin plastic coverings over the small depressions on the sides of the last eight compartments (adonitol, lactose, arabinose, sorbitol, Voges-Proskauer, dulcitol/PA, urea, and citrate). These holes will enable aerobic growth in these eight compartments.
8. Replace the caps at both ends.
9. Incubate at 35° to 37° C for 18 to 24 hours with the Enterotube II lying on its flat surface. *When incubating several tubes together, allow space between them to allow for air circulation.*

Second Period

Reading Results

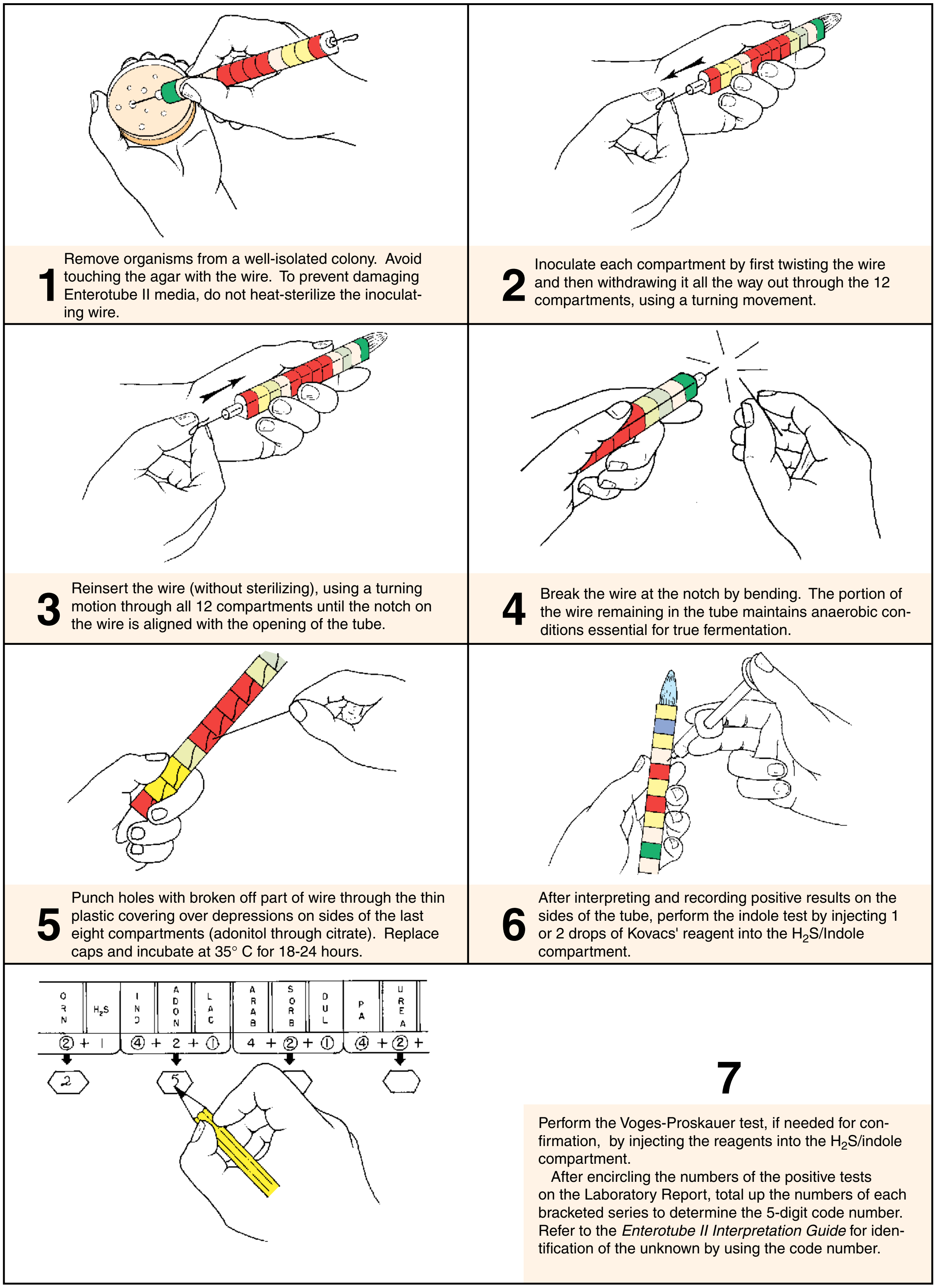
Reading the results on the Enterotube may be done in one of two ways: (1) by simply comparing the results with information on Chart IV, Appendix D, or (2) by finding the five-digit code number you compute for your unknown in the *Enterotube II Interpretation Guide.* Of the two methods, the latter is much preferred. The chart in the appendix should be used *only* if the *Interpretation Guide* is not available.

Whether or not the *Interpretation Guide* is available, these three steps will be performed during this period to complete this experiment: (1) positive test results must *first* be recorded on the Laboratory Report, (2) the indole test, a presumptive test, is performed on compartment 4, and (3) confirmatory tests, if needed, are performed. The Voges-Proskauer test falls in the latter category. Proceed as follows:

Materials:

Enterotube II, inoculated and incubated
Kovacs' reagent
10% KOH with 0.3% creatine solution
5% alpha-naphthol in absolute ethyl alcohol
syringes with needles, or disposable Pasteur pipettes
test-tube rack
Enterotube II Results Pad (optional)
coding manual: *Enterotube II Interpretation Guide*

1. Compare the colors of each compartment of your Enterotube II with the lower tube illustrated in figure 53.1.
2. With a pencil, mark a small plus (+) or minus (−) near each compartment symbol on the white label on the side of the tube.
3. Consult table 53.1 for information as to the significance of each compartment label.
4. Record the results of the tests on the Laboratory Report. *All results must be recorded before doing the indole test.*
5. Record results on the Laboratory Report. **Important:** If at this point you discover that your unknown is GLU-negative, proceed no further with the Enterotube II because your unknown is not one of the Enterobacteriaceae. Your unknown may be *Acinetobacter* sp. or *Pseudomonas maltophilia.* If an Oxi/Ferm Tube is available, try it, using the procedure outlined in the next exercise.
6. **Indole Test:** Perform the indole test as follows:
   a. Place the Enterotube II into a test-tube rack with the GLU-GAS compartment pointing upward.
   b. Inject one or two drops of Kovacs' reagent onto the surface of the medium in the H_2S/indole compartment. This may be done with a syringe and needle through the thin Mylar plastic film that covers the flat surface, or with a disposable Pasteur pipette through a small hole made in the Mylar film with a hot inoculating needle.



1 Remove organisms from a well-isolated colony. Avoid touching the agar with the wire. To prevent damaging Enterotube II media, do not heat-sterilize the inoculating wire.

2 Inoculate each compartment by first twisting the wire and then withdrawing it all the way out through the 12 compartments, using a turning movement.

3 Reinsert the wire (without sterilizing), using a turning motion through all 12 compartments until the notch on the wire is aligned with the opening of the tube.

4 Break the wire at the notch by bending. The portion of the wire remaining in the tube maintains anaerobic conditions essential for true fermentation.

5 Punch holes with broken off part of wire through the thin plastic covering over depressions on sides of the last eight compartments (adonitol through citrate). Replace caps and incubate at 35° C for 18-24 hours.

6 After interpreting and recording positive results on the sides of the tube, perform the indole test by injecting 1 or 2 drops of Kovacs' reagent into the H_2S/Indole compartment.

7 Perform the Voges-Proskauer test, if needed for confirmation, by injecting the reagents into the H_2S/indole compartment.

After encircling the numbers of the positive tests on the Laboratory Report, total up the numbers of each bracketed series to determine the 5-digit code number. Refer to the *Enterotube II Interpretation Guide* for identification of the unknown by using the code number.

Figure 53.2 The Enterotube II procedure

c. A positive test is indicated by the development of a **red color** on the surface of the medium or Mylar film within 10 seconds.

7. **Voges-Proskauer Test:** Since this test is used as a confirmatory test, it should be performed *only* when called for in the *Enterotube II Interpretation Guide.* If it is called for, perform the test in the following manner:
   a. Use a syringe or Pasteur pipette to inject two drops of potassium hydroxide containing creatine into the V-P section.
   b. Inject three drops of 5% alpha-naphthol.
   c. A positive test is indicated by a **red color** within 10 minutes.
8. Record the indole and V-P results on the Laboratory Report.

Laboratory Report

Determine the name of your unknown by following the instructions on the Laboratory Report. Note that two methods of making the final determination are given.

Table 53.1 Biochemical Reactions of Enterotube II

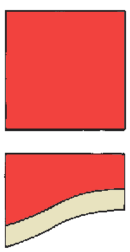
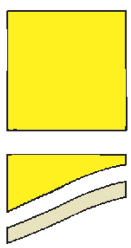
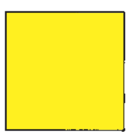
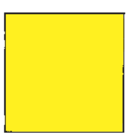
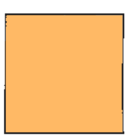
SYMBOL	UNINOCULATED COLOR	REACTED COLOR	TYPE OF REACTION
GLU-GAS			**Glucose (GLU)** The end products of bacterial fermentation of glucose are either acid or acid and gas. The shift in pH due to the production of acid is indicated by a color change from red (alkaline) to yellow (acidic). Any degree of yellow should be interpreted as a positive reaction; orange should be considered negative. **Gas Production (GAS)** Complete separation of the wax overlay from the surface of the glucose medium occurs when gas is produced. The amount of separation between the medium and overlay will vary with the strain of bacteria.
LYS			**Lysine Decarboxylase** Bacterial decarboxylation of lysine, which results in the formation of the alkaline end product cadaverine, is indicated by a change in the color of the indicator from pale yellow (acidic) to purple (alkaline). Any degree of purple should be interpreted as a positive reaction. The medium remains yellow if decarboxylation of lysine does not occur.
ORN			**Ornithine Decarboxylase** Bacterial decarboxylation of ornithine causes the alkaline end product putrescine to be produced. The acidic (yellow) nature of the medium is converted to purple as alkalinity occurs. Any degree of purple should be interpreted as a positive reaction. The medium remains yellow if decarboxylation of ornithine does not occur.
H_2S/IND			**H_2S Production** Hydrogen sulfide, liberated by bacteria that reduce sulfur-containing compounds such as peptones and sodium thiosulfate, reacts with the iron salts in the medium to form a black precipitate of ferric sulfide usually along the line of inoculation. Some **Proteus** and **Providencia** strains may produce a diffuse brown coloration in this medium, which should not be confused with true H_2S production. **Indole Formation** The production of indole from the metabolism of tryptophan by the bacterial enzyme tryptophanase is detected by the development of a pink to red color after the addition of Kovac's reagent.

Table 53.1 Biochemical Reactions of Enterotube II (continued)

SYMBOL	UNINOCULATED COLOR	REACTED COLOR	TYPE OF REACTION
ADON			**Adonitol** Bacterial fermentation of adonitol, which results in the formation of acidic end products, is indicated by a change in color of the indicator present in the medium from red (alkaline) to yellow (acidic). Any sign of yellow should be interpreted as a positive reaction; orange should be considered negative.
LAC			**Lactose** Bacterial fermentation of lactose, which results in the formation of acidic end products, is indicated by a change in color of the indicator present in the medium from red (alkaline) to yellow (acidic). Any sign of yellow should be interpreted as a positive reaction; orange should be considered negative.
ARAB			**Arabinose** Bacterial fermentation of arabinose, which results in the formation of acidic end products, is indicated by a change in color from red (alkaline) to yellow (acidic). Any sign of yellow should be interpreted as a positive reaction; orange should be considered negative.
SORB			**Sorbitol** Bacterial fermentation of sorbitol, which results in the formation of acidic end products, is indicated by a change in color from red (alkaline) to yellow (acidic). Any sign of yellow should be interpreted as a positive reaction; orange should be considered negative.
V.P.			**Voges-Proskauer** Acetylmethylcarbinol (acetoin) is an intermediate in the production of butylene glycol from glucose fermentation. The presence of acetoin is indicated by the development of a red color within 20 minutes. Most positive reactions are evident within 10 minutes.
DUL-PA			**Dulcitol** Bacterial fermentation of dulcitol, which results in the formation of acidic end products, is indicated by a change in color of the indicator present in the medium from green (alkaline) to yellow or pale yellow (acidic). **Phenylalanine Deaminase** This test detects the formation of pyruvic acid from the deamination of phenylalanine. The pyruvic acid formed reacts with a ferric salt in the medium to produce a characteristic black to smoky gray color.
UREA			**Urea** The production of urease by some bacteria hydrolyzes urea in this medium to produce ammonia, which causes a shift in pH from yellow (acidic) to reddish-purple (alkaline). This test is strongly positive for **Proteus** in 6 hours and weakly positive for **Klebsiella** and some **Enterobacter** species in 24 hours.
CIT			**Citrate** Organisms that are able to utilize the citrate in this medium as their sole source of carbon produce alkaline metabolites that change the color of the indicator from green (acidic) to deep blue (alkaline). Any degree of blue should be considered positive.

Courtesy of Becton-Dickinson, Cockeysville, Maryland.

54 O/F Gram-Negative Rods Identification: The Oxi/Ferm Tube II System

The Oxi/Ferm Tube II, produced by Becton-Dickinson, takes care of the identification of the oxidase-positive, gram-negative bacteria that cannot be identified by using the Enterotube II system. The two multitest systems were developed to work together. If an unknown gram-negative rod is oxidase-negative, the Enterotube II is used. If the organism is oxidase-positive, the Oxi/Ferm Tube II must be used. Whenever an oxidase-negative gram-negative rod turns out to be glucose-negative on the Enterotube II test, one must move on to use the Oxi/Ferm Tube II.

The Oxi/Ferm Tube II system is intended for the identification of non-fastidious species of oxidative-fermentative gram-negative rods from clinical specimens. This includes the following genera: *Aeromonas, Plesiomonas, Vibrio, Achromobacter, Alcaligenes, Bordetella, Moraxella,* and *Pasteurella.* Some other gram-negative bacteria can also be identified with additional biochemical tests. The system incorporates 12 different conventional media that can be inoculated simultaneously in a moment's time with a minimum of equipment. A total of 14 physiological tests are performed.

Like the Enterotube II system, the Oxi/Ferm Tube II has an inoculating wire that extends through all 12 compartments of the entire tube. To inoculate the media, one simply picks up some organisms on the end of the wire and pulls the wire through each of the chambers in a rotating action.

After incubation, the results are recorded and Kovacs' reagent is injected into one of the compartments to perform the indole test. Positive reactions are given numerical values that are totaled to arrive at a five-digit code. By looking up the code in an Oxi/Ferm *Biocode Manual,* one can quickly determine the name of the unknown and any tests that might be needed to confirm the identification.

Figure 54.1 illustrates an uninoculated tube and a tube with all positive reactions. Figure 54.2 illustrates the entire procedure for utilizing this system. A minimum of two periods are required to use this system. Proceed as follows:

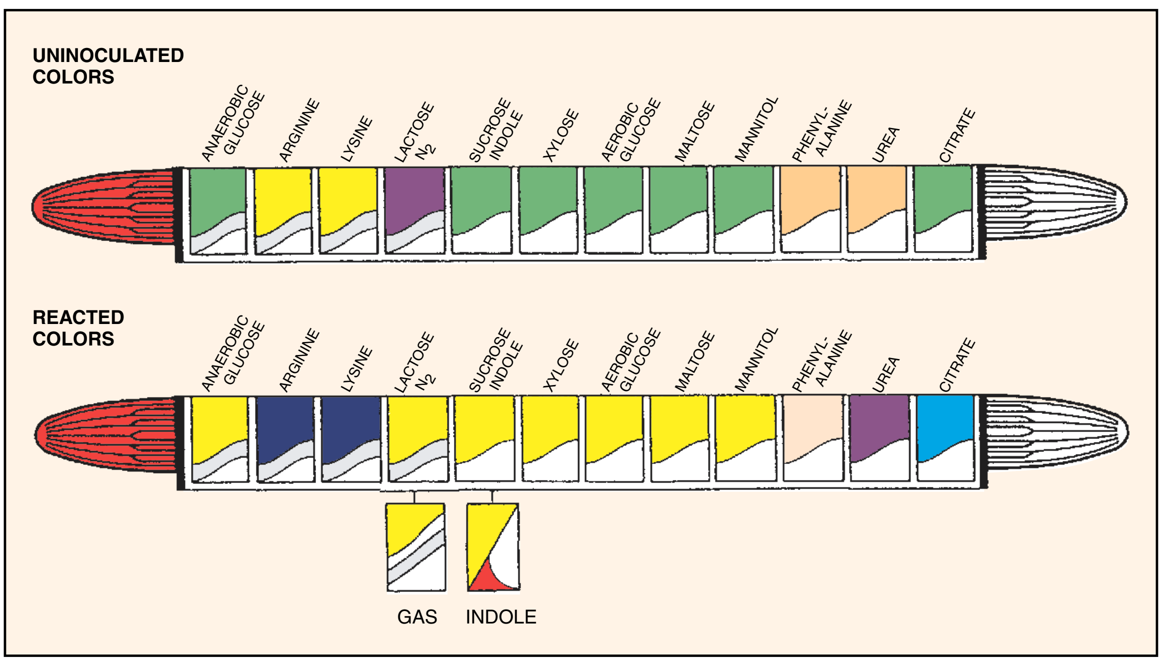
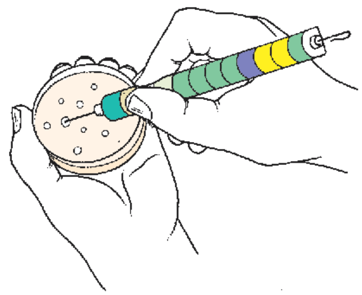
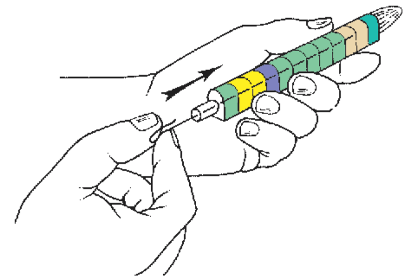


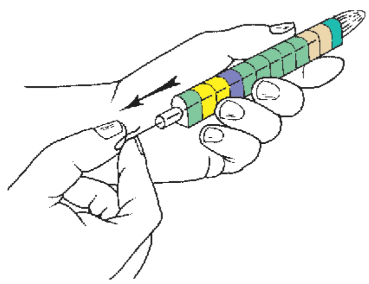
Figure 54.1 Oxi/Ferm Tube II color differences between uninoculated and positive tests



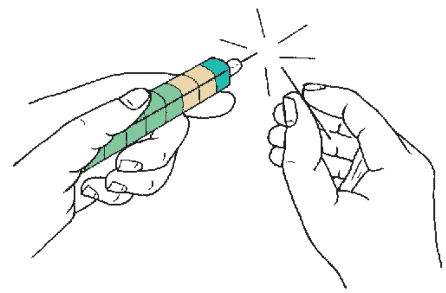
1 Remove organisms from a well-isolated colony. Avoid touching the agar with the wire. To prevent damaging Enterotube II media, do not heat-sterilize the inoculating wire.



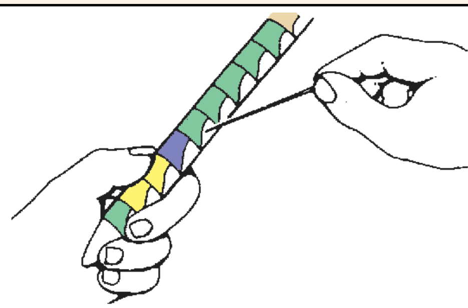
2 Inoculate each compartment by first twisting the wire and then withdrawing it all the way out through the 12 compartments, using a turning movement.



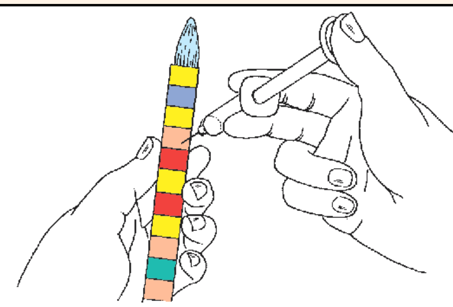
3 Reinsert the wire (without sterilizing), using a turning motion through all 12 compartments until the notch on the wire is aligned with the opening of the tube.



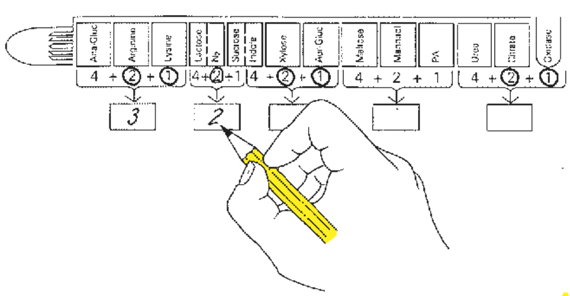
4 Break the wire at the notch by bending. The portion of the wire remaining in the tube maintains anaerobic conditions essential for true fermentation.



5 Punch holes with broken-off part of wire through the thin plastic covering over depressions on sides of the last eight compartments (sucrose/indole through citrate). Replace caps and incubate at 35° C for 18-24 hours.



6 After interpreting and recording positive results on the sides of the tube, perform the indole test by injecting 1 or 2 drops of Kovacs' reagent into the sucrose/indole compartment.



7 After encircling the numbers of the positive tests on the Laboratory Report, total up the numbers of each bracketed series to determine the 5-digit code number. Refer to the *Biocode Manual* for identification of the unknown by using the code number.

Figure 54.2 The Oxi/Ferm Tube II procedure

First Period

Inoculation and Incubation

The Oxi/Ferm Tube II must be inoculated with a large inoculum from a well-isolated colony. Culture purity, of course, is of paramount importance. If there is any doubt of purity, a TSA plate should be inoculated and incubated at 35° C for 24 hours, followed by 24 hours incubation at room temperature. If no growth occurs on TSA, but growth does occur on blood agar, the organism has special growth requirements. *Such organisms are too fastidious and cannot be identified with the Oxi/Ferm Tube II.*

Materials:

culture plate of unknown
1 Oxi/Ferm Tube II
1 plate of trypticase soy agar (TSA) (for purity check, if needed)

1. Write your initials or unknown number on the side of the tube.
2. Unscrew both caps from the Oxi/Ferm Tube II. The tip of the inoculating end is under the white cap.
3. *Without heat-sterilizing* the exposed inoculating wire, insert it into a well-isolated colony. Do not puncture the agar.
4. Inoculate each chamber by first twisting the wire and then withdrawing it through all 12 compartments. Rotate the wire as you pull it through. See illustration 2, figure 54.2.
5. If a purity check of the culture is necessary, streak a Petri plate of TSA with the inoculating wire that has just been pulled through the tube. **Do not flame.**
6. Again, *without sterilizing,* reinsert the wire, and with a turning motion, force it through all 12 compartments until the notch on the wire is aligned with the opening of the tube. (The notch is about 1⅝″ from the handle end of the wire.) The tip of the wire should be visible in the citrate compartment. See illustration 3, figure 54.2.
7. Break the wire at the notch by bending, as noted in step 4, figure 54.2. The portion of the wire remaining in the tube maintains anaerobic conditions essential for true fermentation.
8. With the retained portion of the needle, punch holes through the thin plastic coverings over the small depressions on the sides of the last eight compartments (sucrose/indole, xylose, aerobic glucose, maltose, mannitol, phenylalanine, urea, and citrate). These holes will enable aerobic growth in these eight compartments.
9. Replace both caps on the tube.
10. Incubate at 35° to 37° C for 24 hours, with the tube lying on its flat surface or upright. At the end of 24 hours inspect the tube to check results and continue incubation for another 24 hours. The 24-hour check may be needed for doing confirmatory tests as required in the *Biocode Manual.* Occasionally, an Oxi/Ferm Tube II should be incubated longer than 48 hours.

Second Period

Evaluation of Tests

During this period you will record the results of the various tests on your Oxi/Ferm Tube II, do an indole test, tabulate your results, use the *Biocode Manual,* and perform any confirmatory tests called for. Proceed as follows:

Materials:

Oxi/Ferm Tube II, inoculated and incubated
Kovacs' reagent
syringes with needles, or disposable Pasteur pipettes
Becton-Dickinson *Biocode Manual* (a booklet)

1. Compare the colors of each compartment of your Oxi/Ferm Tube II with the lower tube illustrated in figure 54.1.
2. With a pencil, mark a small plus (+) or minus (−) near each compartment symbol on the white label on the side of the tube.
3. Consult Table 54.1 for information as to the significance of each compartment label.
4. Record the results of all the tests on the Laboratory Report. *All results must be recorded before doing the indole test.*
5. **Indole Test** (illustration 6, figure 54.2): Do an indole test by injecting two or three drops of Kovacs' reagent through the flat, plastic surface into the sucrose/indole compartment. Release the reagent onto the inside flat surface and allow it to drop down onto the agar.

   If a Pasteur pipette is used instead of a syringe needle, it will be necessary to form a small hole in the Mylar film with a hot inoculating needle to admit the tip of the Pasteur pipette.

   A positive test is indicated by the development of a **red color** on the surface of the medium or Mylar film within 10 seconds.
6. Record the results of the indole test on the Laboratory Report.

Laboratory Report

Follow the instructions on the Laboratory Report for determining the five-digit code. Use the *Biocode Manual* booklet for identifying your unknown.

Table 54.1 Biochemical Reactions of the Oxi/Ferm Tube II

Reaction	Negative	Positive	Special Remarks
Anaerobic Glucose			Positive fermentation is shown by change in color from green (neutral) to yellow (acid). Most oxidative-fermentative, gram-negative rods are negative.
Arginine Dihydrolase			Decarboxylation of arginine results in the formation of alkaline end products that changes bromcresol purple from yellow (acid) to purple (alkaline). Grey is negative.
Lysine			Decarboxylation of lysine results in the formation of alkaline end products that changes bromcresol purple from yellow (acid) to purple (alkaline). Grey is negative.
Lactose			Fermentation of lactose changes the color of the medium from red (neutral) to yellow (acid). Most O/F gram-negative rods are negative.
N_2 Gas–production			Gas production causes separation of wax overlay from medium. Occasionally, the gas will also cause separation of the agar from the compartment wall.
Sucrose			Bacterial oxidation of sucrose causes a change in color from green (neutral) to yellow (acid).
Indole			The bacterial enzyme tryptophanase metabolizes tryptophan to produce indole. Detection is by adding Kovacs' reagent to the compartment 48 hours after incubation.
Xylose			Bacterial oxidation of xylose causes a color change of green (neutral) to yellow (acid).
Aerobic Glucose			Bacterial oxidation of glucose causes a color change of green (neutral) to yellow (acid).
Maltose			Bacterial oxidation of maltose causes a color change of green (neutral) to yellow (acid).
Mannitol			Bacterial oxidation of this carbohydrate is evidenced by a change in color from green (neutral) to yellow (acid).
Phenylalanine			Pyruvic acid is formed by deamination of phenylalanine. The pyruvic acid reacts with a ferric salt to produce a brownish tinge.
Urea			The production of ammonia by the action of urease on urea increases the alkalinity of the medium. The phenol red in this medium changes from beige (acid) to pink or purple. Pale pink should be considered negative.
Citrate			Organisms that grow on this medium are able to utilize citrate as their sole source of carbon. Utilization of citrate raises the alkalinity of the medium. The color changes from green (neutral) to blue (alkaline).

55 Staphylococcus Identification: The API Staph-Ident System

The **API Staph-Ident System,** produced by Analytab Products of Plainview, New York, was developed to provide a rapid (5-hour) method for identifying 13 of the most clinically important species of staphylococci. This system consists of 10 microcupules that contain dehydrated substrates and/or nutrient media. Except for the coagulase test, all the tests that are needed for the identification of staphylococci are included on the strip.

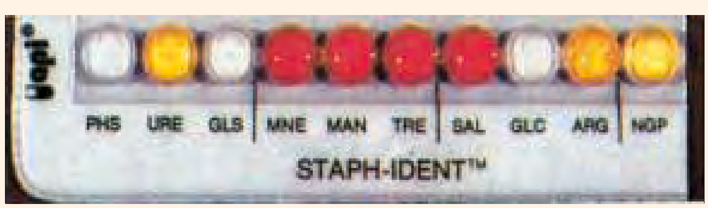
Figure 55.1 illustrates two inoculated strips: the lower one just after inoculation and the upper one with all positive reactions. Note that the appearance of each microcupule undergoes a pronounced color change when a positive reaction occurs.

Figure 55.2 illustrates the overall procedure. The first step is to make a saline suspension of the organism from an isolated colony. A Staph-Ident strip is then placed in a tray that has a small amount of water added to it to provide humidity during incubation. Next, a sterile Pasteur pipette is used to dispense two to three drops of the bacterial suspension to each microcupule. The inoculated tray is then covered and incubated aerobically at 35° to 37° C for 5 hours. After incubation, a few drops of Staph-Ident reagent are added to the tenth microcupule and the results are read immediately. Finally, a four-digit profile is computed that is used to determine the species from a chart in Appendix D.

As simple as this system might seem, there are a few limitations that one must keep in mind. Final species determination by a competent microbiologist must take into consideration other factors such as the source of the specimen, the catalase reaction, colony characteristics, and antimicrobial susceptibility pattern. Very often there are confirmatory tests that must also be made.

If you have been working with an unknown that appears to be one of the staphylococci, use this system to confirm your conclusions. If you have already done the coagulase test and have learned that your organism is coagulase-negative, this system will enable you to identify one of the numerous coagulase-negative species that are not identifiable by the procedures in Exercise 78.

ALL TESTS: POSITIVE



JUST INOCULATED: ALL NEGATIVE

Courtesy of Analytab Products, Plainview, N.Y.

Figure 55.1 Positive and negative results on API Staph-Ident test strips

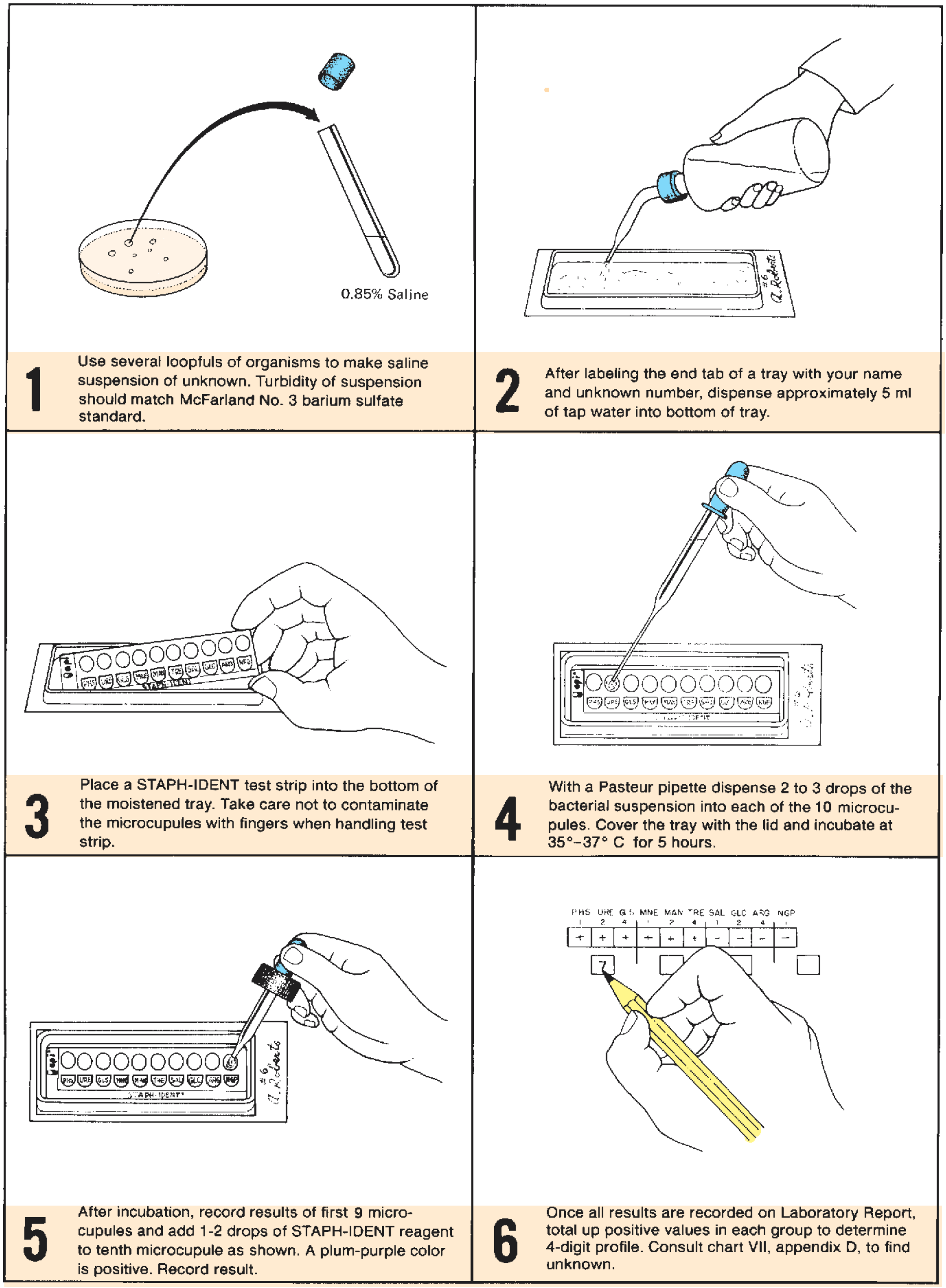


Figure 55.2 The API Staph-Ident procedure

First Period

(Inoculations and Coagulase Test)

Before setting up this experiment, take into consideration that it must be completed at the end of 5 hours. Holding the test strips overnight is not recommended.

Materials:

API Staph-Ident test strip
API incubation tray and cover
blood agar plate culture of unknown (must not have been incubated over 30 hours)
blood agar plate (if needed for purity check)
serological tube of 2 ml sterile saline
test-tube rack
sterile swabs (optional in step 2 below)
squeeze bottle of tap water
tubes containing McFarland No. 3 ($BaSO_4$) standard (see Appendix B)
sterile Pasteur pipette (5 ml size)

1. If the **coagulase test** has not been performed, refer to Exercise 78, page 260, for the procedure and perform it on your unknown.
2. Prepare a saline suspension of your unknown by transferring organisms to a tube of sterile saline from one or more colonies with a loop or sterile swab. Turbidity of the suspension should match a tube of No. 3 McFarland barium sulfate standard.

   Important: Do not allow the bacterial suspension to go unused for any great length of time. Suspensions older than 15 minutes become less effective.
3. Label the end strip of the tray with your name and unknown number. See illustration 2, figure 55.2.
4. Dispense about 5 ml of tap water into the bottom of the tray with a squeeze bottle. Note that the bottom of the tray has numerous depressions to accept the water.
5. Remove the API test strip from its sealed envelope and place the strip in the bottom of the tray.
6. After shaking the saline suspension to disperse the organisms, fill a sterile Pasteur pipette with the bacterial suspension.
7. Inoculate each of the microcupules with two or three drops of the suspension. If a purity check is necessary, use the excess suspension to inoculate another blood agar plate.
8. Place the plastic lid on the tray and incubate the strip aerobically for 5 hours at 35° to 37° C.

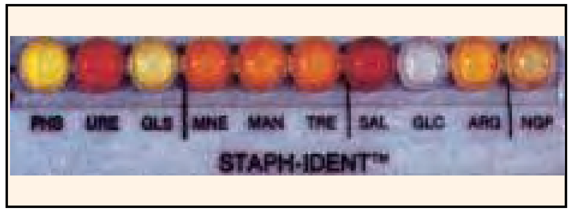


Figure 55.3 Test results of a strip inoculated with *S. aureus.*
(Courtesy of Analytab Products)

Second Period

(Five Hours Later)

During this period the results will be recorded on the Laboratory Report, the profile number will be determined, and the unknown will be identified by looking up the number on the *Staph-Ident Profile Register* (or Chart VII, Appendix D).

Materials:

API Staph-Ident test strip (incubated 5 hours)
1 bottle of Staph-Ident reagent (room temperature)
Staph-Ident Profile Register

1. After 5 hours incubation, refer to Chart V, Appendix D, to interpret and record the results of the first nine microcupules (PHS through ARG).
2. Record the results on the Profile Determination Table on the Laboratory Report. Chart VI, Appendix D, reveals the biochemistry involved in these tests.
3. Add one or two drops of **Staph-Ident reagent** to the NGP microcupule. Allow 30 seconds for the color change to occur.

   A positive test results in a change of color to plum-purple. Record the results of this test.
4. Construct the profile number according to the instructions on the Laboratory Report and determine the name of your unknown.

   If no recent *Profile Register* is available, use Chart VII, Appendix D. Since the *API Register* is constantly being updated, the one in the appendix may be out of date.

Disposal

Once all the information has been recorded be sure to place the entire incubation unit in a receptacle that is to be autoclaved.

Part 10 Microbiology of Soil

With ideal temperature and moisture conditions, soils provide excellent culture media for many kinds of microorganisms. This is especially true of cultivated and improved soils. In many different ways these organisms contribute to the fertility of the very medium they inhabit. The action of certain autotrophic protists on minerals produces substances, organic and inorganic, that are available to plants. Maintaining a proper balance of available nitrogen to photosynthetic plants is one of the most important activities of some forms of bacteria. Several experiments in this unit pertain to the cycling of nitrogen between the soil, water, and atmosphere. The decomposition of lifeless plant and animal tissues returns materials to soils in a form that is reusable by plants.

In addition to studying phases of the nitrogen cycle, an attempt will be made in Exercise 57 to isolate antibiotic producers from soil samples. This study will concentrate primarily on *Actinomyces*-like colonies.

56 Microbial Population Counts of Soil

Soils contain enormous numbers and kinds of microorganisms. In addition to the multitudes of bacteria, there are protozoans, yeasts, molds, algae, and microscopic worms in unbelievable numbers. Types that predominate will depend on the composition of the soil, moisture, pH, and other related environmental factors. No one technique can be used for counting all organisms since such great variability in types exists.

In this exercise we will use the plate count procedure that was used in Exercise 23 to determine the numbers of bacteria, actinomycetes, and molds. It will be necessary to use different kinds of media for each group of organisms. For economy of time and materials, the class will be divided into three groups.

Materials:

1 bottle (50 ml) of nutrient agar (⅓ of class)
1 bottle (50 ml) of glucose peptone acid agar (⅓ of class)
1 bottle (50 ml) of glycerol yeast extract agar (⅓ of class)
3 sterile water blanks (99 ml) (per pair of students)
4 sterile Petri plates per student
1.1 ml dilution pipettes
soil sample

1. Liquefy and cool to 50° C a bottle of medium to be used for the organisms that you will attempt to count. The chart below indicates your assignment:
2. Label four Petri plates according to type of organisms and dilutions. Since the numbers of each type will vary, different dilutions are necessary.

Bacteria	Actinomycetes	Molds
1:10,000	1:1000	1:100
1:100,000	1:10,000	1:1000
1:1,000,000	1:100,000	1:10,000
1:10,000,000	1:1,000,000	1:100,000

3. Label three 99 ml sterile water blanks as you did in Exercise 23.
4. Add 1 gram of soil to blank A, shake vigorously for 5 minutes, and carry out the dilution of blanks B and C.
5. With 1.1 ml pipettes, distribute the proper amounts of water from the blanks to the plates for your final dilutions. See figure 23.1. For the 1:1000 and 1:100 plates, you will need 0.1 ml and 1.0 ml, respectively, from blank A.
6. Pour the appropriate medium into each plate and allow them to cool.
7. Incubate the plates in your locker for 3 to 7 days.
8. Count the colonies, using the procedures outlined in Exercise 23. Record the results on the first portion of Laboratory Report 56, 57.

Student Number	Organisms	Medium
1, 4, 7, 10, 13, 16, 19, 22, 25, 28	Bacteria	Nutrient agar
2, 5, 8, 11, 14, 17, 20, 23, 26, 29	Actinomycetes	Glycerol yeast extract agar
3, 6, 9, 12, 15, 18, 21, 24, 27, 30	Molds	Glucose peptone acid agar

Isolation of an Antibiotic Producer from Soil 57

The constant search of soils throughout the world has yielded an abundance of antibiotics of great value for the treatment of many infectious diseases. Pharmaceutical companies are in constant search for new strains of bacteria, molds, and *Actinomyces* that can be used for antibiotic production. Although many organisms in soil produce antibiotics, only a small portion of new antibiotics are suitable for medical use. In this experiment an attempt will be made to isolate an antibiotic-producing *Actinomyces* from soil. Students will work in pairs.

First Period

(Primary Isolation)

Unless the organisms in a soil sample are thinned out sufficiently, the isolation of potential antibiotic producers is nearly impossible. As indicated in figure 57.1, it will be necessary to use a series of six dilution tubes to produce a final soil dilution of 10^{-6}. Proceed as follows:

Materials:

per pair of students:

- 6 large test tubes
- 1 bottle of physiological saline solution
- 3 Petri plates of glycerol yeast extract agar
- L-shaped glass rod
- beaker of alcohol
- 6 1 ml pipettes
- 1 10 ml pipette

1. Label six test tubes 1 through 6, and with a 10 ml pipette, dispense 9 ml of saline into each tube.
2. Weigh out 1 g of soil and deposit it into tube 1.
3. Vortex mix tube 1 until all soil is well dispersed throughout the tube.

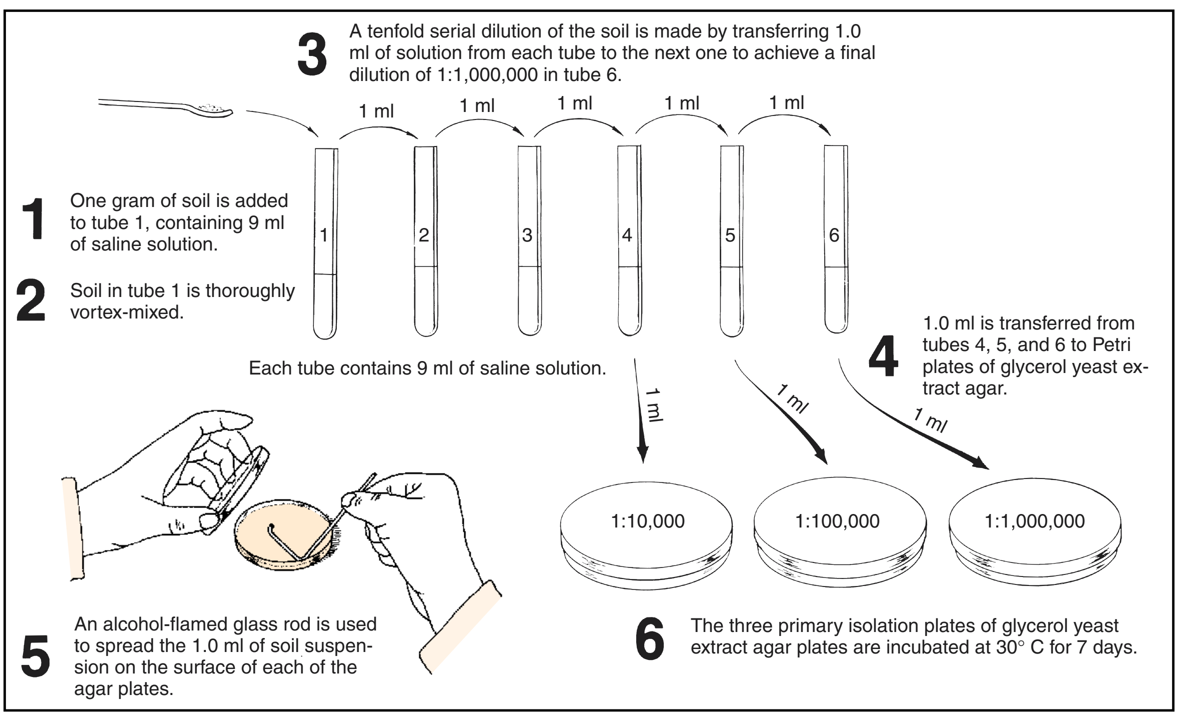


Figure 57.1 Primary isolation of antibiotic-producing *Actinomyces*

4. Make a tenfold dilution from tube 1 through tube 6 by transferring 1 ml from tube to tube. Use a fresh pipette for each transfer and be sure to pipette-mix thoroughly before each transfer.
5. Label three Petri plates with your initials and the dilutions to be deposited into them.
6. From each of the last three tubes transfer 1 ml to a plate of glycerol yeast extract agar.
7. Spread the organisms over the agar surfaces on each plate with an L-shaped glass rod that has been sterilized each time in alcohol and open flame. Be sure to cool rod before using.

CAUTION

Keep Bunsen burner flame away from beaker of alcohol. Alcohol fumes are ignitable. Be sure to flame the glass rod when finished.

8. Incubate the plates at 30° C for 7 days.

Second Period

(Colony Selection and Inoculation)

The objective in this laboratory period will be to select *Actinomyces*-like colonies that may be antibiotic producers. The organisms will be streaked on nutrient agar plates that have been seeded with *Staphylococcus epidermidis.* After incubation we will look for evidence of antibiosis. Students will continue to work in pairs. Figure 57.2 illustrates the procedure.

Materials:

per pair of students:

- 4 trypticase soy agar pours (liquefied)
- 4 sterile Petri plates
- TSB culture of *Staphylococcus epidermidis*
- 1 ml pipette
- 3 primary isolate plates from previous period
- water bath at student station (50° C)

1. Place four liquefied agar pours in water bath (50° C) to prevent solidification, and then inoculate each one with 1 ml of *S. epidermidis.*
2. Label the Petri plates with your initials and date.
3. Pour the contents of each inoculated tube into Petri plates. Allow agar to cool and solidify.
4. Examine the three primary isolation plates for the presence of *Actinomyces*-like colonies. They have

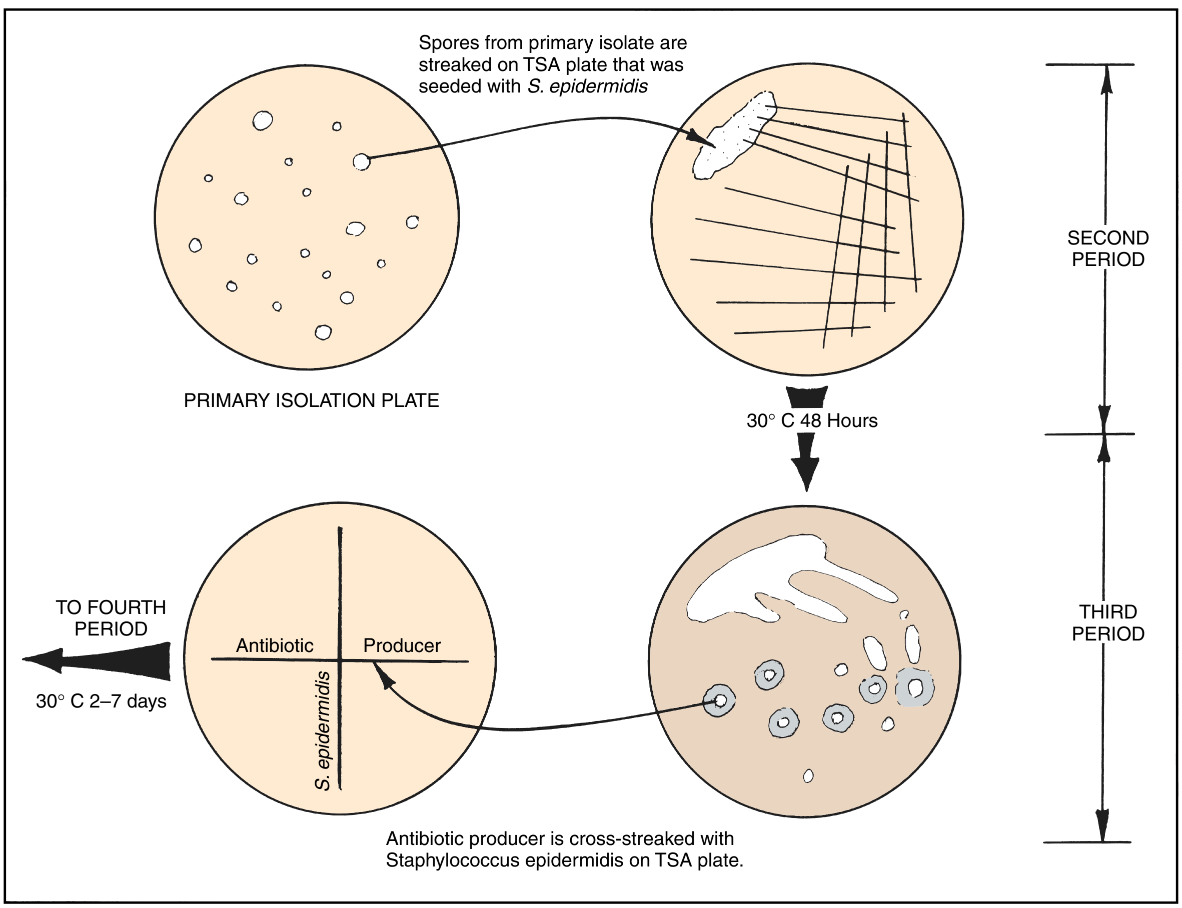


Figure 57.2 Second and third period inoculations

a dusty appearance due to the presence of spores. They may be white or colored. Your instructor will assist in the selection of colonies.

5. Using a sterile inoculating needle, scrape spores from *Actinomyces*-like colonies on the primary isolation plates to inoculate the seeded TSA plates. Use inoculum from a different colony for each of the four plates.
6. Incubate the plates at 30° C until the next laboratory period.

Third and Fourth Periods

(Evidence of Antibiosis and Confirmation)

Examine the four plates you streaked during the last laboratory period. If you see evidence of antibiosis (inhibition of *S. epidermidis* growth), proceed as follows to confirm results.

Materials:

1 Petri plate of trypticase soy agar
TSB culture of *S. epidermidis*

If antibiosis is present, make two streaks on the TSA plate as shown in figure 57.2. Make a straight line streak first with spores from the *Actinomyces* colony, using a sterile inoculating needle. Cross-streak with organisms from a culture of *S. epidermidis.* Incubate at 30° C until the next period.

Laboratory Report

After examining the cross-streaked plate during the fourth period, record your results on the Laboratory Report and answer all the questions.

58 The Nitrogen Cycle

The next three exercises have one thing in common: they all pertain to the nitrogen cycle. This exercise is presented here to unify these experiments as you study the different phases of the nitrogen cycle. There is no laboratory report for this exercise.

As pointed out in Exercise 20, nitrogen is one of the essential elements needed by all living organisms. Although nearly 80% of the atmosphere consists of molecular nitrogen, very few life-forms are able to utilize it in its free state. Instead, most organisms can utilize it only if it is combined ("fixed") with another element such as oxygen or hydrogen. Nitrates (NO_3^-), nitrites (NO_2^-), ammonium (NH_4^+), or organic nitrogenous compounds (proteins and nucleic acids) are the principal forms of fixed nitrogen.

Most plants are able to utilize nitrates and ammonia. Animals, on the other hand, derive their nitrogen from plants and other animals in the form of organic compounds. Microorganisms, however, vary considerably in their nitrogen uptake in that they may get it from all of the sources listed above, plus free nitrogen.

Figure 58.1 illustrates the four phases of the nitrogen cycle: ammonification, nitrification, nitrogen fixation, and denitrification. A discussion of each phase follows.

Ammonification

Most of the nitrogen in soil exists in the form of organic molecules, mostly proteins and nucleic acids that are derived from the decomposition of dead plant and animal tissue. When an organism dies, its proteins are attacked by proteases of soil bacteria to produce polypeptides and amino acids. The amino groups on the amino acids are then removed by a process called **deamination** and converted into ammonia (NH_3). This production of ammonia is called **ammonification.** In addition to the ammonification of protein and nucleic acids of dead animals and plants, other wastes such as urea and uric acid from animal wastes go through the ammonification process. Bacteria and plants that are able to assimilate ammonia convert it into amino acids needed for their own enzyme and protoplasm construction.

$$\text{Amino Acids} \xrightarrow{\text{Many Bacteria}} \text{Ammonia}$$

Nitrification

The next sequence of reactions in the nitrogen cycle involves the oxidation of the nitrogen in the ammonium ion to produce nitrite. This step is followed by the oxidation of nitrites to produce nitrates. This two-step process is called **nitrification.** Note in the reaction below that the first stage is controlled by autotrophs of the genera *Nitrosomonas* and *Nitrosococcus.* The second stage is performed by members of the genera *Nitrobacter* and *Nitrococcus.* Although there are other nitrifying bacteria in soil that can perform these conversions, they are insignificant contributors to this process.

$$\underset{\text{Ammonium ion}}{NH_4^+} \xrightarrow[O_2]{\text{Nitrosomonas, Nitrosococcus}} \underset{\text{Nitrite ion}}{NO_2^-} \xrightarrow[O_2]{\text{Nitrobacter, Nitrococcus}} \underset{\text{Nitrate ion}}{NO_3^-}$$

Nitrogen Fixation

The conversion of atmospheric nitrogen to ammonia is called **nitrogen fixation.** This process, which is illustrated on the right side of figure 58.1, is performed by three groups of microorganisms: (1) free-living bacteria, (2) cyanobacteria, and (3) symbiotic bacteria in root nodules of leguminous plants.

Of the various free-living bacteria, the most beneficial ones belong to genus *Azotobacter.* Since these organisms are strict aerobes, they function efficiently in well-aerated garden soils. Due to the fact that soils usually lack abundant sources of carbohydrates, most of the other free-living bacteria, such as *Clostridium* and *Klebsiella,* fail to contribute as much fixed nitrogen.

Cyanobacteria, such as *Nostoc* and *Anabaena,* which do not require a carbohydrate source for energy, are excellent fixers of atmospheric nitrogen. These chlorophyll-packed organisms usually carry their nitrogen-fixing enzymes in specialized structures called *heterocysts.* The fact that these organisms are so productive in nitrogen fixation explains why they often contribute to organic pollution of freshwater ponds and lakes.

The symbiotic nitrogen-fixing bacteria are the most important contributors to soil enrichment. They develop in root nodules of leguminous plants, such as peas, beans, peanuts, clover, and alfalfa. The principal genera are *Rhizobium* and *Bradyrhizobium.* They are symbiotic in that they produce nourishment for the host plant and the host provides anaerobic conditions and nutrients for the bacteria. Farmers utilize this bacterial relationship through crop rotation to pump literally millions of tons of fixed nitrogen into their soils annually.

Denitrification

Under anaerobic conditions, some microbes can utilize nitrates as electron acceptors to metabolize other organic substances. This conversion of nitrates to free nitrogen is called **denitrification.** The denitrification process takes place as follows:

NO_3^-	→	NO_2^-	→	NO	→	N_2O	→	N_2
Nitrate Ion		Nitrite Ion		Nitric Oxide		Nitrous Oxide		Dinitrogen

Pseudomonas aeruginosa and *Paracoccus denitrificans* are examples of two species that can bring about the denitrification of nitrates and nitrites. Since the denitrification process occurs in waterlogged soils where there is a deficiency of oxygen, farmers minimize nutrient loss from soil by constant cultivation to promote aeration.

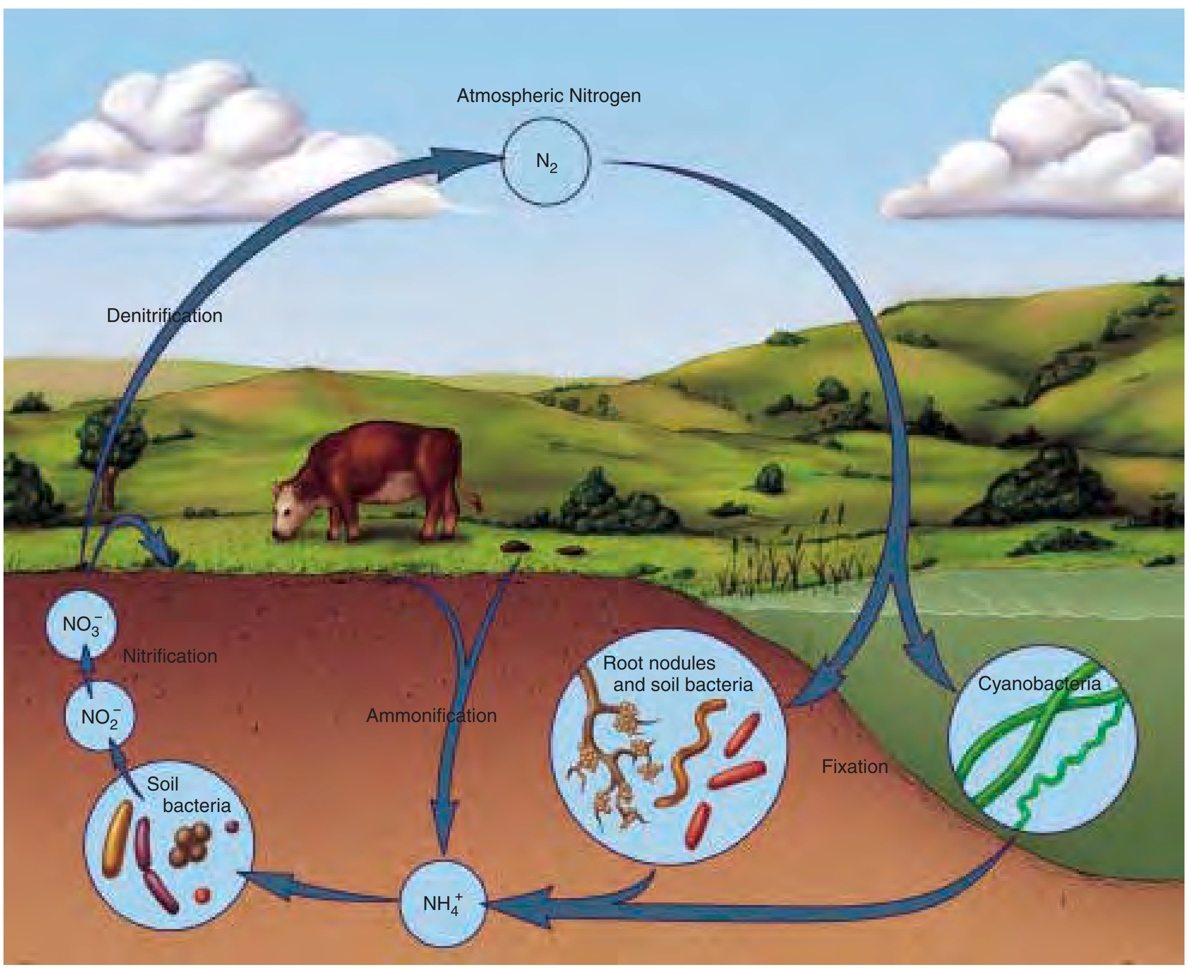


Figure 58.1 The nitrogen cycle

59 Nitrogen-Fixing Bacteria

Among the most beneficial microorganisms of the soil are those that are able to convert gaseous nitrogen of the air to "fixed forms" of nitrogen that can be utilized by other bacteria and plants. Without these nitrogen-fixers, life on this planet would probably disappear within a relatively short period of time.

The utilization of free nitrogen gas by fixation can be accomplished by organisms that are able to produce the essential enzyme *nitrogenase.* This enzyme, in the presence of traces of molybdenum, enables the organisms to combine atmospheric nitrogen with other elements to form organic compounds in living cells. In organic combinations nitrogen is more reduced than when it is free. From these organic compounds, upon their decomposition, the nitrogen is liberated in a fixed form, available to plants either directly or through further microbial action.

The most important nitrogen-fixers belong to two families: **Azotobacteraceae** and **Rhizobiaceae.** Other organisms of less importance that have this ability are a few strains of *Klebsiella,* some species of *Clostridium,* the cyanobacteria, and photosynthetic bacteria.

In this exercise we will concern ourselves with two activities: the isolation of *Azotobacter* from garden soil and the demonstration of *Rhizobium* in root nodules of legumes.

Azotobacteraceae

Bergey's Manual of Systematic Bacteriology, volume 1, section 4, lists two genera of bacteria in family Azotobacteraceae that fix nitrogen as free-living organisms under aerobic conditions: *Azotobacter* and *Azomonas.* The basic difference between these two genera is that *Azotobacter* produces drought-resistant cysts and *Azomonas* does not. Aside from the presence or absence of cysts, these two genera are very similar. Both are large gram-negative motile rods that may be ovoid or coccoidal in shape (pleomorphic). Catalase is produced by both genera. There are six species of *Azotobacter* and three species of *Azomonas.*

Figure 59.1 illustrates the overall procedure that we will use for isolating Azotobacteraceae from garden soil. Note that a small amount of rich garden soil is added to a bottle of nitrogen-free medium that contains glucose as a carbon source. The bottle of medium is incubated in a horizontal position for 4 to 7 days at 30° C.

After incubation, a wet mount slide is made from surface growth to see if typical azotobacter-like organisms are present. If organisms are present, an agar plate of the same medium, less iron, is used to streak out for isolated colonies. After another 4 to 7 days incubation, colonies on the plate are studied and more slides are made in an attempt to identify the isolates.

The N_2-free medium used here contains glucose for a carbon source and is completely lacking in nitrogen. It is selective in that only organisms that can use nitrogen from the air and use the carbon in glucose will grow on it. All species of *Azotobacter* and *Azomonas* are able to grow on it. The metallic ion molybdenum is included to activate the enzyme nitrogenase, which is involved in this process.

First Period (Enrichment)

Proceed as follows to inoculate a bottle of the nitrogen-free glucose medium with a sample of garden soil.

Materials:

1 bottle (50 ml) N_2-free glucose medium (Thompson-Skerman)
rich garden soil (neutral or alkaline)
spatula

1. With a small spatula, put about 1 gm of soil into the bottle of medium. Cap the bottle and shake it sufficiently to mix the soil and medium.
2. Loosen the cap slightly and incubate the bottle at 30° C for 4 to 7 days. Since the organisms are strict aerobes, it is best to incubate the bottle horizontally to provide maximum surface exposure to air.

Second Period (Plating Out)

During this period a slide will be made to make certain that organisms have grown on the medium. If the culture has been successful, a streak plate will be made on nitrogen-free, iron-free agar. Proceed as follows:

Materials:

microscope slides and cover glasses
microscope with phase-contrast optics
1 agar plate of nitrogen-free, iron-free glucose medium

1. After 4 to 7 days incubation, carefully move the bottle of medium to your desktop *without agitating the culture.*
2. Make a wet mount slide with a few loopfuls from the surface of the medium and examine under oil immersion, preferably with phase-contrast optics. Look for large ovoid to rod-shaped organisms, singly and in pairs.

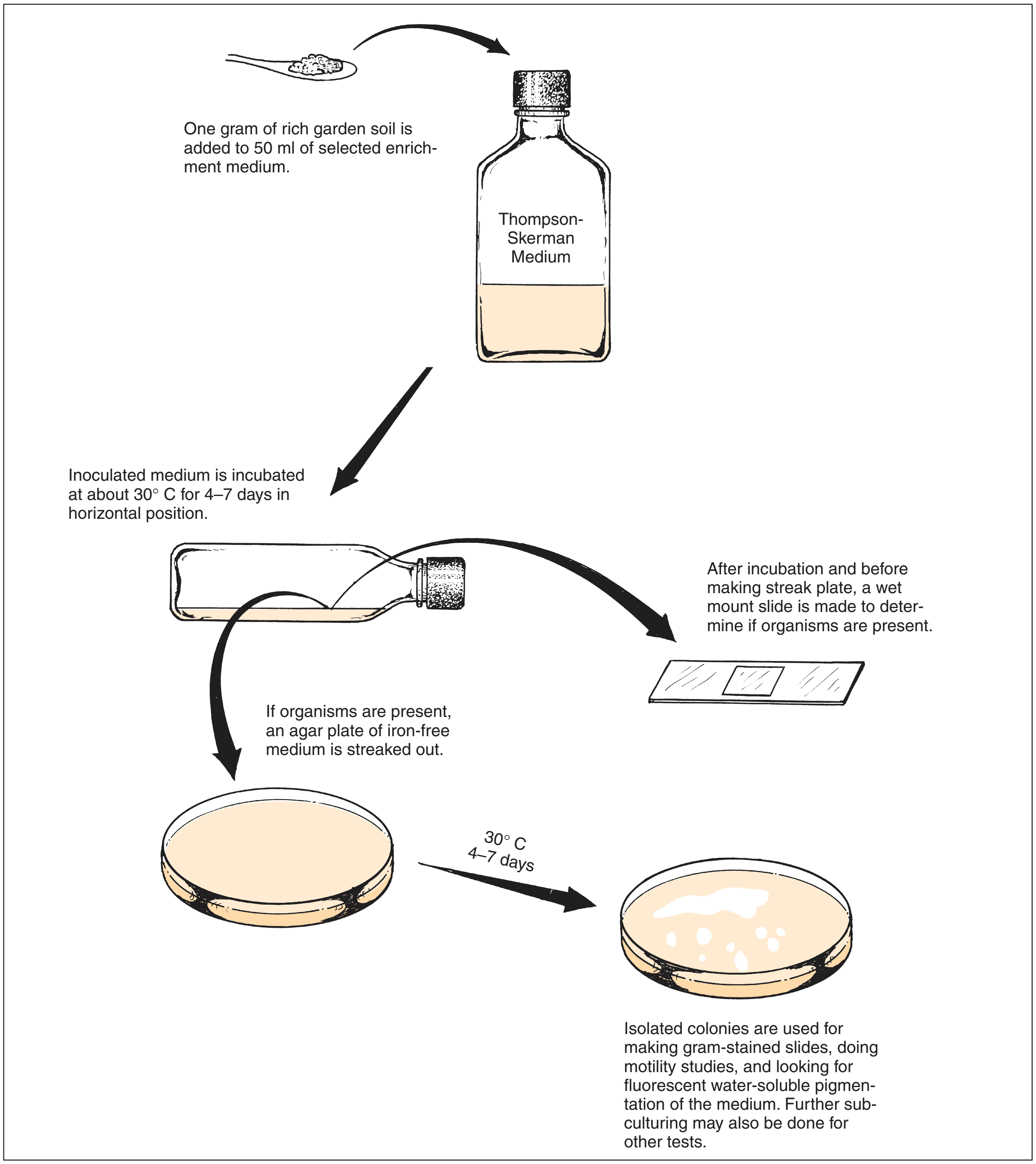


Figure 59.1 Enrichment and isolation procedure for *Azotobacter* and *Azomonas*

3. If azotobacter-like organisms are seen, note whether or not they are motile and if cysts are present. Cysts look much like endospores in that they are refractile. Since cysts often take 2 weeks to form, they may not be seen.
4. If the presence of azotobacter-like organisms is confirmed, streak an agar plate of nitrogen-free, iron-free medium, using a good isolation streak pattern. Ferrous sulfate has been left out of this medium to facilitate the detection of water-soluble pigments.
5. Incubate the plate at 30° C for 4 or 5 days. A longer period of incubation is desirable for cyst formation.

Third Period (Identification)

Azotobacter chroococcum is the type species of genus *Azotobacter.* The cells are 1.5–2.0 micrometers in diameter and pleomorphic, ranging from rods to coccoidal in shape. They occur singly, in pairs, and in irregular clumps. Motility exists with peritrichous flagella. Drought-resistant cysts are produced. They are strict aerobes. Catalase is produced and starch is hydrolyzed. Morphologically, the other five species of this genus look very much like this organism.

Azomonas agilis is the type species of genus *Azomonas.* Except for the absence of cysts, this species and the other two species in this genus are morphologically very similar to *Azotobacter chroococcum.* Practically all of them produce water-soluble fluorescent pigments.

Differentiation of the six species of the genus *Azotobacter* and three species of *Azomonas* is based primarily on the presence or absence of motility, the type of water-soluble pigment produced, and carbon source utilization. Table 59.1 reveals how the organisms can be differentiated. For presumptive identification, use the following character information to identify your isolate.

Materials:

agar plate from previous period
ultraviolet lamp

Motility Note in table 59.1 that four species of *Azotobacter* and all three species of *Azomonas* are motile.

Pigmentation Although these organisms produce both water-soluble and water-insoluble pigments, only the water-soluble ones (those capable of diffus-

Table 59.1 Differential characteristics of the Azotobacteraceae

					Water-Soluble Pigments					
		Cysts	Motility	Long Filaments	Brown-Black	Brown-Black to Red-Violet	Red-Violet	Green	Yellow-Green Fluorescent	Blue-White Fluorescent
Azotobacter	A. chroococcum	+	+	–	–	–	–	–	–	–
	A. vinelandii	+	+	–	–	–	d	d	+	–
	A. beijerinckii	+	–	–	–	–	–	–	–	–
	A. nigricans	+	–	–	d	+	d	–	–	–
	A. armeniacus	+	+	–	–	+	+	–	–	–
	A. paspali	+	+	+	–	–	+	–	+	–
Azomonas	A. agilis	–	+	–	–	–	–	–	+	+
	A. insignis	–	+	–	d[1]	–	d	–	d	–
	A. macrocytogenes	–	+	–	–	–	–	–	d	d

d = 11%–89% positive
d[1] = 11%–89% positive on benzoate

From *Bergey's Manual of Systematic Bacteriology*, volume 1, section 4 ing into an agar medium) are important from the standpoint of species differentiation.

Note in table 59.1 that two of the water-soluble pigments are fluorescent: one is yellow-green and the other is blue-white. To observe fluorescence the cultures must be exposed to ultraviolet light (wavelength 364 nm) in a darkened room. The characteristics of pigment production in each species may be limited by certain factors, as indicated below:

Brown-black: If the colonies produce this hue of diffusible pigment without becoming red-violet, the organism is *A. nigricans.* Although the table indicates that *A. insignis* can produce the brown-black pigment, it can do so only if the medium contains benzoate.

Brown-black to red-violet: As indicated in the table, *A. nigricans* and *A. armeniacus* are the only genera that produce this type of pigment. Motility is a good way to differentiate these two species.

Red-violet: Although table 59.1 reveals that five species can produce this color of diffusible pigment, one (*A. insignis*) cannot produce it on the medium we used. A red-violet isolate is unlikely to be *A. paspali* because this organism has been isolated from the rhizosphere of only one species of grass (*Paspalum notatum*). Thus, isolates that produce this pigment are probably one of the other three in the table.

Green: Note that only *A. vinelandii* can produce this water-soluble pigment; however, only 11%–89% of them produce it.

Yellow-green fluorescent: *A. vinelandii, A. paspali,* and all species of *Azomonas* are able to produce this pigment on the medium we used. Check for fluorescence with an ultraviolet lamp in a darkened room.

Blue-white fluorescent: Note in table 59.1 that two species of *Azomonas* can produce this type of diffusible pigment; no *Azotobacter* are able to produce it. Check for fluorescence with an ultraviolet lamp in a darkened room.

Carbon Source The medium we used in this experiment contains 1% glucose, which can be utilized by all *Azotobacter* and *Azomonas.* Selectivity can be achieved by replacing the glucose with rhamnose, caproate, caprylate, *meso*-inositol, mannitol, malonate, or several other carbon sources. If more precise differentiation is desirable, the student is referred to Tables 4.48 and 4.49 on pages 231 and 232 in *Bergey's Manual,* volume 1.

Laboratory Report

Record your observations and conclusions for the Azotobacteraceae on the Laboratory Report.

Rhizobiaceae

Although the free-living Azotobacteraceae are beneficial nitrogen-fixers, their contribution to nitrogen enrichment of the soil is limited due to the fact that they would rather utilize NH_3 in soil than fix nitrogen. In other words, if ammonia is present in the soil, nitrogen fixation by these organisms is suppressed. By contrast, the symbiotic nitrogen-fixers of genus *Rhizobium,* family Rhizobiaceae, are the principal nitrogen enrichers of soil.

Bergey's Manual lists three genera in family Rhizobiaceae: *Rhizobium, Bradyrhizobium,* and *Agrobacterium.* Although the three genera are related, only genus *Rhizobium* fixes nitrogen. This genus of symbiotic nitrogen-fixers contains only three species. Differentiation of these species relies primarily on plant inoculation tests. A partial list of the host plants for each species is as follows:

R. leguminosarum: peas, vetch, lentils, beans, scarlet runner, and clover
R. meliloti: sweet clover, alfalfa, and fenugreek
R. loti: trefoil, lupines, kidney vetch, chickpea, mimosa, and a few others

All three of these species are gram-negative pleomorphic rods (bacteroids), often X-, Y-, star-, and club-shaped; some exhibit branching. Refractile granules are usually observed with phase-contrast optics. All are aerobic and motile. Our study of *Rhizobium* will be of crushed root nodules from whatever legume is available.

Materials:

washed nodules from the root of a legume
methylene blue stain
microscope slides

1. Place a nodule on a clean microscope slide and crush it by pressing another slide over it. Produce a thin smear by sliding the top slide over the lower one.
2. After air-drying and fixing with heat, stain the smear with **methylene blue** for **30 seconds.**
3. Examine under oil immersion and draw some of the organisms on the Laboratory Report. Look for typical bacteroids of various configurations.

Laboratory Report

Complete the Laboratory Report for this exercise.

60 Ammonification in Soil

As indicated in our discussion of the nitrogen cycle in Exercise 58, the nitrogen in most plants and animals exists in the form of protein. When these organisms die, the protein is broken down to amino acids, which, in turn, are deaminated to liberate ammonia. This process of the production of ammonia from organic compounds is called **ammonification.** Since most bacteria and plants can assimilate ammonia, this is a very important step in the nitrogen cycle. The majority of bacteria in soil are able to take part in this process.

To demonstrate the existence of this process we will inoculate peptone broth with a sample of soil, incubate it for a few days, and test for ammonia production. After a total of 7 days' incubation it will be tested again to see if the amount of ammonia has increased.

First Period

(Inoculation)

Materials:

2 tubes of peptone broth
rich garden soil

1. Inoculate one tube of peptone broth with a loopful of soil. Save the other tube for a control.
2. Incubate the tube at room temperature for 3–4 days and 7 days.

Second and Third Periods

(Ammonia Detection)

After 3 or 4 days, test the medium for ammonia with the following procedure. Repeat these tests again after a total of 7 days of incubation.

Materials:

Nessler's reagent
bromthymol blue and indicator chart spot plate

1. Deposit a drop of **Nessler's reagent** into two separate depressions of a spot plate.
2. Add a loopful of the inoculated peptone broth to one depression and a loopful from the sterile uninoculated tube in the other. Interpretation of ammonia presence is as follows:

   Faint yellow color—small amount of ammonia
   Deep yellow—more ammonia
   Brown precipitate—large amount of ammonia

3. Check the pH of the two tubes by placing several loopfuls of each in separate depressions on the spot plate and adding 1 drop of **bromthymol blue** to each one. Compare the color with a color chart or set of indicator tubes to determine the pH.
4. Record results on the Laboratory Report.

Isolation of a Denitrifier from Soil: Using Nitrate Agar

61

Denitrification is defined as the reduction of nitrate (NO_3) to gaseous dinitrogen (N_2). The consequence of this process is the loss of fixed nitrogen from the soil and water. As far as we know today, the only organisms that are able to denitrify fixed nitrogen are the prokaryotes. Although the percentage of prokaryotes that can perform this phenomenon is not very high, that which they are able to accomplish is truly extensive. The four steps in the denitrification process are as follows:

$$\overline{\mathrm{N0}}_3 \xrightarrow{1} \overline{\mathrm{N0}}_2 \xrightarrow{2} \mathrm{N0} \xrightarrow{3} \mathrm{N_20} \xrightarrow{4} \mathrm{N_2}$$

Nitrate Ion	Nitrite Ion	Nitric Oxide	Nitrous Oxide	Dinitrogen Gas

That denitrification is extremely important in ecological and geochemical terms is undeniable. A summary of the effects of denitrification on ecology is as follows:

- Without the existence of denitrification, the nitrogen in our atmosphere would become completely depleted within a very short period of time. Prokaryotic denitrification is essentially the only source of nitrogen in our atmosphere.
- Denitrification is responsible for the extensive depletion of fixed nitrogen in fertilizers that are put into the soil by farmers. (It has been estimated that somewhere between 5% and 80% of fixed nitrogen is removed from soils by this process.)
- Denitrification plays a major role in the return of N_2 to the atmosphere from fixed nitrogen that exists in runoff water of rivers into the ocean.
- Denitrification is the most practical means of reducing fixed nitrogen from sewage in sewage-treatment plants.
- Nitrous oxide generated in the lower atmosphere diffuses upward to the stratosphere where it is converted to nitric oxide by a photochemical reaction. Result: nitric oxide reacts with ozone to bring about ozone depletion, which threatens our principal barrier against ultraviolet damage to all living organisms. (It should be noted here that the significance of the cumulative effects of industrial, automotive, and other factors on nitrous oxide depletion of the ozone layer is highly controversial.)

The essential function of denitrification to organisms is the generation of ATP. Although some organisms can carry the reaction completely from nitrate to dinitrogen, there are many organisms that are able to act only at stages 1, 2, or 3. If an organism can work from one stage to another without final production of dinitrogen, it is not considered to be a denitrifier.

As far as we know at this time, organisms that can convert nitrate to ammonia do not generate ATP from the production of ammonia; rather, ATP is produced only when the nitrate is first converted to nitrite. This process has been referred to as *nitrate respiration.* This reaction, as seen in the metabolism of *E. coli,* appears to be a means of detoxification of nitrite by conversion to ammonia.

Habitats of Denitrifiers

Although most denitrifiers grow only in an anaerobic environment, they are not all restricted to such places. Those that grow elsewhere have alternative mechanisms such as aerobic respiration, photosynthesis, or fermentation to satisfy their ATP needs.

The most favorable environments for these organisms are heavily fertilized agricultural soils and sewage where nitrogenous compounds abound in considerable quantity. However, denitrifying prokaryotes have been isolated from soils in the Arctic and Antarctic, as well as from sediments in freshwater, brackish water, and salt water. A thermophilic denitrifier has even been isolated from a hot spring. It is obvious, thus, that these organisms are ubiquitous.

Organisms

Lists that have been compiled of denitrifiers reveal that almost all groups of bacteria contain denitrifiers. Typical groups are the phototrophic bacteria, gliding bacteria, spiral and curved bacteria, gram-negative aerobic bacteria, gram-negative cocci, chemolithic sulfur bacteria, gram-positive spore formers, and gram-positive non–spore formers.

Of all the groups listed, the gram-negative aerobic group appears to have the largest number of denitrifiers, with genus *Pseudomonas* predominating.

Procedure

To isolate denitrifiers from a soil sample, the following conditions must be met in the growth medium:

- Some nitrate must be available, which will provide the only terminal electron acceptor for the generation of ATP.
- A carbon source must be present that cannot be fermented by denitrifiers that have a fermentative metabolism. Being unable to ferment the carbon they are forced to use nitrate or nitrogenous oxide for ATP generation.
- Some peptone must be present to provide essential amino acids needed by some denitrifiers.

Once we get an organism that grows on a medium with these characteristics, the next step is to demonstrate the ability of the organism to generate visible nitrogen gas. An isolate that grows on nitrate media and generates gas can be presumed to be a denitrifier. It is these principles that govern the procedure that we will follow here. Figure 61.1 illustrates the procedure that involves a minimum of three laboratory periods.

First Period

Note that the water used in the blender contains 0.1% Tween 80. Tween 80 is a surface active agent that lowers the surface tension around bacteria to improve dispersion of the organisms. The nitrate agar used in the Petri plate is essentially nutrient agar to which 0.5% KNO_3 is added.

Materials:

blenders
fresh soil sample
90 ml distilled water with 0.1% Tween 80
graduate
1 ml pipette
1 Petri plate of nitrate agar
GasPak anaerobic jar, generator envelopes, and generator strips

1. Add 10 grams of soil to 90 ml of water that contains Tween 80.
2. Blend for 2 minutes.
3. Label the bottom of a nitrate agar plate with your name and date of inoculation.
4. Pipette 1.0 ml of the blended mix onto the surface of a plate of nitrate agar.
5. Spread the inoculum over the surface of the agar with a bent glass rod.
6. Incubate the plate, inverted, at 30° C for 3 to 5 days in a GasPak anaerobic jar.

Second Period

During this period, nitrate agar plates will be examined to select colonies that have developed during the incubation period. Since the presence of growth doesn't necessarily mean that the organism is a denitrifier, it will be necessary to see if any of the isolates are nitrogen gas producers; thus, Durham tube nitrate broths must be inoculated and incubated anaerobically. Nitrate broth consists of nutrient broth plus 0.5% KNO_3.

Materials:

nitrate agar plates with colonies
3 Durham tubes of nitrate broth
GasPak anaerobic jar, generator envelopes, and generator strips

1. Examine the nitrate agar plate. Look for colonies that might be *Pseudomonas aeruginosa,* which produces a soluble pigment into the medium. *P. fluorescens* is also a denitrifier.
2. Select three different colonies to inoculate separate tubes.
   Note: Keep a record of the appearance of the colonies transferred to the tubes.
3. Incubate the tubes at 30° C for 3 to 5 days in a GasPak anaerobic jar.

Third Period

Although most denitrifiers grow only in an anaerobic environment, they are not all restricted to such places. Those that grow elsewhere have alternative mechanisms such as aerobic respiration, photosynthesis, or fermentation to satisfy their ATP needs.

Materials:

3 Durham tubes from last period
1 Petri plate of sterile nitrate agar
GasPak anaerobic jar, generator envelopes, and generator strips

1. Record on the Laboratory Report whether you have positive or negative results on the three Durham tubes. The presence of gas is presumptive evidence that a denitrifier has been isolated.
2. If you plan to carry this experiment on further to make more specific identification of an isolate, make a streak plate from the positive tube and proceed as indicated in figure 61.1.

Fourth Period

This period of inoculations is in preparation of trying to do a definitive identification of a denitrifier. Note in figure 61.1 that from an isolated colony a nutrient broth is inoculated and a gram-stained slide is made. After incubation, the broth culture can be used as a stock culture for doing further tests to identify your isolate. The slide will reveal the morphological nature of your organism.

Laboratory Report

Complete the Laboratory Report for this exercise.

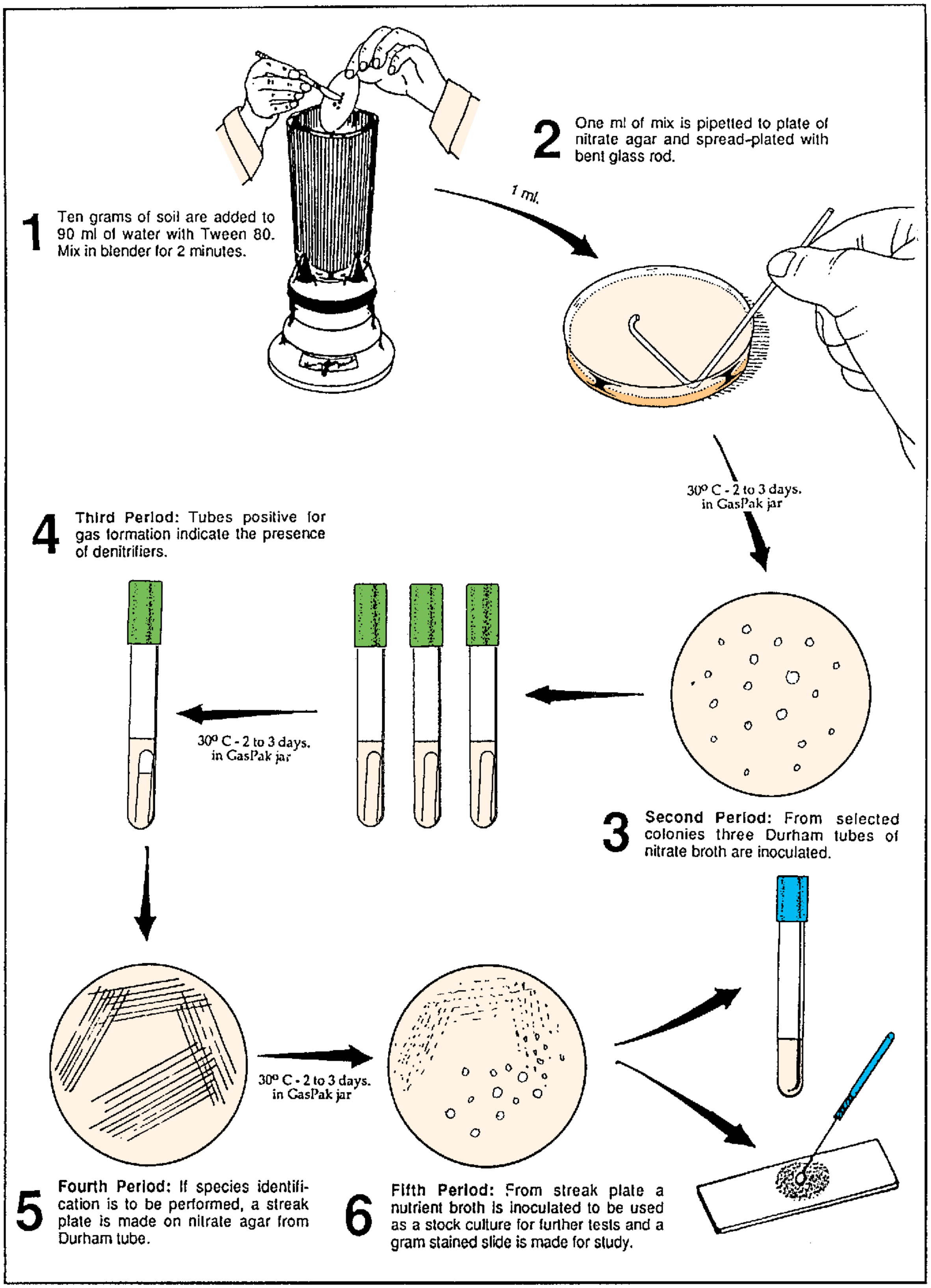


Figure 61.1 Procedure for isolating a denitrifier

Isolation of a Denitrifier from Soil: Using an Enrichment Medium

62

If you have already performed the experiment in Exercise 61, much of the introductory information that follows has already been discussed. For students who have not done Exercise 61, however, this information is critical to understanding this experiment.

Denitrification is defined as the reduction of nitrate (NO_3) to gaseous dinitrogen (N_2). The consequence of this process is the loss of fixed nitrogen from soil and water. As far as we know today, the only organisms that are able to denitrify fixed nitrogen are the prokaryotes. Although the percentage of prokaryotes that can perform this phenomenon is not very high, that which they are able to accomplish is truly extensive. The four steps in the denitrification process are as follows:

$$\overset{-}{N0_3} \xrightarrow{1} \overset{-}{N0_2} \xrightarrow{2} N0 \xrightarrow{3} N_20 \xrightarrow{4} N_2$$

Nitrate Ion	Nitrite Ion	Nitric Oxide	Nitrous Oxide	Dinitrogen Gas

That denitrification is extremely important in ecological and geochemical terms is undeniable. A summary of the effects of denitrification on ecology is as follows:

- Without the existence of denitrification, the nitrogen in our atmosphere would become completely depleted within a very short period of time. Prokaryotic denitrification is essentially the only source of nitrogen in our atmosphere.
- Denitrification is responsible for the extensive depletion of fixed nitrogen in fertilizers that are put into the soil by farmers. (It has been estimated that somewhere between 5% and 80% of fixed nitrogen is removed from soils by this process.)
- Denitrification plays a major role in the return of N_2 to the atmosphere from fixed nitrogen that exists in runoff water of rivers in the ocean.
- Denitrification is the most practical means of reducing fixed nitrogen from sewage in sewage-treatment plants.
- Nitrous oxide generated in the lower atmosphere diffuses upward to the stratosphere where it is converted to nitric oxide by a photochemical reaction. Result: nitric oxide reacts with ozone to bring about ozone depletion, which threatens our principal barrier against ultraviolet damage to all living organisms. (It should be noted here that the significance of the cumulative effects of industrial, automotive, and other factors on nitrous oxide depletion of the ozone layer is highly controversial.)

The essential function of denitrification to organisms is the generation of ATP. Although some organisms can carry the reaction completely from nitrate to dinitrogen, there are many organisms that are able to act only at stages 1, 2, or 3. If an organism can work from one stage to another without final production of dinitrogen, it is not considered to be a denitrifier.

As far as we know at this time, organisms that can convert nitrate to ammonia do not generate ATP from the production of ammonia; rather, ATP is produced only when the nitrate is first converted to nitrite. This process has been referred to as *nitrate respiration.* This reaction, as seen in the metabolism of *E. coli,* appears to be a means of detoxification of nitrite by conversion to ammonia.

Habitats of Denitrifiers

Although most denitrifiers grow only in an anaerobic environment, they are not all restricted to such places. Those that grow elsewhere have alternative mechanisms such as aerobic respiration, photosynthesis, or fermentation to satisfy their ATP needs.

The most favorable environments for these organisms are heavily fertilized agricultural soils and sewage where nitrogenous compounds abound in considerable quantity. However, denitrifying prokaryotes have been isolated from soils in the Arctic and Antarctic, as well as from sediments in freshwater, brackish water, and salt water. A thermophilic denitrifier has even been isolated from a hot spring. It is obvious, thus, that these organisms are ubiquitous.

Organisms

Lists that have been compiled of denitrifiers reveal that almost all groups of bacteria contain denitrifiers. Typical groups are the phototrophic bacteria, gliding bacteria, spiral and curved bacteria, gram-negative aerobic bacteria, gram-negative cocci, chemolithic sulfur bacteria, gram-positive spore formers, and gram-positive non–spore formers.

Of all the groups listed, the gram-negative aerobic group appears to have the largest number of denitrifiers, with genus *Pseudomonas* predominating.

Paracoccus denitrificans In our experiment here we will focus on isolating *Paracoccus denitrificans,* a member of this gram-negative aerobic group. According to *Bergey's Manual,* the characteristics of this denitrifier are as follows:

Cells may be spherical or short rods, gram-negative, and nonmotile. They are aerobic, having a strictly respiratory type of metabolism. Anaerobic growth does occur, however, if nitrate, nitrite, or nitrous oxide are available as terminal electron acceptors. Under anaerobic conditions, nitrate is reduced to nitrous oxide and dinitrogen gas. Colonies on nutrient agar are 2 to 3 mm in diameter, usually circular, entire, smooth, glistening white, and opaque.

To isolate *P. denitrificans* we will use an enrichment culture technique, which employs a **nitrate succinate–mineral salts medium.** This medium contains sodium succinate, potassium nitrate, and basic mineral salts. Being able to oxidize the succinate and use nitrate as a terminal electron acceptor to produce nitrogen gas, *P. denitrificans* will grow very well.

Procedure

Figure 62.1 illustrates the procedure for the first two laboratory sessions. This phase of the experiment will yield a mixed culture of *P. denitrificans* and other soil bacteria. To get a pure culture of *P. denitrificans* we will follow the procedure in figure 62.2. Proceed as follows:

First Period

Materials:

fresh soil sample
flask containing 200 ml nitrate succinate–mineral salts broth
sterile glass-stoppered bottle (60 ml size)
Petri dish top or bottom

1. Add 1 g of soil to the nitrate succinate-mineral salts broth. Shake the flask vigorously and allow the soil contents to settle.
2. Carefully decant some of the supernatant into a glass-stoppered bottle, filling it to total capacity.
3. Insert the stopper into the bottle in such a way that the medium in the neck of the bottle is expelled.
4. Place the bottle in the top or bottom of a Petri dish to collect any liquid that is expelled during incubation.
5. Incubate the bottle at 30° C until the next laboratory session.

Second Period

Materials:

flask of sterile nitrate succinate–mineral salts broth
culture in glass-stoppered bottle (from previous lab period)
sterile glass-stoppered bottle (60 ml size)
1 ml pipette
microscope slides and cover glasses
gram-staining kit

1. Examine the bottled culture for the presence of gas. A stream of nitrogen-gas bubbles should be visible extending up from the bottom of the bottle and collecting at the top of the culture. The glass stopper is often displaced by the force of the gas.
2. Prepare a second enrichment by aseptically transferring 1 ml of the initial culture to a sterile second glass-stoppered bottle. Fill the bottle completely and stopper it. Set this bottle aside in a Petri dish cover to incubate at 30° C until the next laboratory period.
3. Prepare a gram-stained slide from the initial culture and examine it under oil immersion.
4. Make a wet mount slide from the initial culture and examine with phase-contrast optics.
5. Record all your observations on the Laboratory Report.

Third Period

Materials:

new culture in glass-stoppered bottle (from previous lab period)
Petri plate with nitrate succinate–mineral salts agar
microscope slides and cover glasses
gram staining kit
GasPak anaerobic jar, generator envelopes, and generator strips

1. Examine the second enrichment bottle, looking for bubbles. Check its clarity, also.
2. Streak a nitrate succinate–mineral salts agar plate from this second enrichment bottle. Set the plate aside to incubate in a GasPak jar at 30° C until the next laboratory period.
3. Prepare a gram-stained slide from the second enrichment culture and examine it under oil immersion.

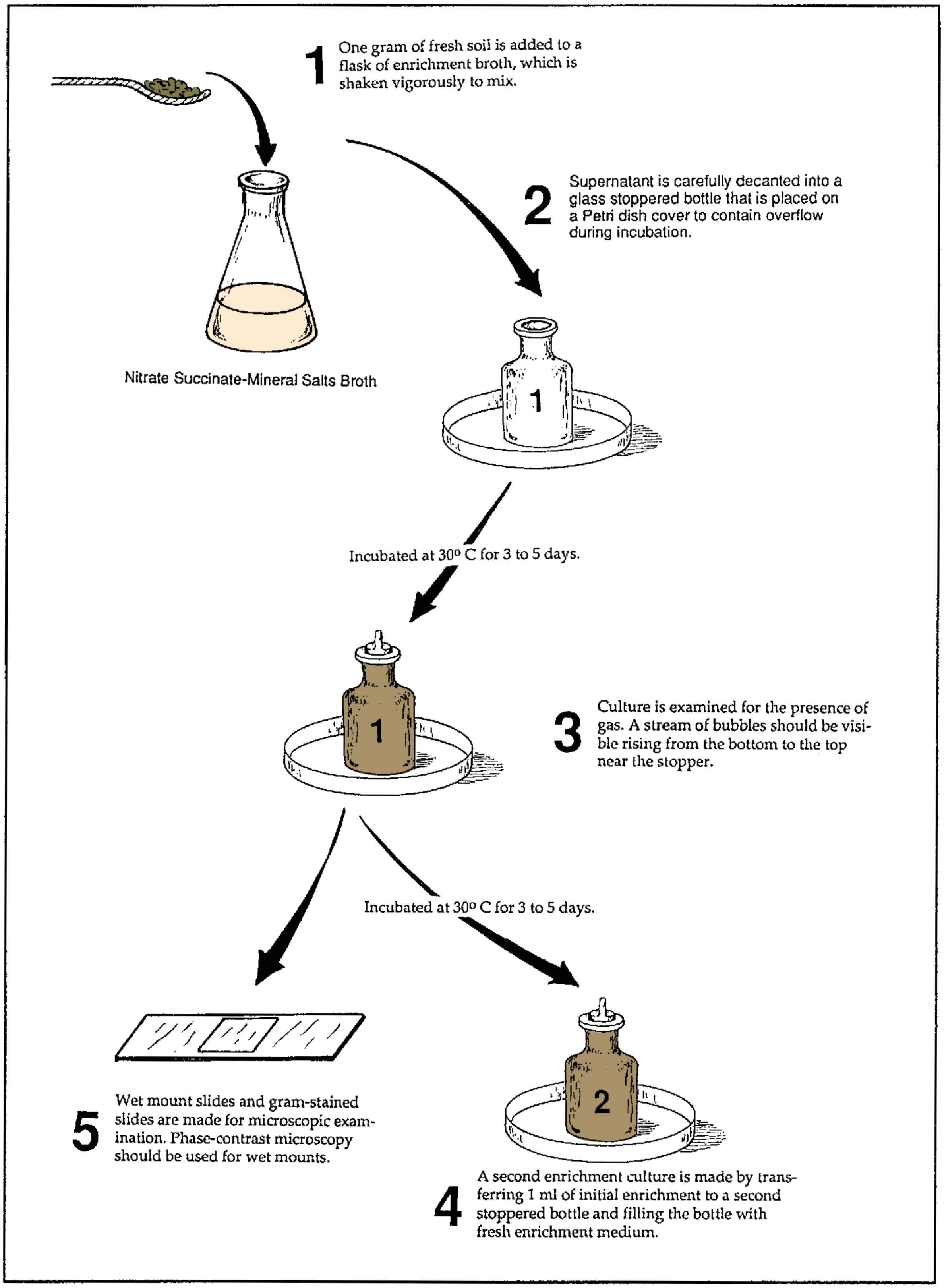


Figure 62.1 Procedure for culturing *Paracoccus denitrificans* from a soil sample

4. Make a wet mount slide from the same cuture and examine with phase-contrast optics.
5. Record all your observations on the Laboratory Report.

Fourth Period

Materials:

Petri plate culture from last period
microscope slides and gram-staining kit

1. Examine the colonies on the streak plate you incubated anaerobically from the last period. Compare the characteristics of the colonies with the characteristics of *Paracoccus denitrificans* that are given on page 218. Do the colonies look like *P. denitrificans?*
2. Make a gram-stained slide from one of the colonies and examine the slide under the microscope. How do the characteristics of the organism match *Bergey's Manual* description? Record your results on the Laboratory Report.

Laboratory Report

Complete the Laboratory Report for this exercise.

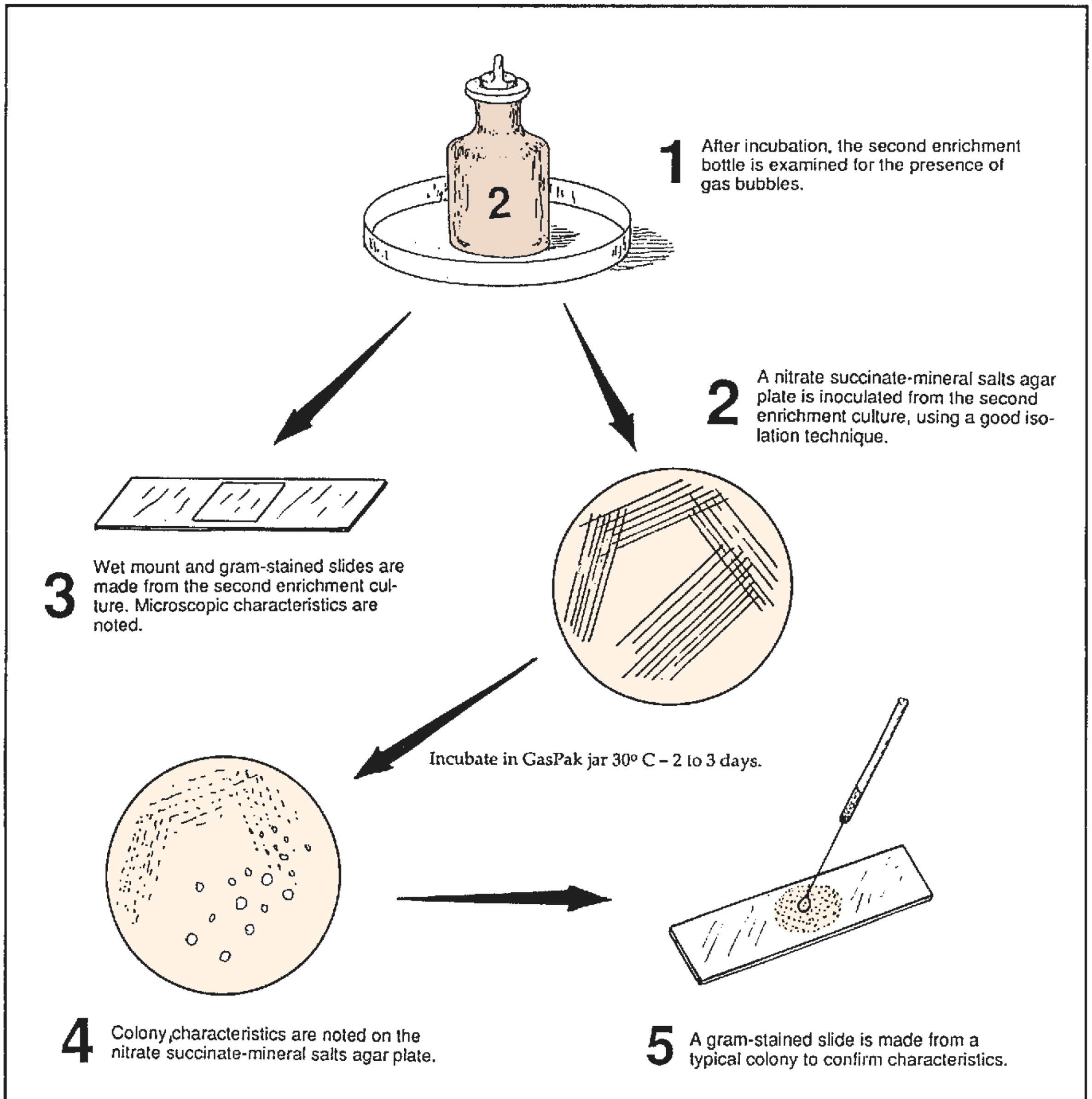


Figure 62.2 Procedure for getting a pure culture out of second enrichment culture

PART 11 Microbiology of Water

The microorganisms of natural waters are extremely diverse. The numbers and types of bacteria present will depend on the amounts of organic matter present, the presence of toxic substances, the water's saline content, and environmental factors such as pH, temperature, and aeration. The largest numbers of heterotrophic forms will exist on the bottoms and banks of rivers and lakes where organic matter predominates. Open water in the center of large bodies of water, free of floating debris, will have small numbers of bacteria. Many species of autotrophic types are present, however, that require only the dissolved inorganic salts and minerals that are present.

The threat to human welfare by contamination of water supplies with sewage is a prime concern of everyone. The enteric diseases such as cholera, typhoid fever, and bacillary dysentery often result in epidemics when water supplies are not properly protected or treated. Thus, our prime concern in this unit is the sanitary phase of water microbiology. The American Public Health Association in its *Standard Methods for the Examination of Water and Wastewater* has outlined acceptable procedures for testing water for sewage contamination. The exercises of this unit are based on the procedures in that book.

63 Bacteriological Examination of Water: Qualitative Tests

Water that contains large numbers of bacteria may be perfectly safe to drink. The important consideration, from a microbiological standpoint, is the kinds of microorganisms that are present. Water from streams and lakes that contain multitudes of autotrophs and saprophytic heterotrophs is potable as long as pathogens for humans are lacking. The intestinal pathogens such as those that cause typhoid fever, cholera, and bacillary dysentery are of prime concern. The fact that human fecal material is carried away by water in sewage systems that often empty into rivers and lakes presents a colossal sanitary problem; thus, constant testing of municipal water supplies for the presence of fecal microorganisms is essential for the maintenance of water purity.

Routine examination of water for the presence of intestinal pathogens would be a tedious and difficult, if not impossible, task. It is much easier to demonstrate the presence of some nonpathogenic intestinal types such as *Escherichia coli* or *Streptococcus faecalis.* Since these organisms are always found in the intestines, and normally are not present in soil or water, it can be assumed that their presence in water indicates that fecal material has contaminated the water supply.

E. coli and *S. faecalis* are classified as good **sewage indicators.** The characteristics that make them good indicators of fecal contamination are (1) they are normally not present in water or soil, (2) they are relatively easy to identify, and (3) they survive a *little* longer in water than enteric pathogens. If they were hardy organisms, surviving a long time in water, they would make any water purity test too sensitive. Since both organisms are non-spore-formers, their survival in water is not extensive.

E. coli and *S. faecalis* are completely different organisms. *E. coli* is a gram-negative non-spore-forming rod; *S. faecalis* is a gram-positive coccus. The former is classified as a coliform; the latter is an enterococcus. Physiologically, they are also completely different.

The series of tests depicted in figure 63.1 is based on tests that will demonstrate the presence of a coliform in water. By definition, a **coliform** is a facultative anaerobe that ferments lactose to produce gas and is a gram-negative, non-spore-forming rod. *Escherichia coli* and *Enterobacter aerogenes* fit this description. Since *S. faecalis* is not a coliform, a completely different set of tests must be used for it.

Note that three different tests are shown in figure 63.1: presumptive, confirmed, and completed. Each test exploits one or more of the characteristics of a coliform. A description of each test follows.

Presumptive Test In the presumptive test a series of 9 or 12 tubes of lactose broth are inoculated with measured amounts of water to see if the water contains any lactose-fermenting bacteria that produce gas. If, after incubation, gas is seen in any of the lactose broths, it is *presumed* that coliforms are present in the water sample. This test is also used to determine the most probable number (MPN) of coliforms present per 100 ml of water.

Confirmed Test In this test, plates of Levine EMB agar or Endo agar are inoculated from positive (gas-producing) tubes to see if the organisms that are producing the gas are gram-negative (another coliform characteristic). Both of these media inhibit the growth of gram-positive bacteria and cause colonies of coliforms to be distinguishable from noncoliforms. On EMB agar coliforms produce small colonies with dark centers (nucleated colonies). On Endo agar coliforms produce reddish colonies. The presence of coliform-like colonies confirms the presence of a lactose-fermenting gram-negative bacterium.

Completed Test In the completed test our concern is to determine if the isolate from the agar plates truly matches our definition of a coliform. Our media for this test include a nutrient agar slant and a Durham tube of lactose broth. If gas is produced in the lactose tube and a slide from the agar slant reveals that we have a gram-negative non-spore-forming rod, we can be certain that we have a coliform.

The completion of these three tests with positive results establishes that coliforms are present; however, there is no certainty that *E. coli* is the coliform present. The organism might be *E. aerogenes.* Of the two, *E. coli* is the better sewage indicator since *E. aerogenes* can be of nonsewage origin. To differentiate these two species, one must

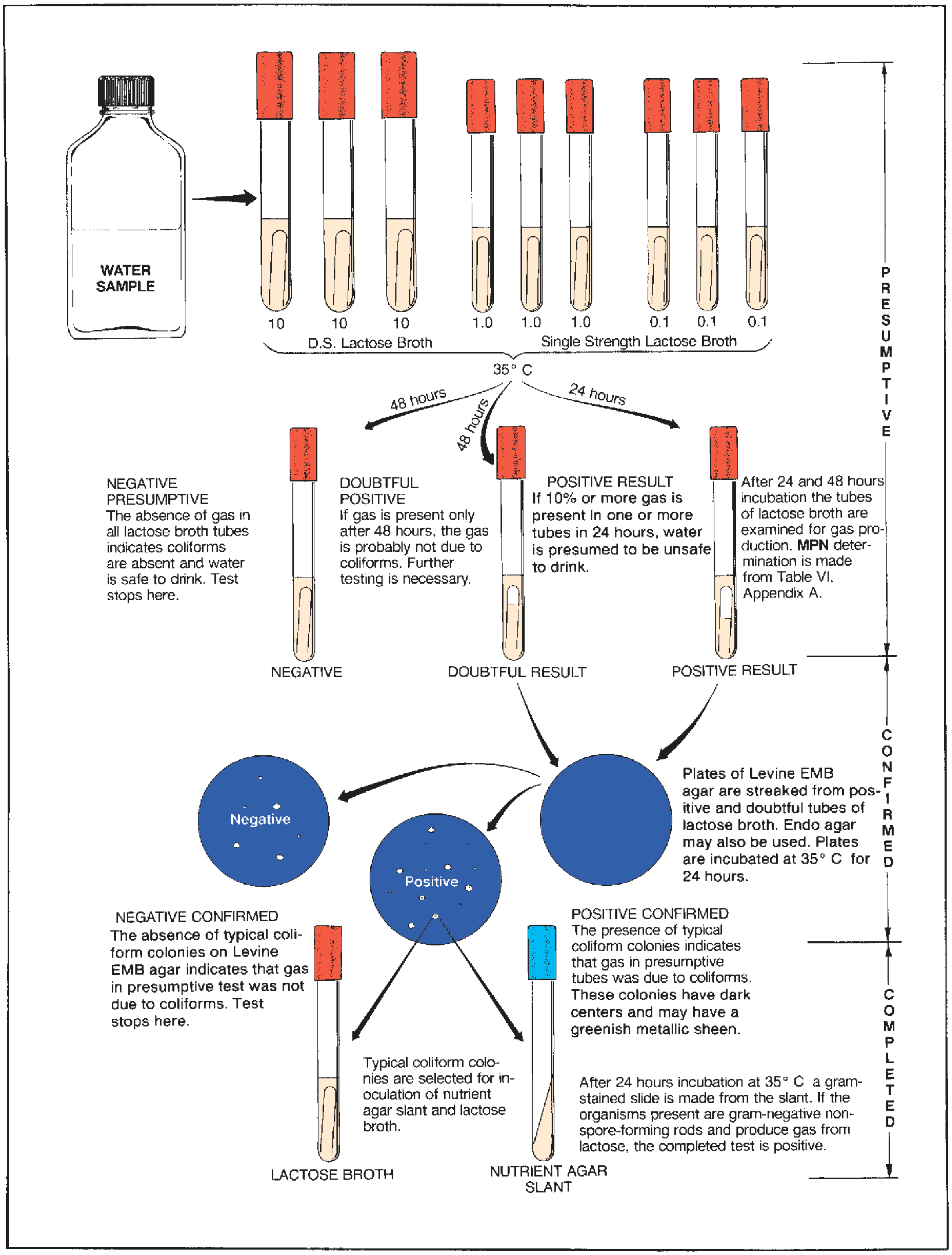


Figure 63.1 Bacteriological analysis of water

perform the **IMViC tests,** which are described on page 175 in Exercise 50.

In this exercise, water will be tested from local ponds, streams, swimming pools, and other sources supplied by students and instructor. Enough known positive samples will be evenly distributed throughout the laboratory so that all students will be able to see positive test results. All three tests in figure 63.1 will be performed. If time permits, the IMViC tests may also be performed.

The Presumptive Test

As stated earlier, the presumptive test is used to determine if gas-producing lactose fermenters are present in a water sample. If clear surface water is being tested, nine tubes of lactose broth will be used as shown in figure 63.1. For turbid surface water an additional three tubes of single strength lactose broth will be inoculated.

In addition to determining the presence or absence of coliforms, we can also use this series of lactose broth tubes to determine the **most probable number** (MPN) of coliforms present in 100 ml of water. A table for determining this value from the number of positive lactose tubes is provided in Appendix A.

Before setting up your test, determine whether your water sample is clear or turbid. Note that a separate set of instructions is provided for each type of water.

Clear Surface Water

If the water sample is relatively clear, proceed as follows:

Materials:

3 Durham tubes of DSLB
6 Durham tubes of SSLB
1 10 ml pipette
1 1 ml pipette
Note: DSLB designates double strength lactose broth. It contains twice as much lactose as SSLB (single strength lactose broth).

1. Set up 3 DSLB and 6 SSLB tubes as illustrated in figure 63.1. Label each tube according to the amount of water that is to be dispensed to it: *10 ml, 1.0 ml,* and *0.1 ml,* respectively.
2. Mix the bottle of water to be tested by shaking 25 times.
3. With a 10 ml pipette, transfer 10 ml of water to each of the DSLB tubes.
4. With a 1.0 ml pipette, transfer 1 ml of water to each of the middle set of tubes, and 0.1 ml to each of the last three SSLB tubes.
5. Incubate the tubes at 35° C for 24 hours.
6. Examine the tubes and record the number of tubes in each set that have 10% gas or more.
7. Determine the MPN by referring to table VI, Appendix A. Consider the following:
   Example: If you had gas in the first three tubes and gas only in one tube of the second series, but none in the last three tubes, your test would be read as 3–1–0. Table VI indicates that the MPN for this reading would be 43. This means that this particular sample of water would have approximately 43 organisms per 100 ml with 95% probability of there being between 7 and 210 organisms. *Keep in mind that the MPN figure of 43 is only a statistical probability figure.*
8. Record the data on the Laboratory Report.

Turbid Surface Water

If your water sample appears to have considerable pollution, do as follows:

Materials:

3 Durham tubes of DSLB
9 Durham tubes of SSLB
1 10 ml pipette
2 1 ml pipettes
1 water blank (99 ml of sterile water)
Note: See comment in previous materials list concerning DSLB and SSLB.

1. Set up three DSLB and nine SSLB tubes in a test-tube rack, with the DSLB tubes on the left.
2. Label the three DSLB tubes *10 ml,* the next three SSLB tubes *1.0 ml,* the next three SSLB tubes *0.1 ml,* and the last three tubes *0.01 ml.*
3. Mix the bottle of water to be tested by shaking 25 times.
4. With a 10 ml pipette, transfer 10 ml of water to each of the DSLB tubes.
5. With a 1.0 ml pipette, transfer 1 ml to each of the next three tubes, and 0.1 ml to each of the third set of tubes.
6. With the same 1 ml pipette, transfer 1 ml of water to the 99 ml blank of sterile water and shake 25 times.
7. *With a fresh 1 ml pipette,* transfer 1.0 ml of water from the blank to the remaining tubes of SSLB. This is equivalent to adding 0.01 ml of full-strength water sample.
8. Incubate the tubes at 35° C for 24 hours.
9. Examine the tubes and record the number of tubes in each set that have 10% gas or more.
10. Determine the MPN by referring to table VI, Appendix A. This table is set up for only 9 tubes. To apply a 12-tube reading to it, do as follows:

a. Select the three consecutive sets of tubes that have at least one tube with no gas.
b. If the first set of tubes (10 ml tubes) are not used, multiply the MPN by 10.

Example: Your tube reading was 3–3–3–1. What is the MPN?

The first set of tubes (10 ml) is ignored and the figures 3–3–1 are applied to the table. The MPN for this series is 460. Multiplying this by 10, the MPN becomes 4600.

Example: Your tube reading was 3–1–2–0. What is the MPN?

The first three numbers are (3–1–2) applied to the table. The MPN is 210. Since the last set of tubes is ignored, 210 is the MPN.

The Confirmed Test

Once it has been established that gas-producing lactose fermenters are present in the water, it is *presumed* to be unsafe. However, gas formation may be due to noncoliform bacteria. Some of these organisms, such as *Clostridium perfringens,* are gram-positive. To confirm the presence of gram-negative lactose fermenters, the next step is to inoculate media such as Levine eosin–methylene blue agar or Endo agar from positive presumptive tubes.

Levine EMB agar contains methylene blue, which inhibits gram-positive bacteria. Gram-negative lactose fermenters (coliforms) that grow on this medium will produce “nucleated colonies” (dark centers). Colonies of *E. coli* and *E. aerogenes* can be differentiated on the basis of size and the presence of a greenish metallic sheen. *E. coli* colonies on this medium are small and have this metallic sheen, whereas *E. aerogenes* colonies usually lack the sheen and are larger. Differentiation in this manner is not completely reliable, however. It should be remembered that *E. coli* is the more reliable sewage indicator since it is not normally present in soil, while *E. aerogenes* has been isolated from soil and grains.

Endo agar contains a fuchsin sulfite indicator that makes identification of lactose fermenters relatively easy. Coliform colonies and the surrounding medium appear red on Endo agar. Nonfermenters of lactose, on the other hand, are colorless and do not affect the color of the medium.

In addition to these two media, there are several other media that can be used for the confirmed test. Brilliant green bile lactose broth, Eijkman’s medium, and EC medium are just a few examples that can be used.

To demonstrate the confirmation of a positive presumptive in this exercise, the class will use Levine EMB agar and Endo agar. One half of the class will use one medium; the other half will use the other medium. Plates will be exchanged for comparisons.

Materials:

1 Petri plate of Levine EMB agar (odd-numbered students)
1 Petri plate of Endo agar (even-numbered students)

1. Select one positive lactose broth tube from the presumptive test and streak a plate of medium according to your assignment. Use a streak method that will produce good isolation of colonies. If all your tubes were negative, borrow a positive tube from another student.
2. Incubate the plate for 24 hours at 35° C.
3. Look for typical coliform colonies on both kinds of media. Record your results on the Laboratory Report. If no coliform colonies are present, the water is considered bacteriologically safe to drink.
   Note: In actual practice, confirmation of all presumptive tubes would be necessary to ensure accuracy of results.

The Completed Test

A final check of the colonies that appear on the confirmatory media is made by inoculating a nutrient agar slant and a Durham tube of lactose broth. After incubation for 24 hours at 35° C, the lactose broth is examined for gas production. A gram-stained slide is made from the slant, and the slide is examined under oil immersion optics.

If the organism proves to be a gram-negative, non-spore-forming rod that ferments lactose, we know that coliforms were present in the tested water sample. If time permits, complete these last tests and record the results on the Laboratory Report.

The IMViC Tests

Review the discussion of the IMViC tests on page 175. The significance of these tests should be much more apparent at this time. Your instructor will indicate whether these tests should also be performed if you have a positive completed test.

64 The Membrane Filter Method

In addition to the multiple tube test, a method utilizing the membrane filter has been recognized by the United States Public Health Service as a reliable method for the detection of coliforms in water. These filter disks are 150 micrometers thick, have pores of 0.45 micrometer diameter, and have 80% area perforation. The precision of manufacture is such that bacteria larger than 0.47 micrometer cannot pass through. Eighty percent area perforation facilitates rapid filtration.

To test a sample of water, the water is passed through one of these filters. All bacteria present in the sample will be retained directly on the filter's surface. The membrane filter is then placed on an absorbent pad saturated with liquid nutrient medium and incubated for 22 to 24 hours. The organisms on the filter disk will form colonies that can be counted under the microscope. If a differential medium such as *m* Endo MF broth is used, coliforms will exhibit a characteristic golden metallic sheen.

The advantages of this method over the multiple tube test are (1) higher degree of reproducibility of results; (2) greater sensitivity since larger volumes of water can be used; and (3) shorter time (one-fourth) for getting results.

Figure 64.1 illustrates the procedure we will use in this experiment.

Materials:

vacuum pump or water faucet aspirators
membrane filter assemblies (sterile)
side-arm flask, 1000 ml size, and rubber hose
sterile graduates (100 ml or 250 ml size)
sterile, plastic Petri dishes, 50 mm dia (Millipore #PD10 047 00)
sterile membrane filter disks (Millipore #HAWG 047 AO)
sterile absorbent disks (packed with filters)
sterile water
5 ml pipettes
bottles of *m* Endo MF broth (50 ml)*
water samples

1. Prepare a small plastic Petri dish as follows:
   a. With a flamed forceps, transfer a sterile absorbent pad to a sterile plastic Petri dish.
   b. Using a 5 ml pipette, transfer 2.0 ml of *m* Endo MF broth to the absorbent pad.
2. Assemble a membrane filtering unit as follows:
   a. *Aseptically* insert the filter holder base into the neck of a 1-liter side-arm flask.
   b. With a flamed forceps, place a sterile membrane filter disk, grid side up, on the filter holder base.
   c. Place the filter funnel on top of the membrane filter disk and secure it to the base with the clamp.
3. Attach the rubber hose to a vacuum source (pump or water aspirator) and pour the appropriate amount of water into the funnel.

   The amount of water used will depend on water quality. No less than 50 ml should be used. Waters with few bacteria and low turbidity permit samples of 200 ml or more. Your instructor will advise you as to the amount of water that you should use. Use a sterile graduate for measuring the water.
4. Rinse the inner sides of the funnel with 20 ml of sterile water.
5. Disconnect the vacuum source, remove the funnel, and carefully transfer the filter disk with sterile forceps to the Petri dish of *m* Endo MF broth. *Keep grid side up.*
6. Incubate at 35° C for 22 to 24 hours. *Don't invert.*
7. After incubation, remove the filter from the dish and dry for 1 hour on absorbent paper.
8. Count the colonies on the disk with low-power magnification, using reflected light. Ignore all colonies that lack the golden metallic sheen. If desired, the disk may be held flat by mounting between two 2″ × 3″ microscope slides after drying. Record your count on the first portion of Laboratory Report 64, 65.

*See Appendix C for special preparation method.

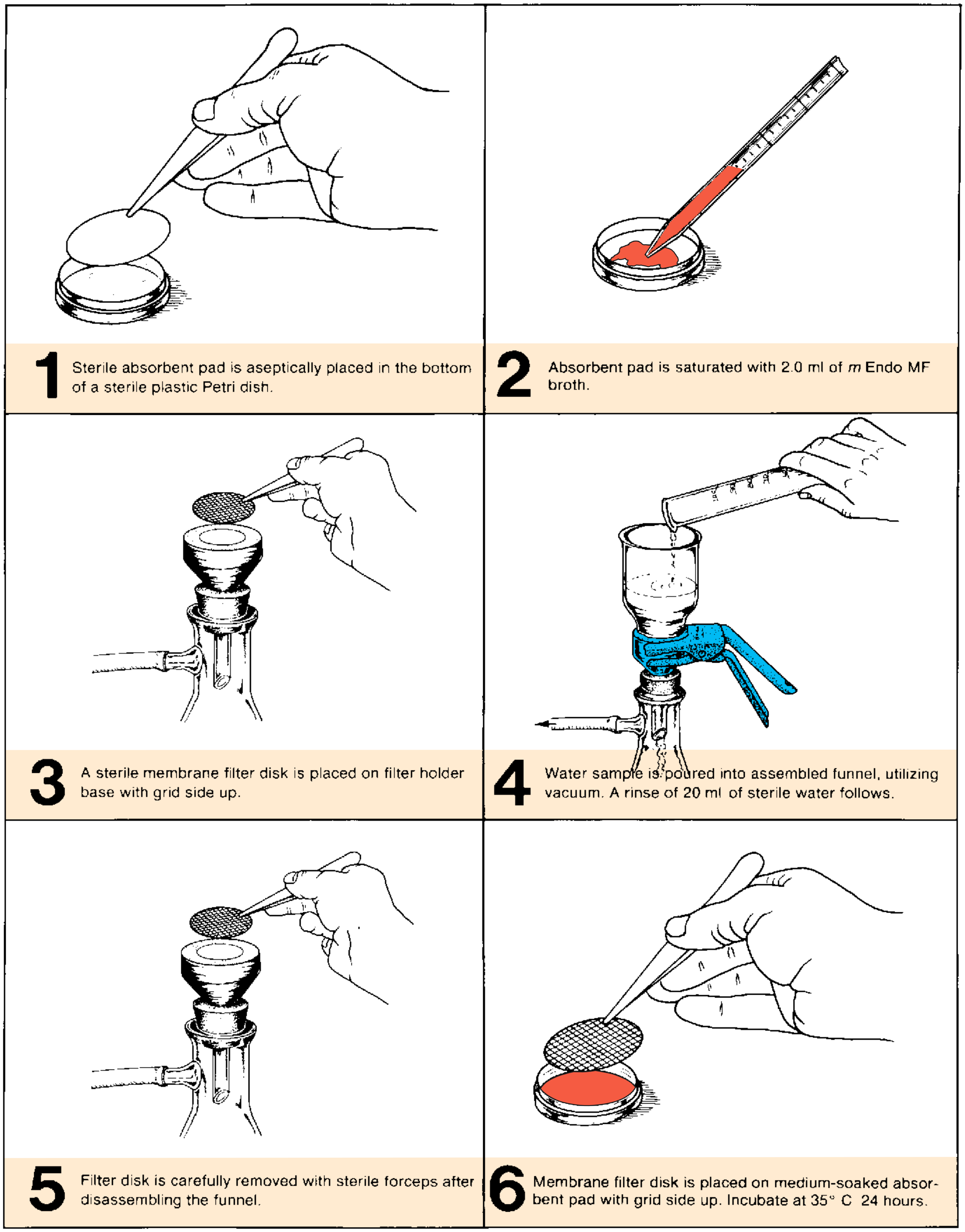


Figure 64.1 Membrane filter routine